

Coding with AI Agent



**[https://github.com/up1/
course-ai-coding-2026](https://github.com/up1/course-ai-coding-2026)**



Coding with AI Agent



Topics

Software development workflows with AI Agent
Spec-Driven Development (SDD)
Model-Driven Development (MDD)
Coding with AI Agent workshop

Requirement

Planning

Coding
Testing

Validate and Testing



"codex is better"

"claude.md"

"i just vibecoded"

"skills"

"ralph"

"agentic"

"obsidian"

"mcp"

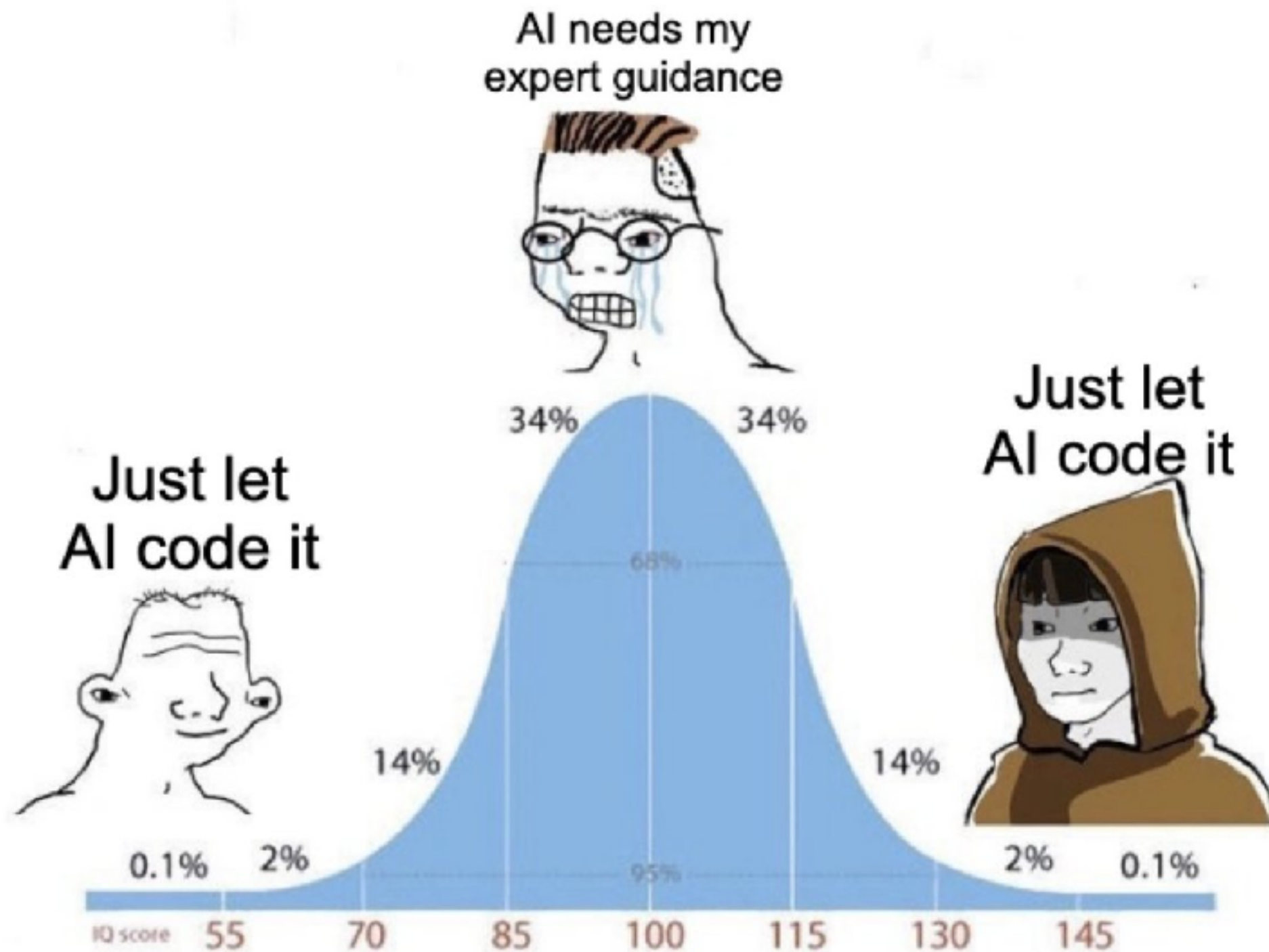
"mac mini"

"clawdbot"

"agent sdk"

"remotion"





AI as
a **Tool**

AI as
an **Assistant**

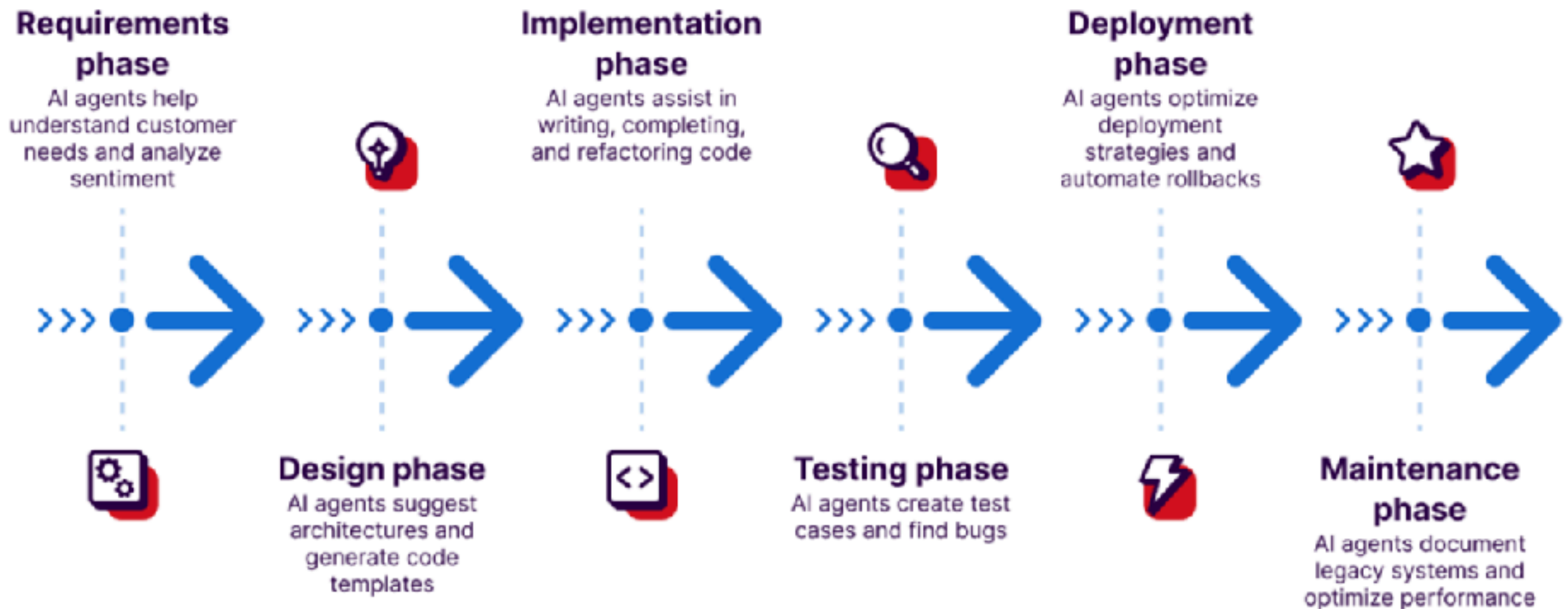
AI as
an **Agent**



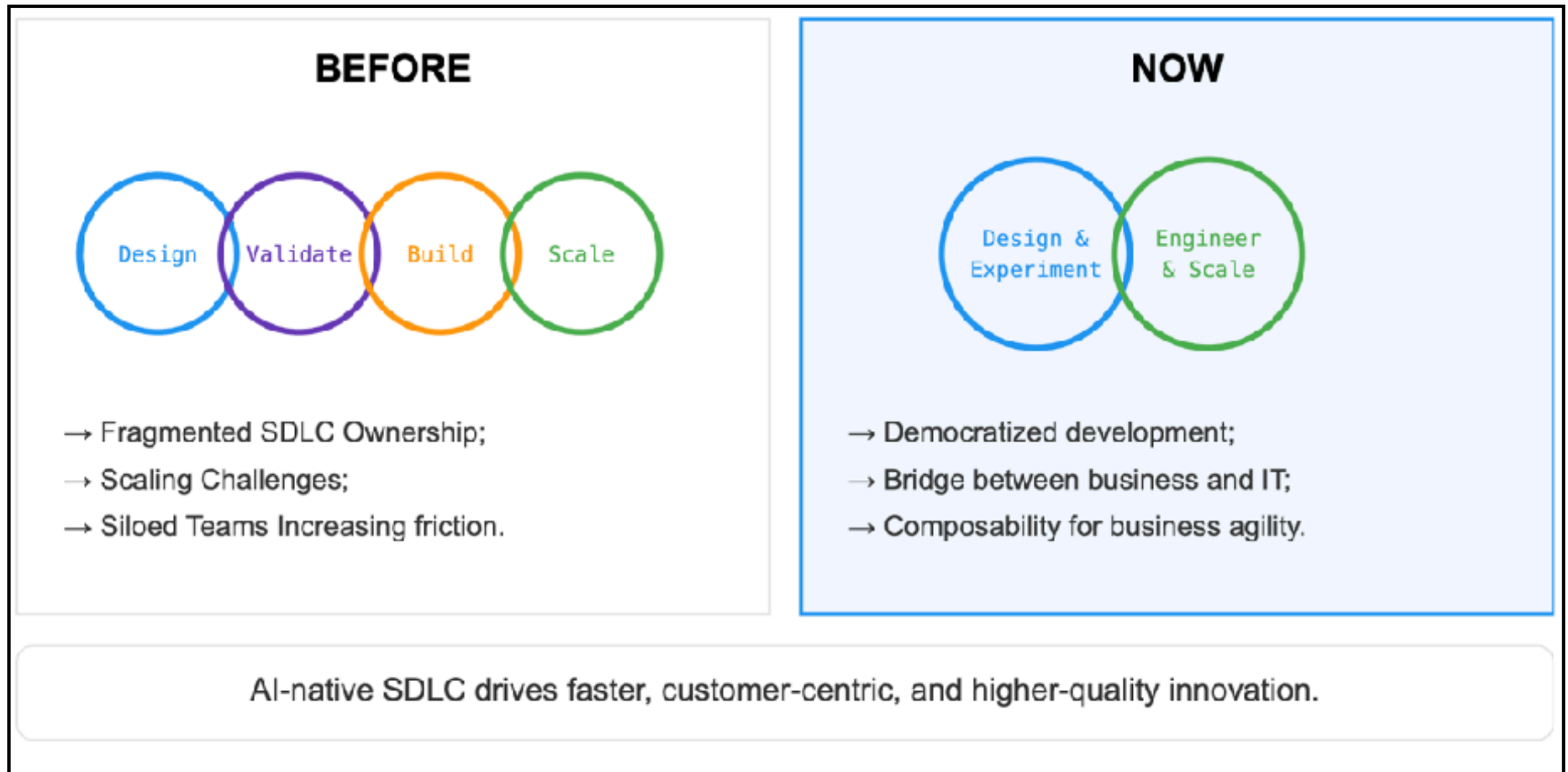
AI is the new normal in software development



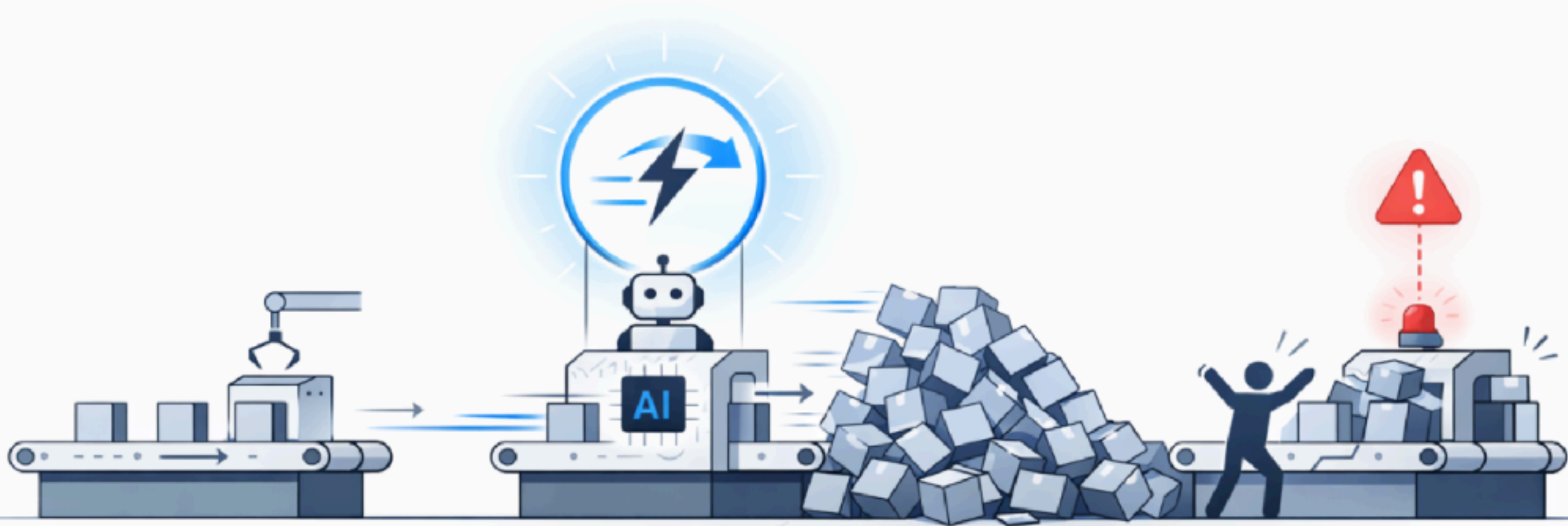
AI in Software Development



AI in Software Development

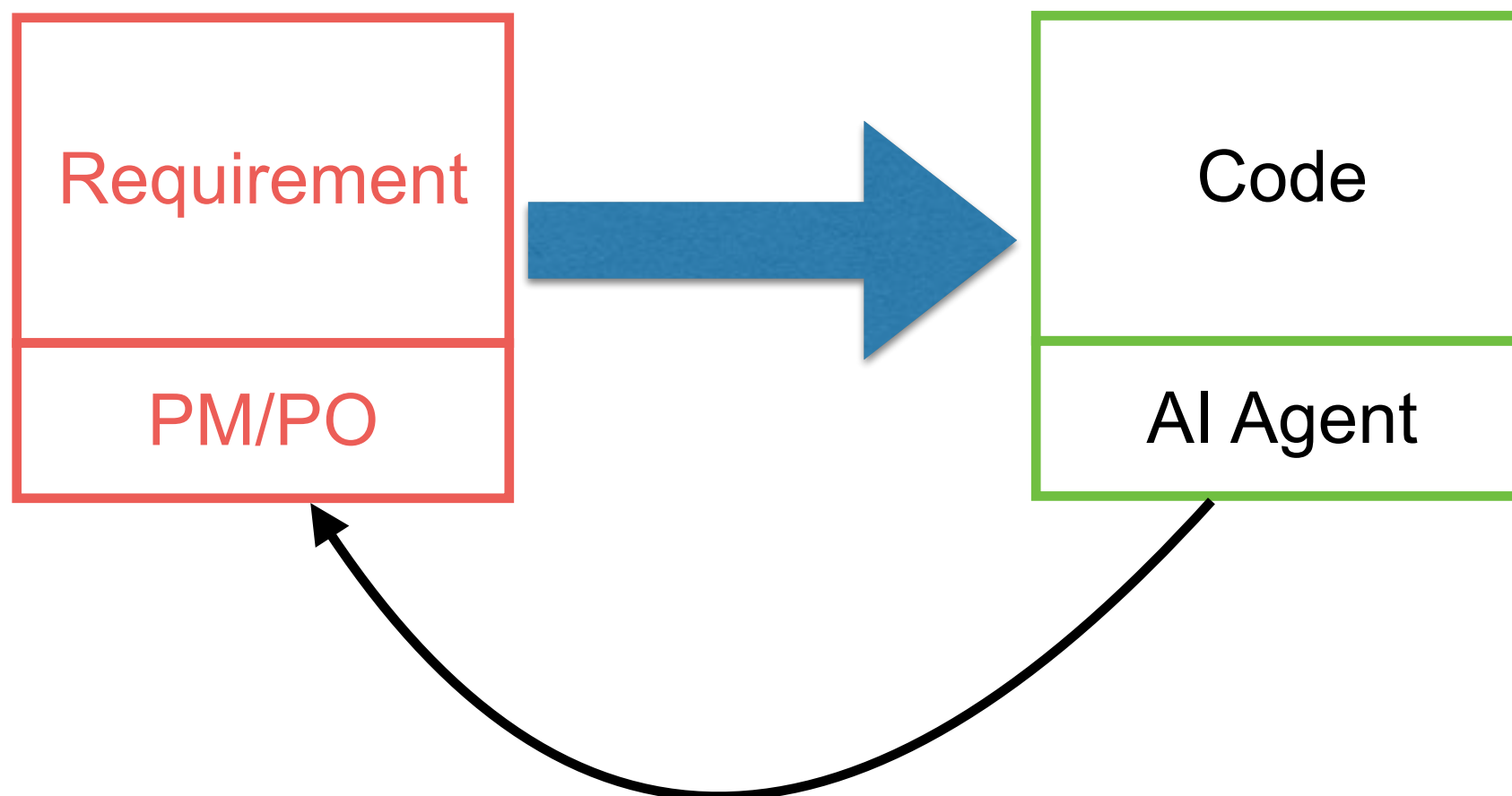


Invest in Real Bottleneck ?



New Bottleneck ?

Decide what to **build**, not code



Get feedback faster

<https://www.theaiopportunities.com/p/is-ai-slowng-down-the-new-bottleneck>



DORA



State of AI-assisted Software Development

Google Cloud

2025

<https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>



AI and Productivity ?

AI Adoption

90% of software development professionals are now using AI at work

Productivity

80% report that AI has increased their productivity

Reliance

65% are relying on AI at least a “moderate amount” at work

Code Quality

59% of survey respondents say that AI has positively impacted their code quality

<https://blog.google/innovation-and-ai/technology/developers-tools/dora-report-2025/>

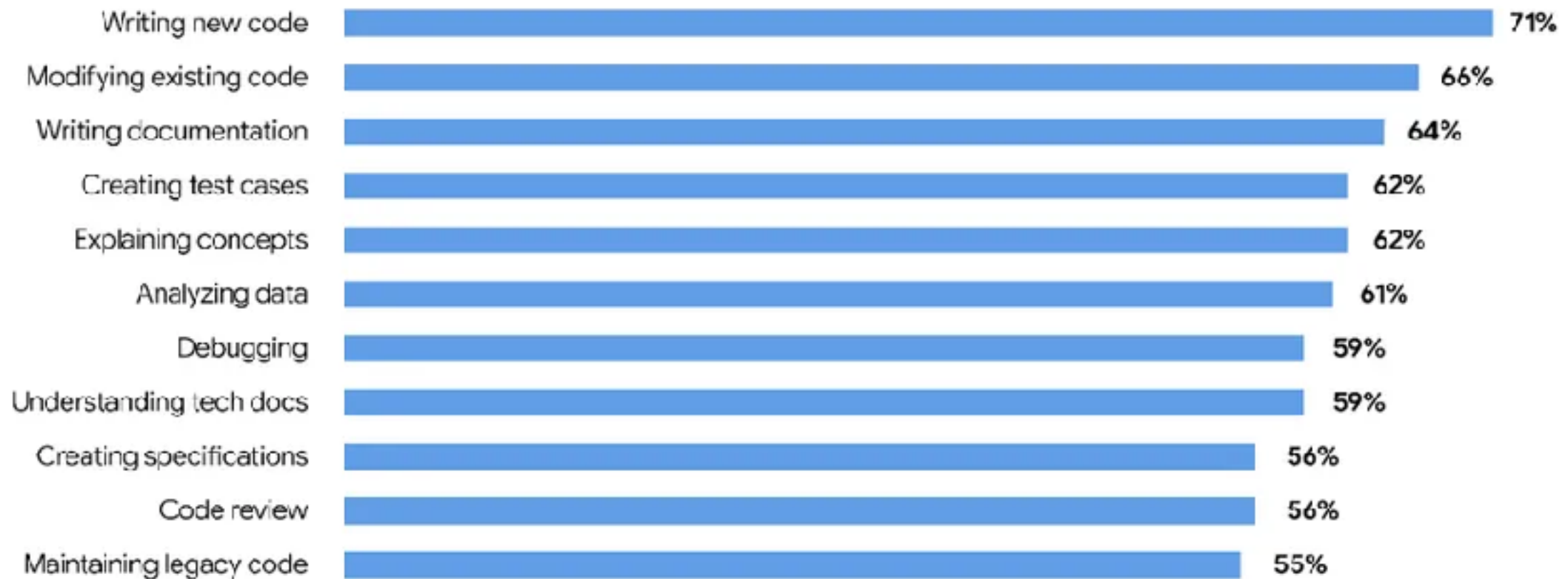


Productivity ?

Reliance on AI by task

DORA
2025

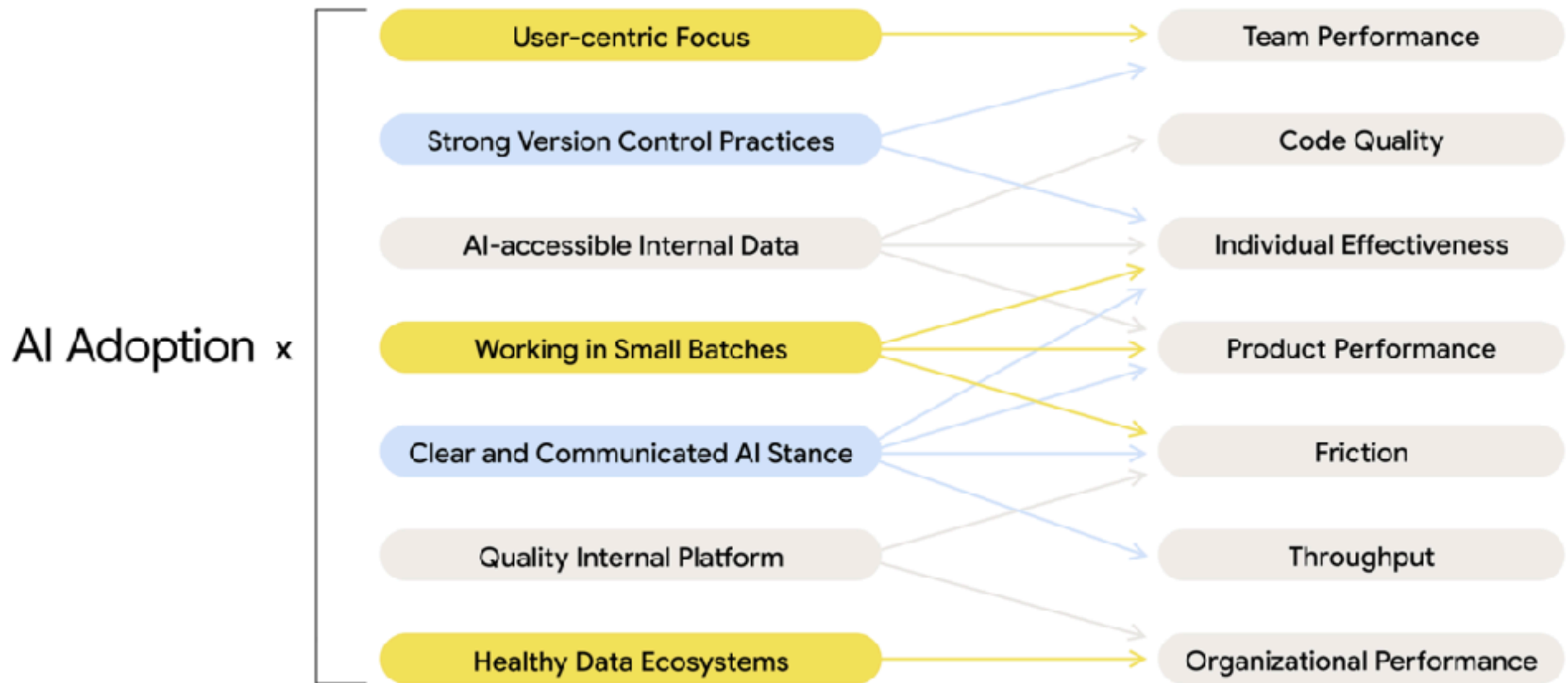
Percentage of task performers who use AI



<https://blog.google/innovation-and-ai/technology/developers-tools/dora-report-2025/>



DORA AI Capabilities Model



<https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>



History of Programming



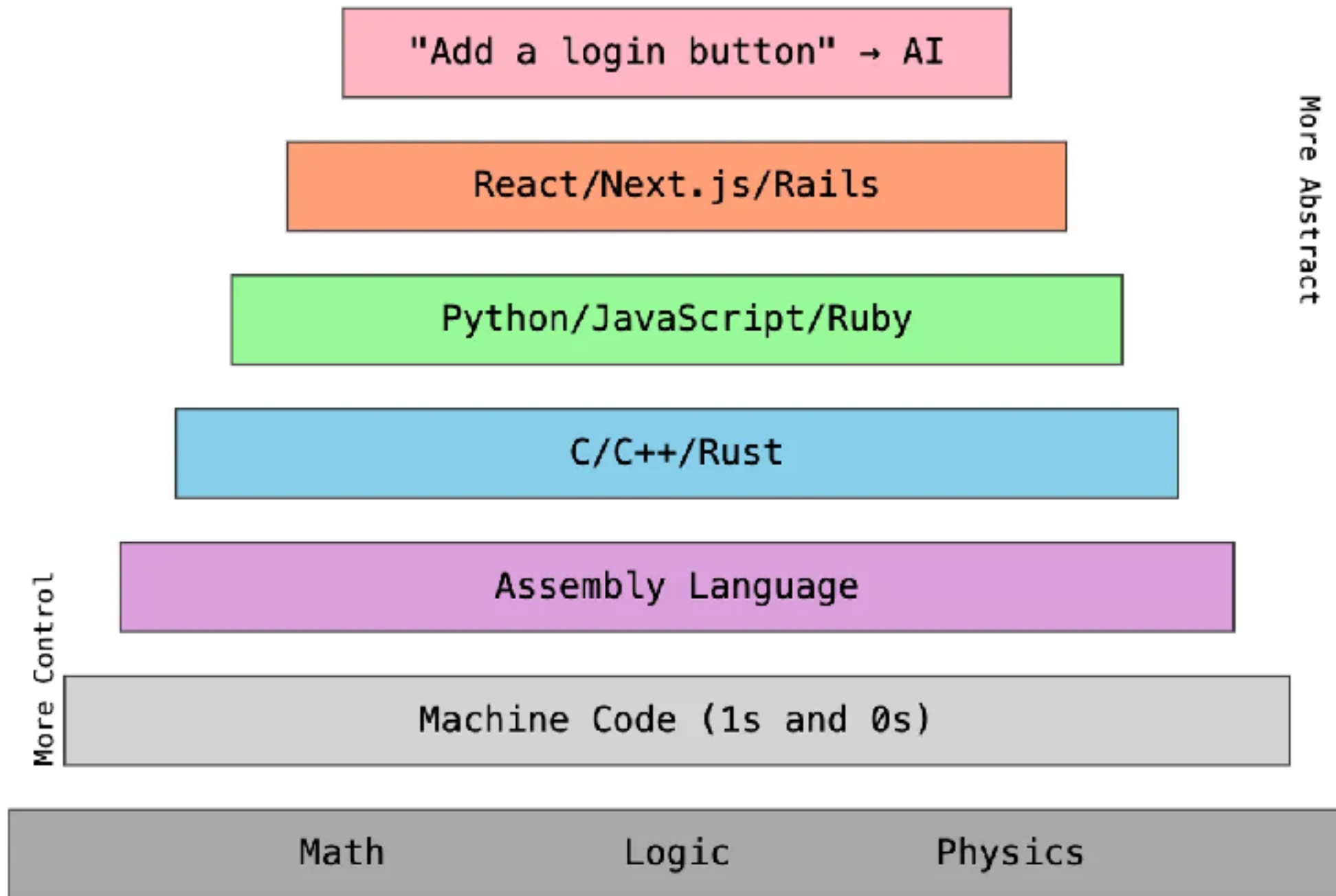
History of Programming



Increase of abstraction



The Modern Tower of Abstraction



<https://cline.bot/blog/from-assembly-to-ai-why-vibe-coding-is-just-another-chapter-in-our-abstraction-story>





Andrej Karpathy ✓

@karpathy



There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

6:17 AM · Feb 3, 2025 · 5.3M Views

<https://x.com/karpathy/status/1886192184808149383>



Vibe Coding

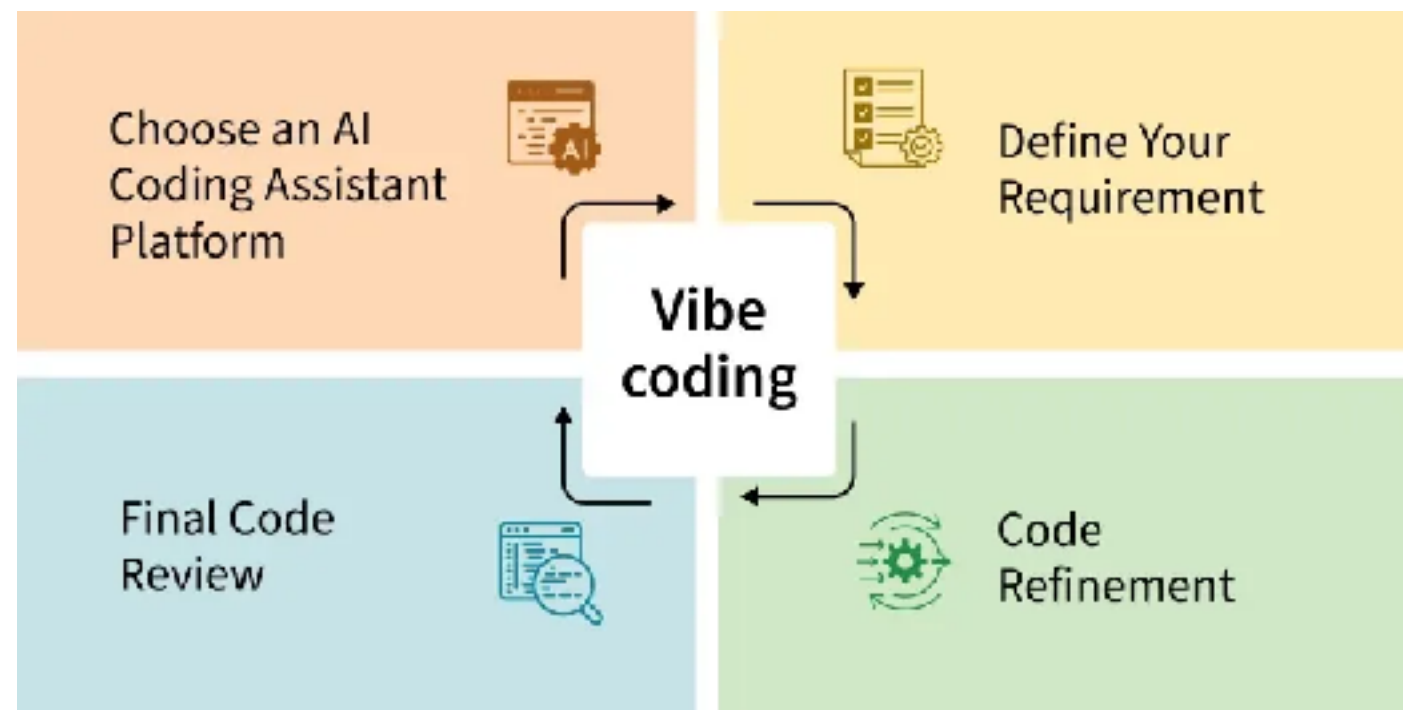
Software development practice that uses artificial intelligence (AI) to generate functional code from **natural language prompts**, accelerating development, and making app building more accessible, especially for those with limited programming experience

<https://cloud.google.com/discover/what-is-vibe-coding>



Types of Vibe Coding ?

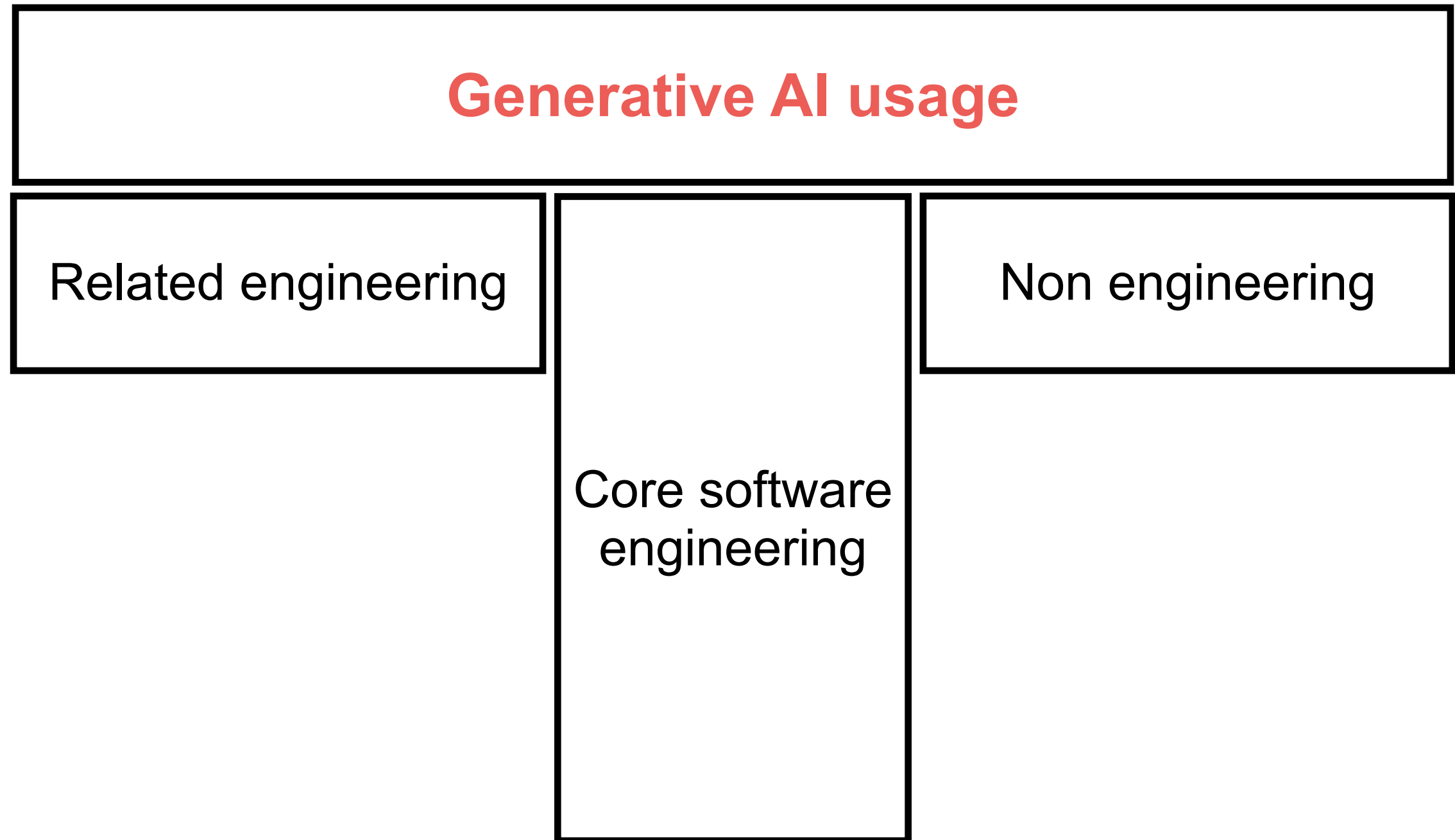
Pure vibe coding
Responsible AI-assist development



<https://cloud.google.com/discover/what-is-vibe-coding>



T-Shape Skills



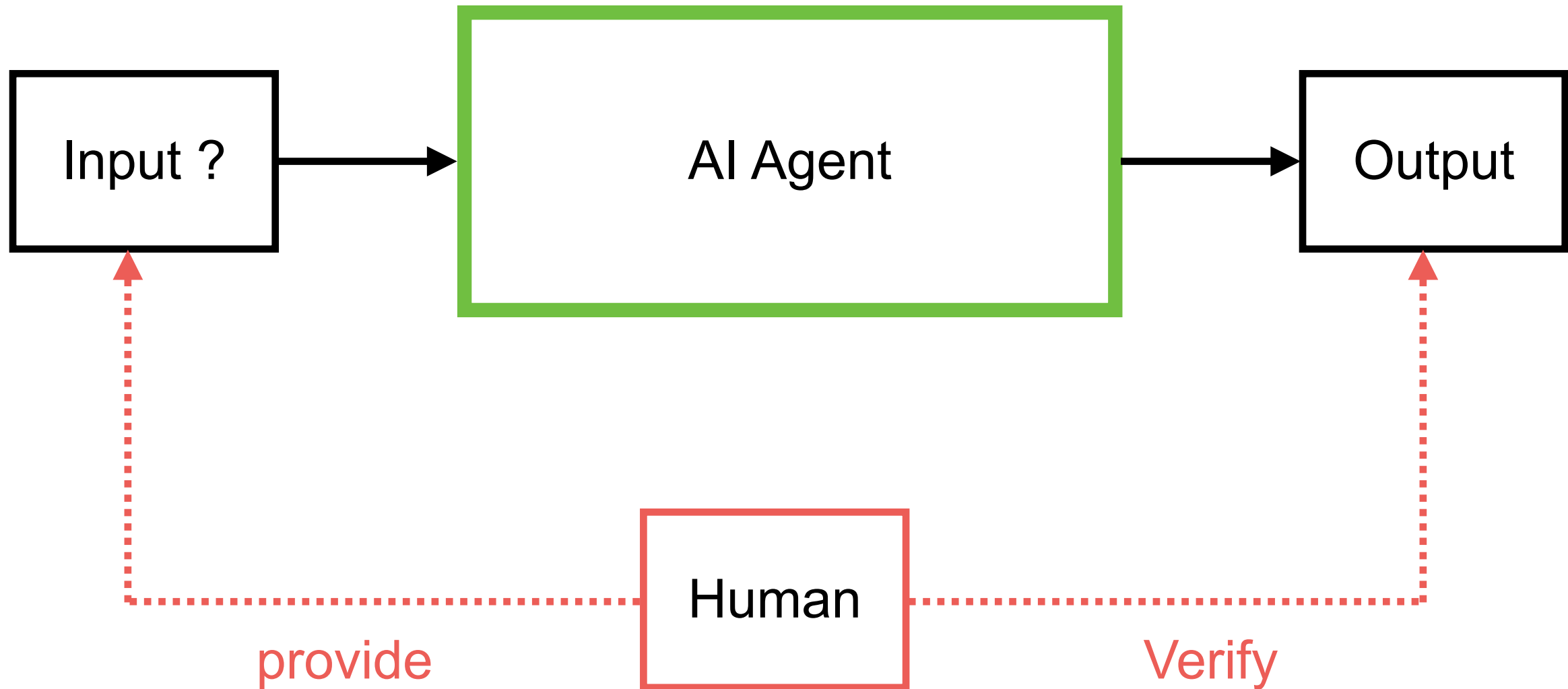
https://www.youtube.com/watch?v=q_Jq4lgYImk



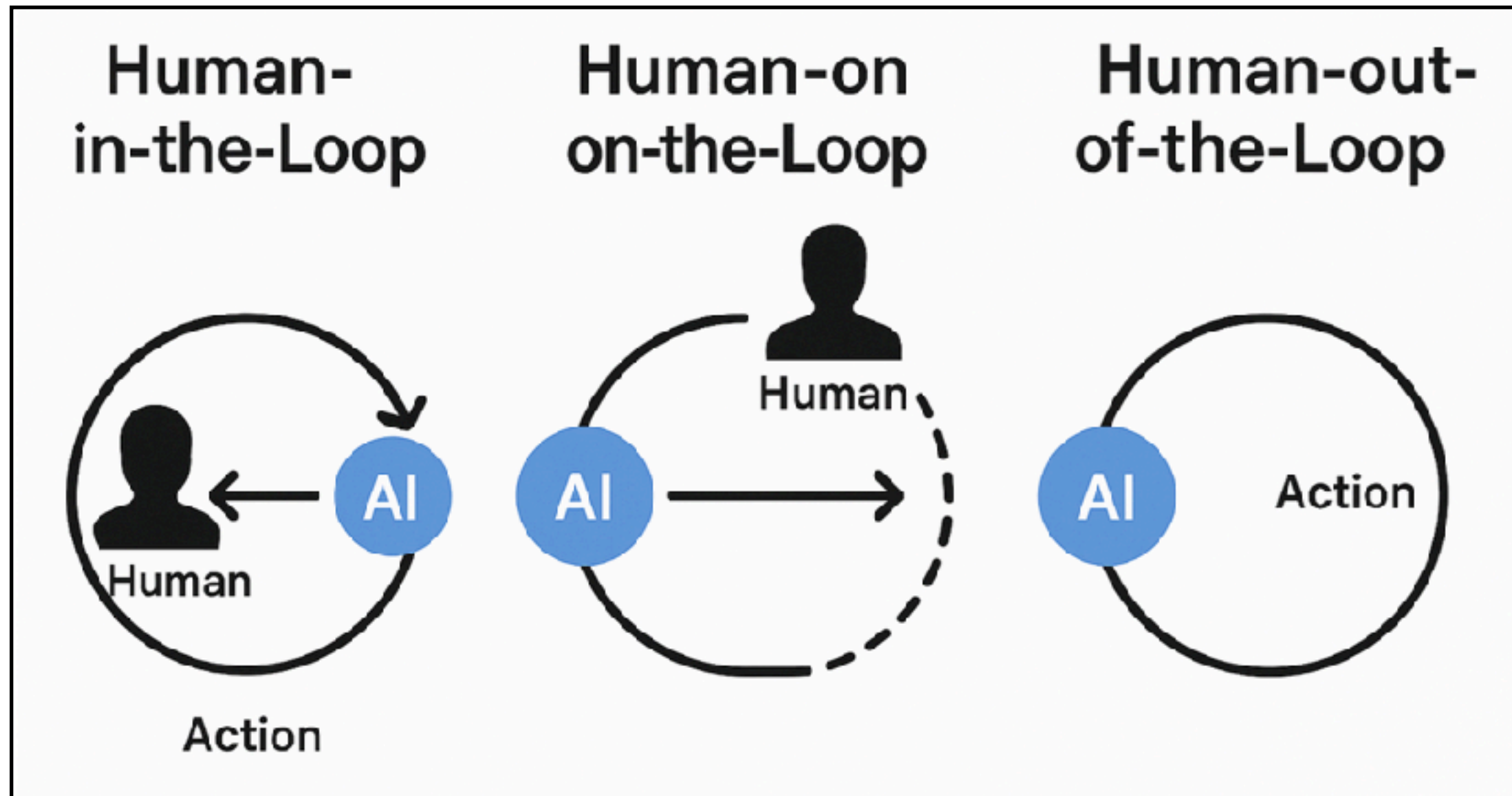
Working with AI Agent !!



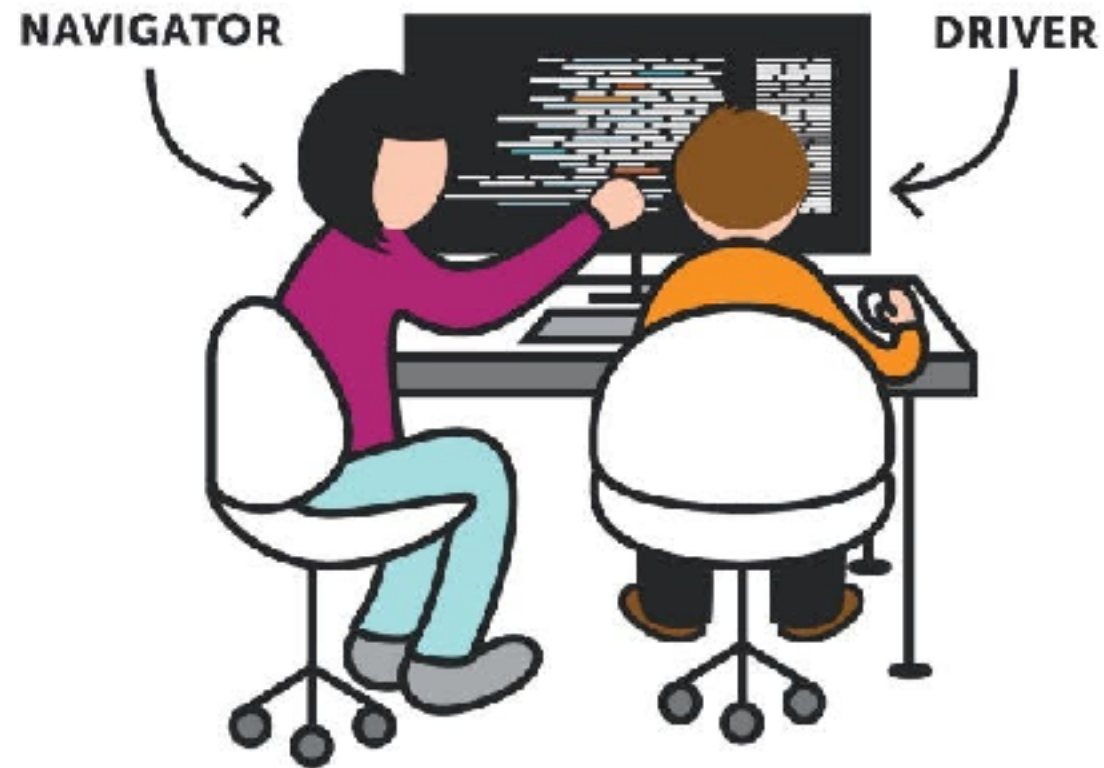
Coding with AI Agent



AI Agent with/without Human



PAIR PROGRAMMING



Reviews, Tests and Understands
with experience



Verify ?

Code quality
Functional requirement
Security review
???

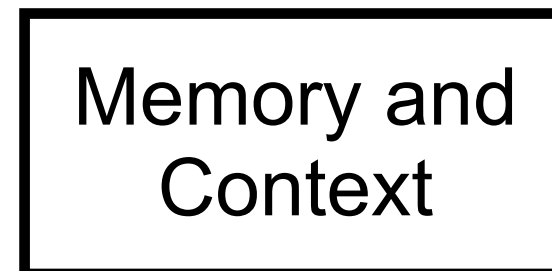
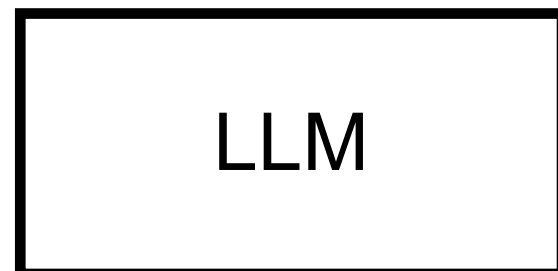
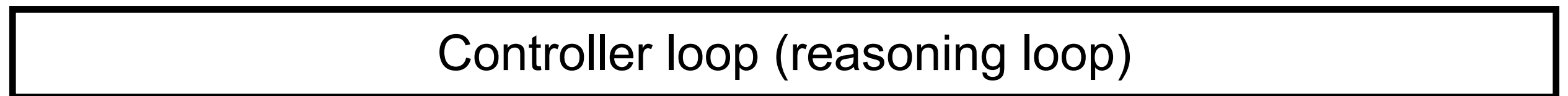


AI draft

Human craft

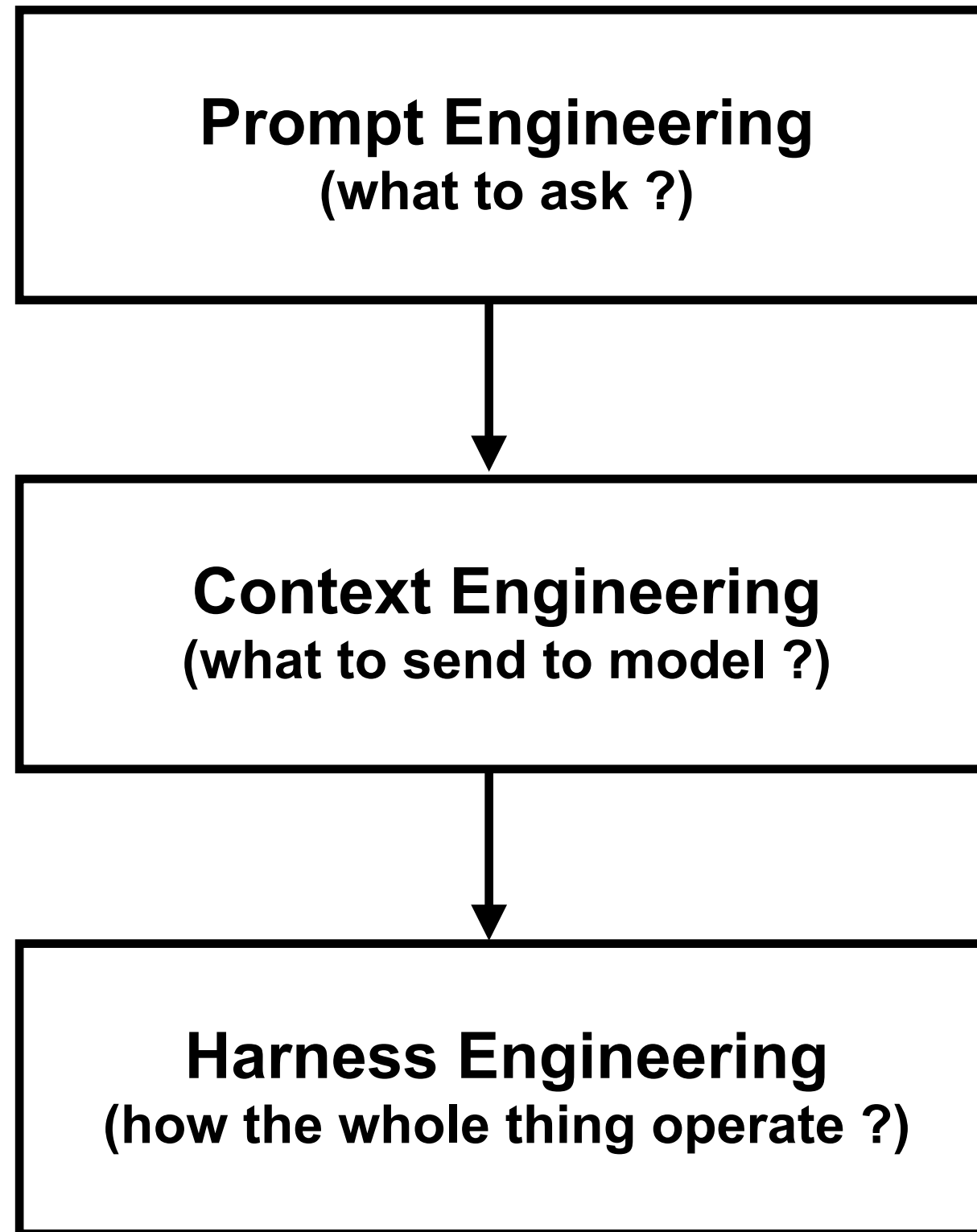


AI Agent for Coding



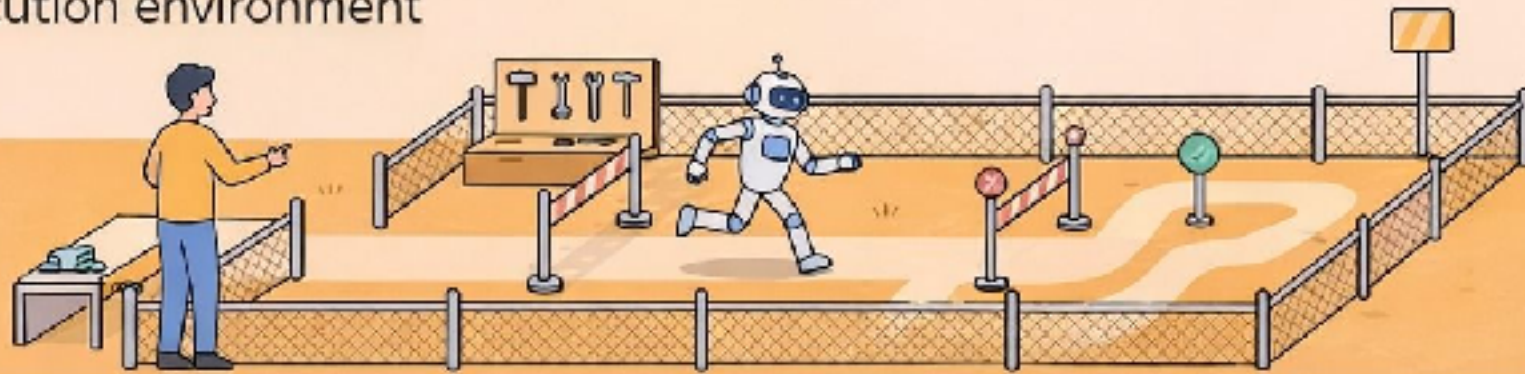
How to work with AI Agent ?





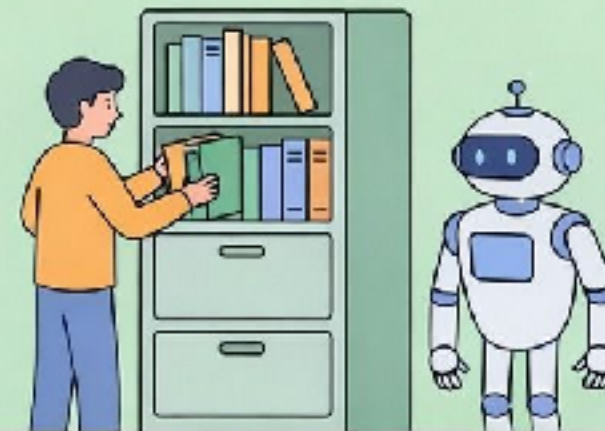
Harness Engineering — Build the World

Design the execution environment



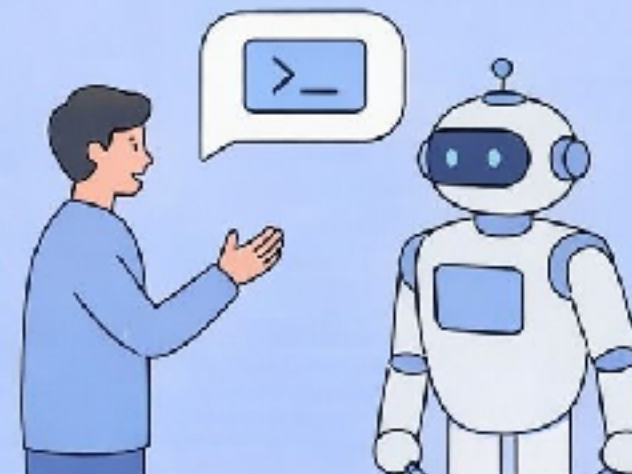
Context Engineering — What to See

Manage visible information



Prompt Engineering — What to Say

Optimize single-turn interactions



<https://milvus.io/blog/harness-engineering-ai-agents.md>



2 Background: From Weights to Context to Harness

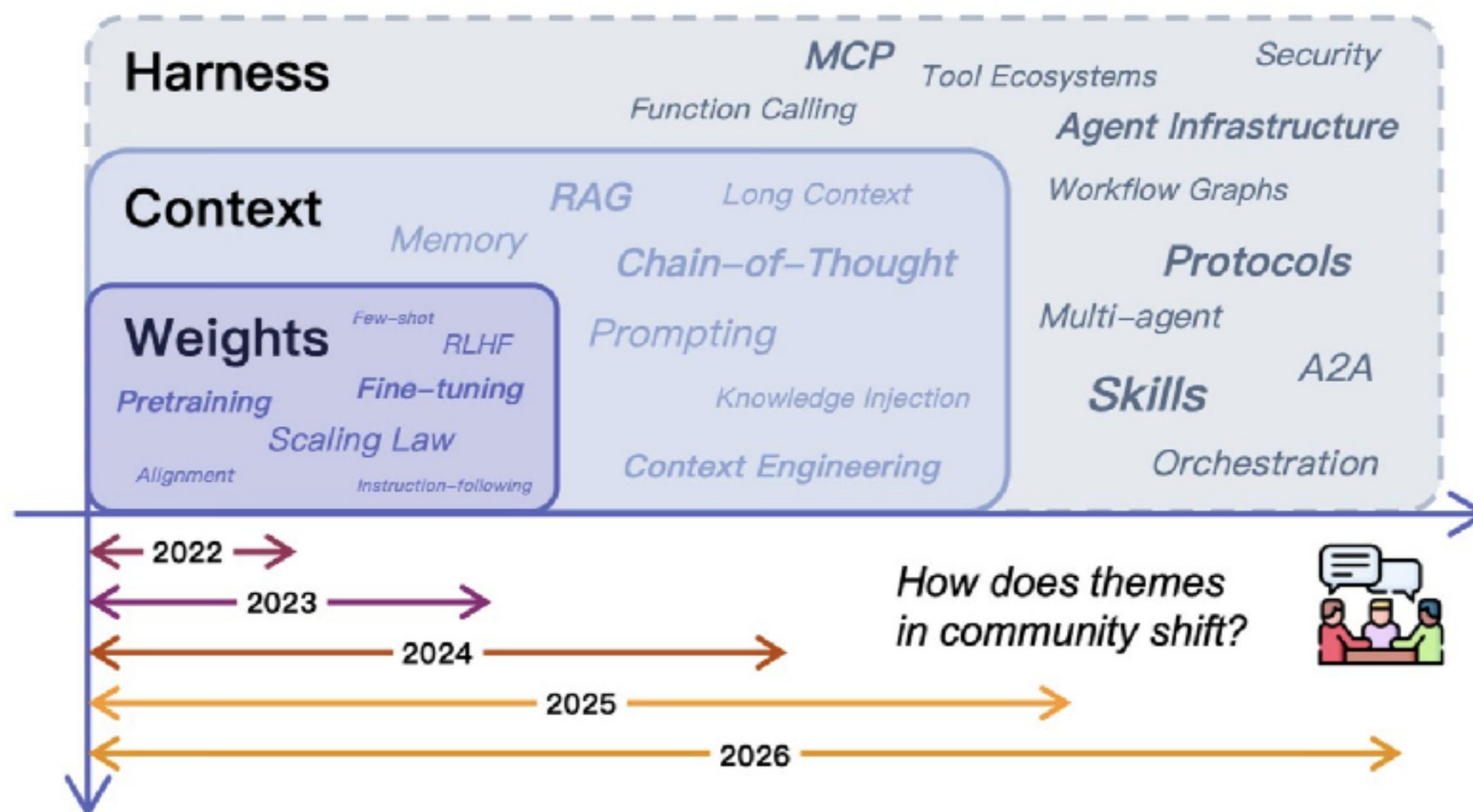
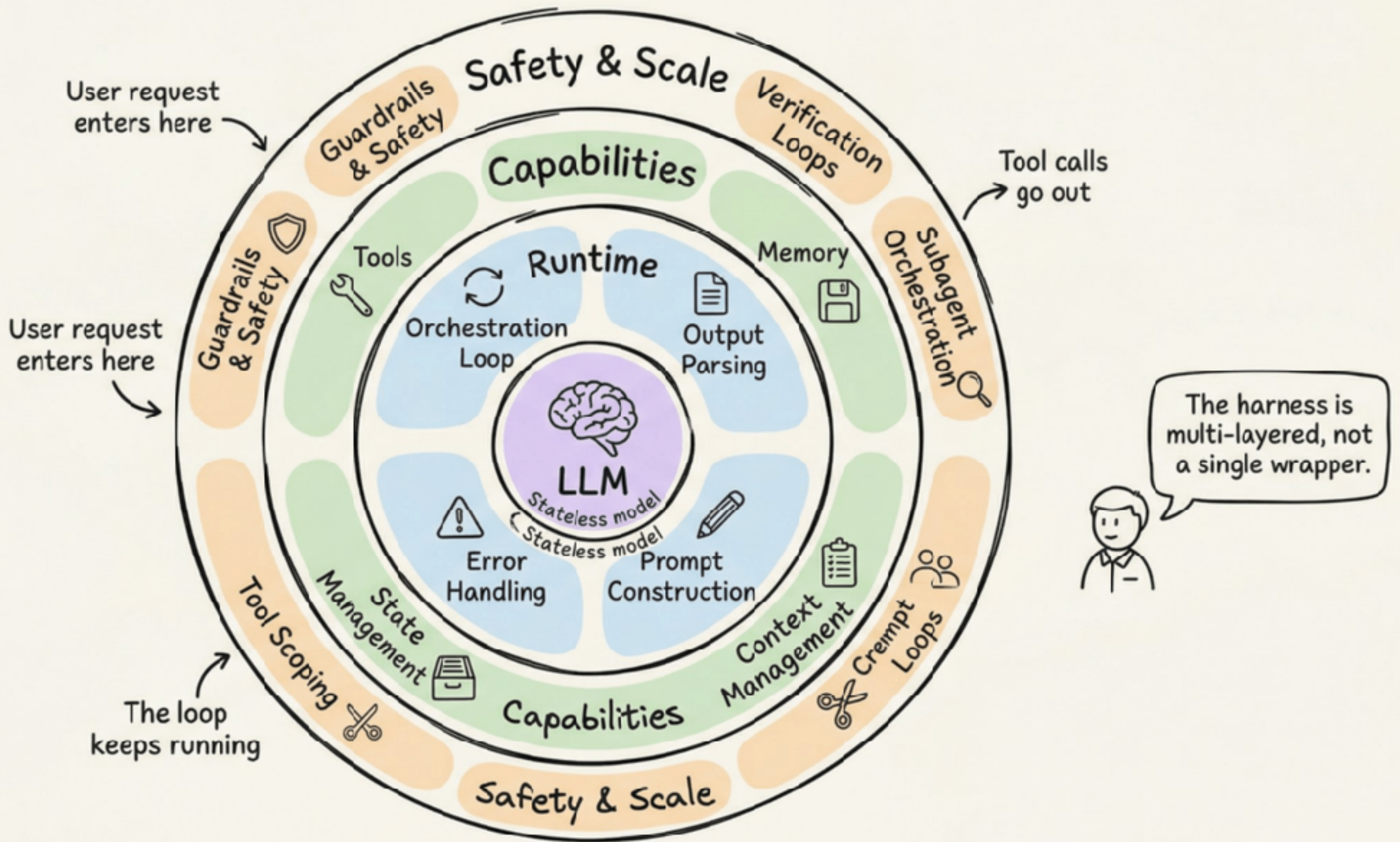


Figure 2 Community theme evolution across three capability layers. The stacked layers—Weights, Context, and Harness—show how the center of gravity in the LLM agent community has shifted outward over time, from parametric knowledge and prompting toward harness-level infrastructure such as tool ecosystems, protocols, skills, and multi-agent orchestration.

<https://arxiv.org/pdf/2604.08224>

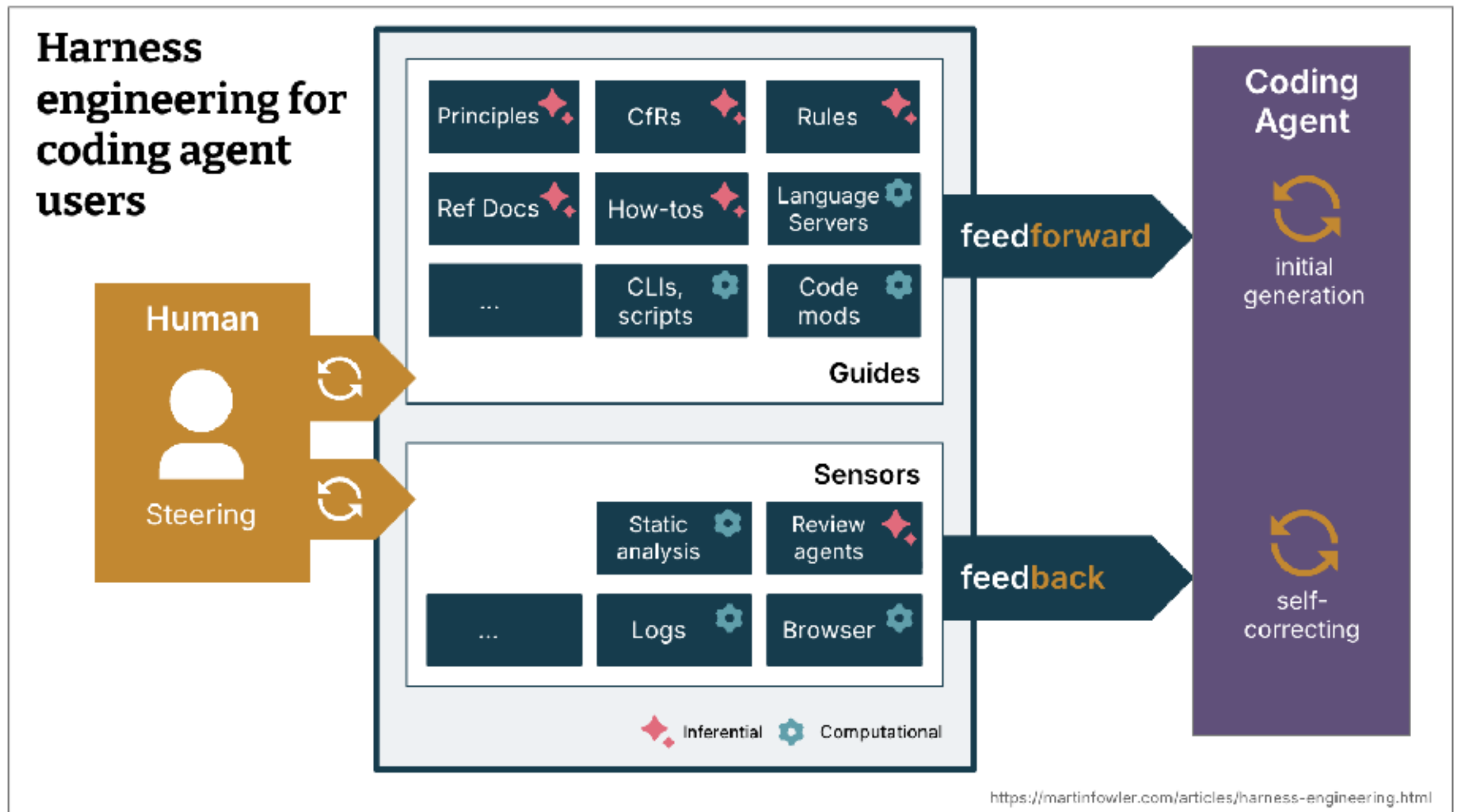




<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>



Harness Engineering



<https://martinfowler.com/articles/harness-engineering.html>



AI Agent = LLM + Harness

Harness = AI Agent - LLM



Workflows with AI Agent ?



Workflows ?

Application level

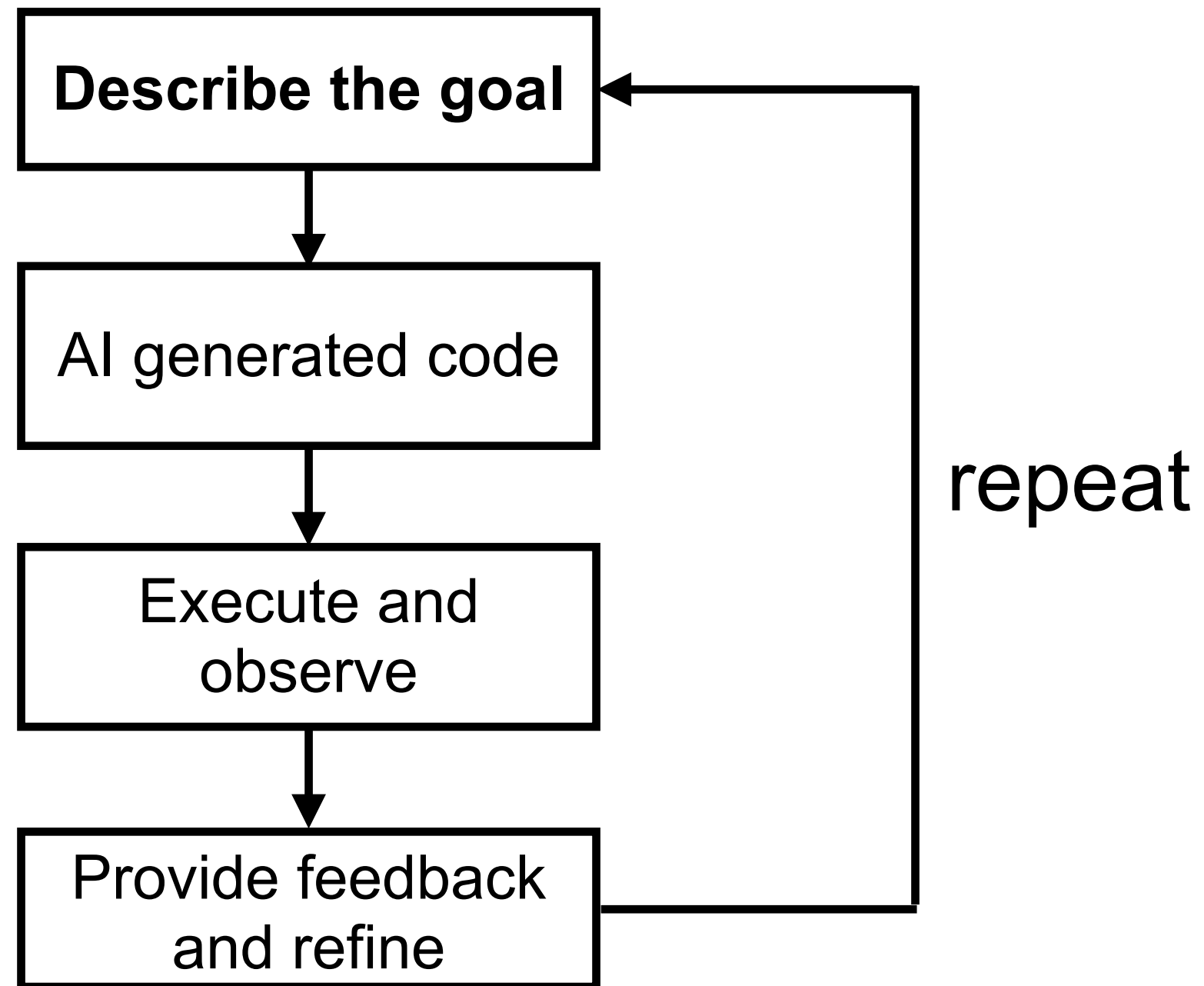
Code level

<https://cloud.google.com/discover/what-is-vibe-coding>



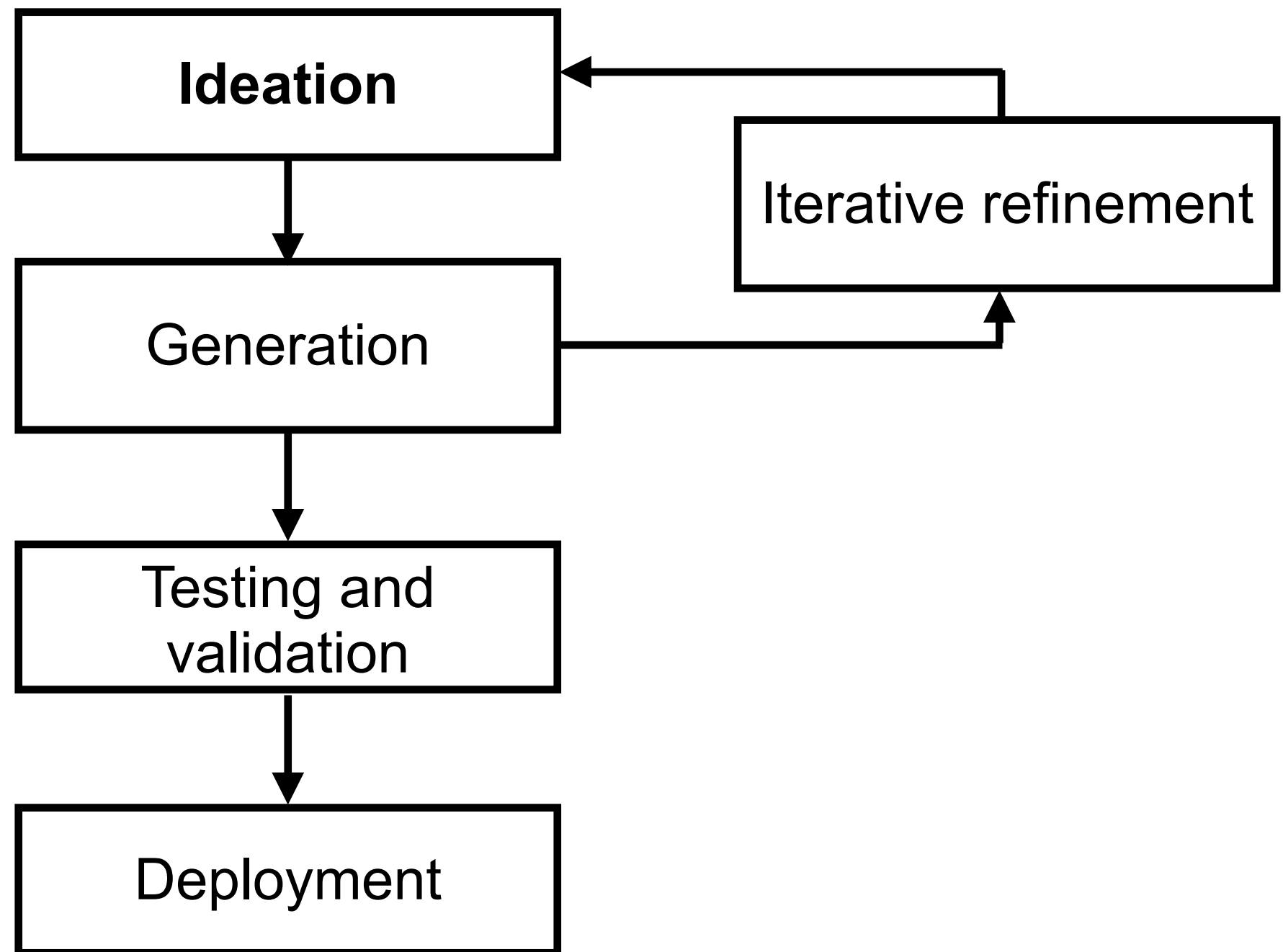
Code level workflow

Conversational loop to create a specific code



Application level workflow

Talking about high-level idea from concept to deployment

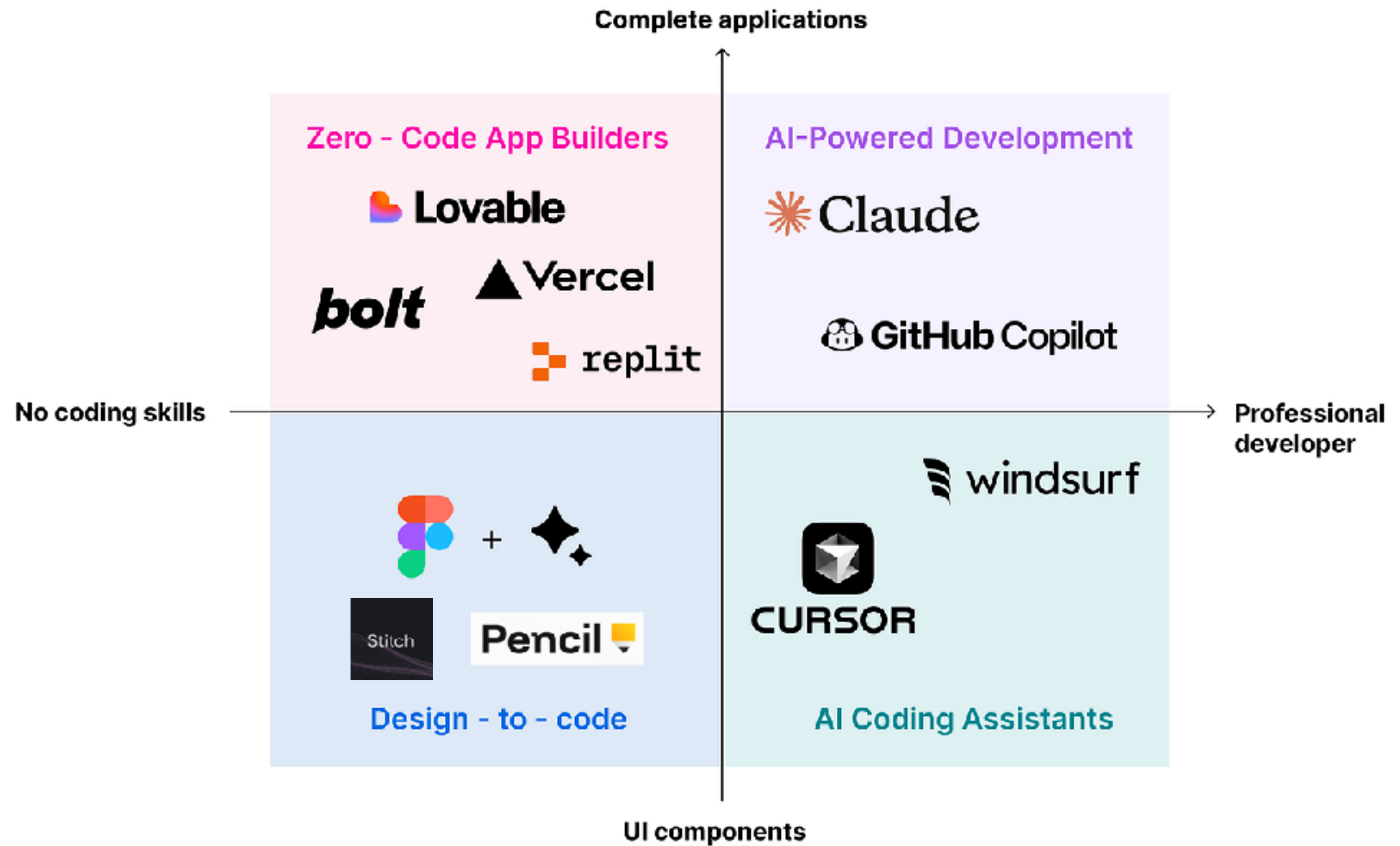


Tools ?

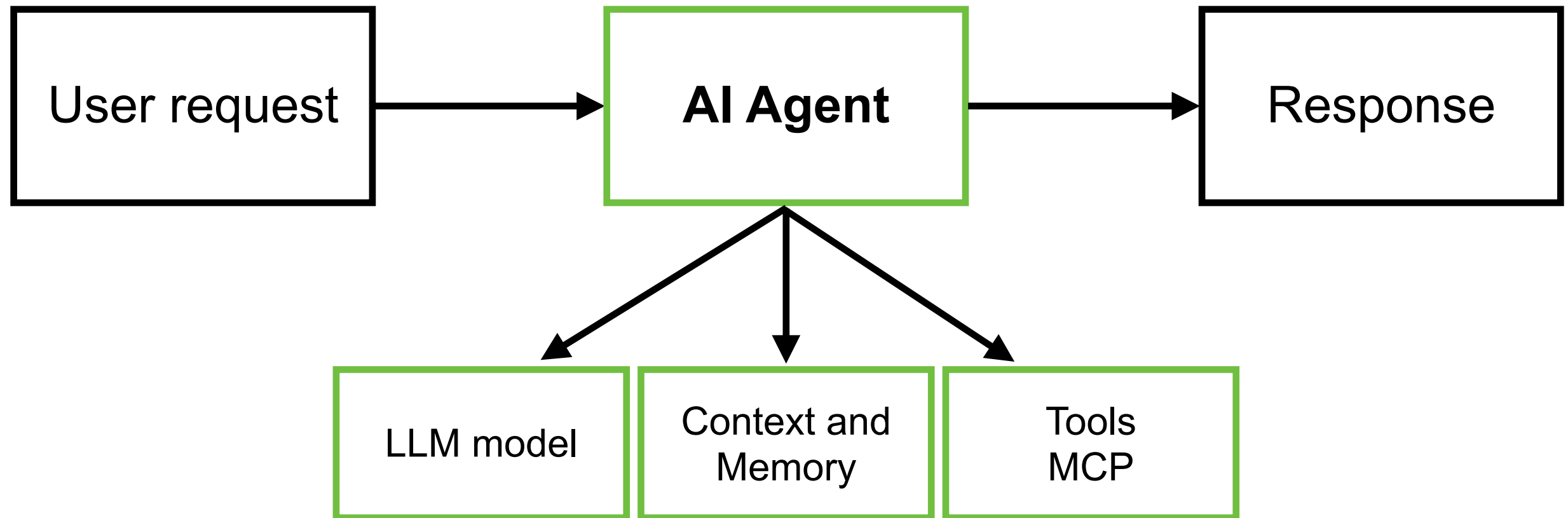


AI Agent for Coding





AI Agent for Coding



How to manage context or knowledge for Agent ?



Token is new oil !!
Token is expensive !!



LLM Model :: GPT 5.5

**GPT-5.5** Default 

Compare Try in Playground

REASONING

Highest

SPEED

Fast

PRICE
\$5 - \$30
Input - Output

INPUT
   
Text, image

OUTPUT
   
Text

GPT-5.5 is our newest frontier model for the most complex professional work. Learn more in our [latest model guide](#). Reasoning effort supports: none, low, medium (default), high and xhigh.

- ✦ 1,050,000 context window
- ➡ 128,000 max output tokens
- 📅 Dec 01, 2025 knowledge cutoff
- 🗨 Reasoning token support

Pricing

Pricing is based on the number of tokens used, or other metrics based on the model type. For tool-specific models, like search and computer use, there's a fee per tool call. See details in the [pricing page](#).

Text tokens Per 1M tokens - Batch API price ☐

Input \$5.00	Cached input \$0.50	Output \$30.00
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<https://developers.openai.com/api/docs/models/gpt-5.5>



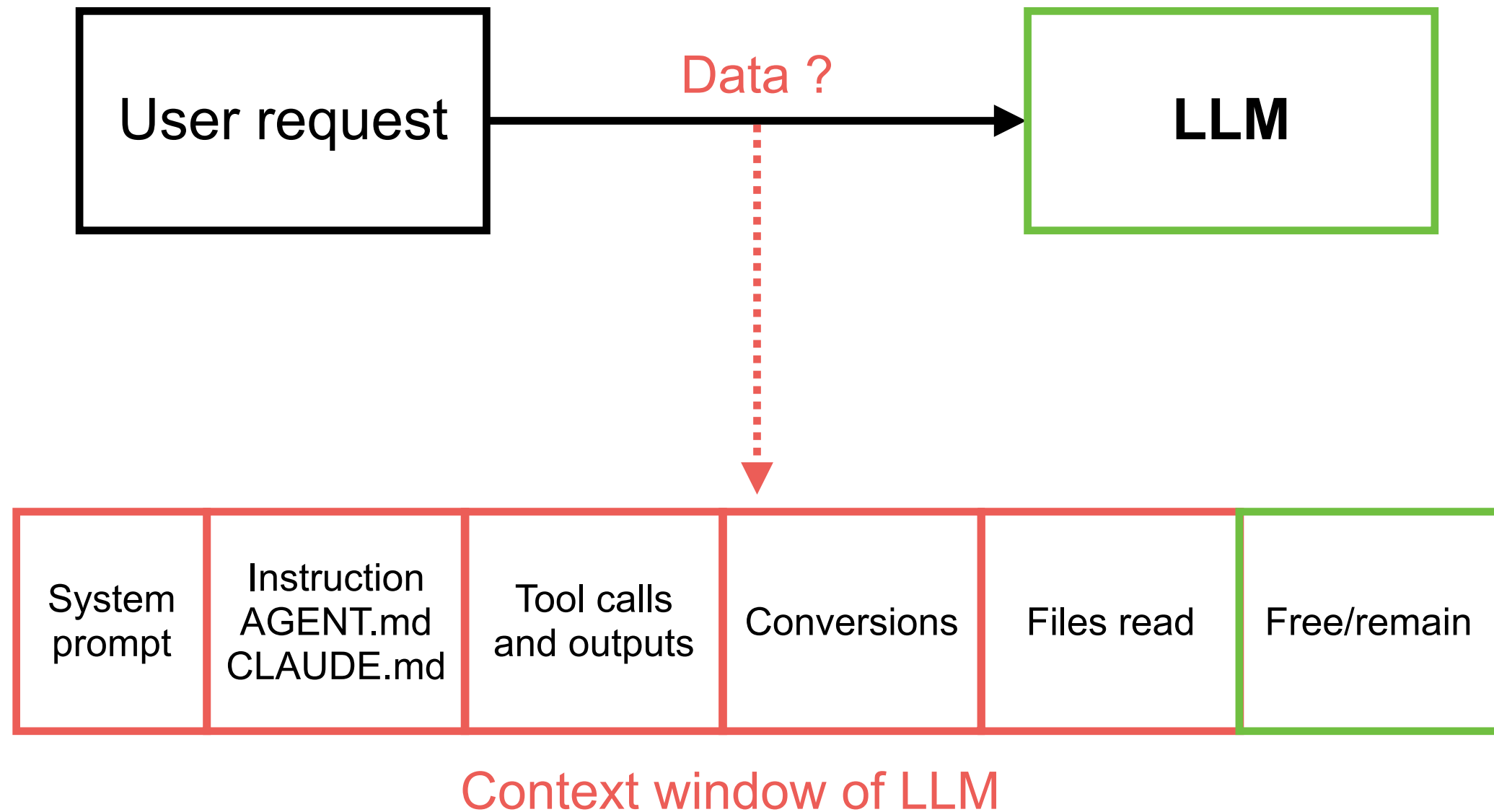
LLM Model :: Anthropic

Model	Base Input Tokens	5m Cache Writes	1h Cache Writes	Cache Hits & Refreshes	Output Tokens
Claude Opus 4.7	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.6	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.5	\$5 / MTok	\$6.25 / MTok	\$10 / MTok	\$0.50 / MTok	\$25 / MTok
Claude Opus 4.1	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Opus 4	\$15 / MTok	\$18.75 / MTok	\$30 / MTok	\$1.50 / MTok	\$75 / MTok
Claude Sonnet 4.6	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Sonnet 4.5	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok
Claude Sonnet 4	\$3 / MTok	\$3.75 / MTok	\$6 / MTok	\$0.30 / MTok	\$15 / MTok

<https://platform.claude.com/docs/en/about-claude/pricing>



Context Window ?



Types of Knowledge/Context ?

Shared context

Static Knowledge

Prompt engineering
Global Instructions, Rules
AGENTS.md
CLAUDE.md
DESIGN.md

Task context

Dynamic Knowledge

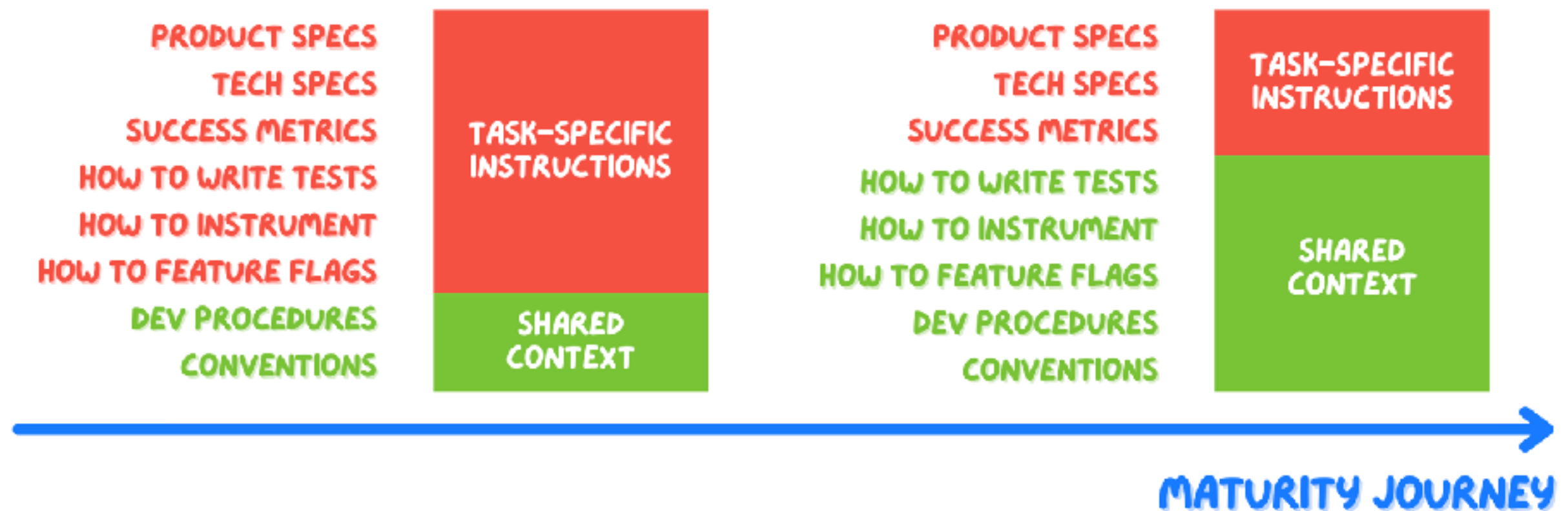
Memory
Graph/Vector database
MCP tools
Specific Instructions
Agent SKILLS

Sub-agent and Modes

Context engineering ?



Types of Knowledge/Context ?

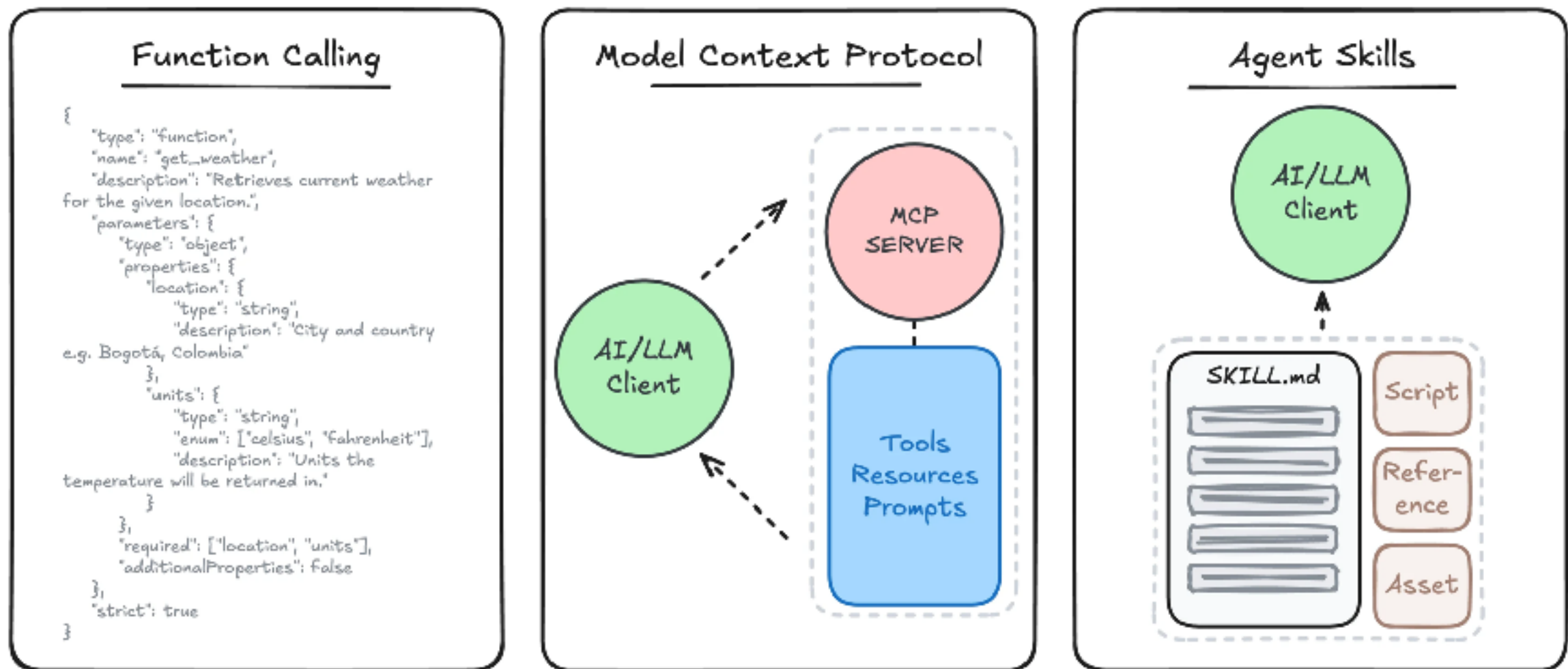


<https://refactoring.fm/p/context-capabilities-and-tech-hubs>



Agent Skills

Standard way to give AI agent new capabilities and expertise



<https://agentskills.io/home>



Vocabulary before start !!



Vocabulary before start !!

Rules

Prompts

MCP servers

Instructions

Commands

Subagents

Modes

Hooks

Agent Skills



Rules and instructions

Static context

Rules are files that contain essential context
Codebase, best practices, common pitfalls
Business requirements

Prevent LLM models from hallucinating
and to ensure specific standards



Commands and Prompts

Static context

Text instructions sent to the agent

Repeat workflow that can be triggered on demand

Allow teams to share standardized workflows
And run complex tasks conditionally

git add -> generate commit message -> git commit



Subagents and Modes

Specific persona or task to specialized agent
Agent, Plan, Ask, Edit mode
Work AI as a team

Make the agent's behavior more deterministic, reliable
and discoverable



Agent Skills

Dynamic context

Create workflow + resources + script

Skills do not bloat the initial context window

Only loaded when the agent determines (relevant with task)

Provide an open standard for sharing capabilities with a team without hitting context limits

<https://agentskills.io/>



Hooks

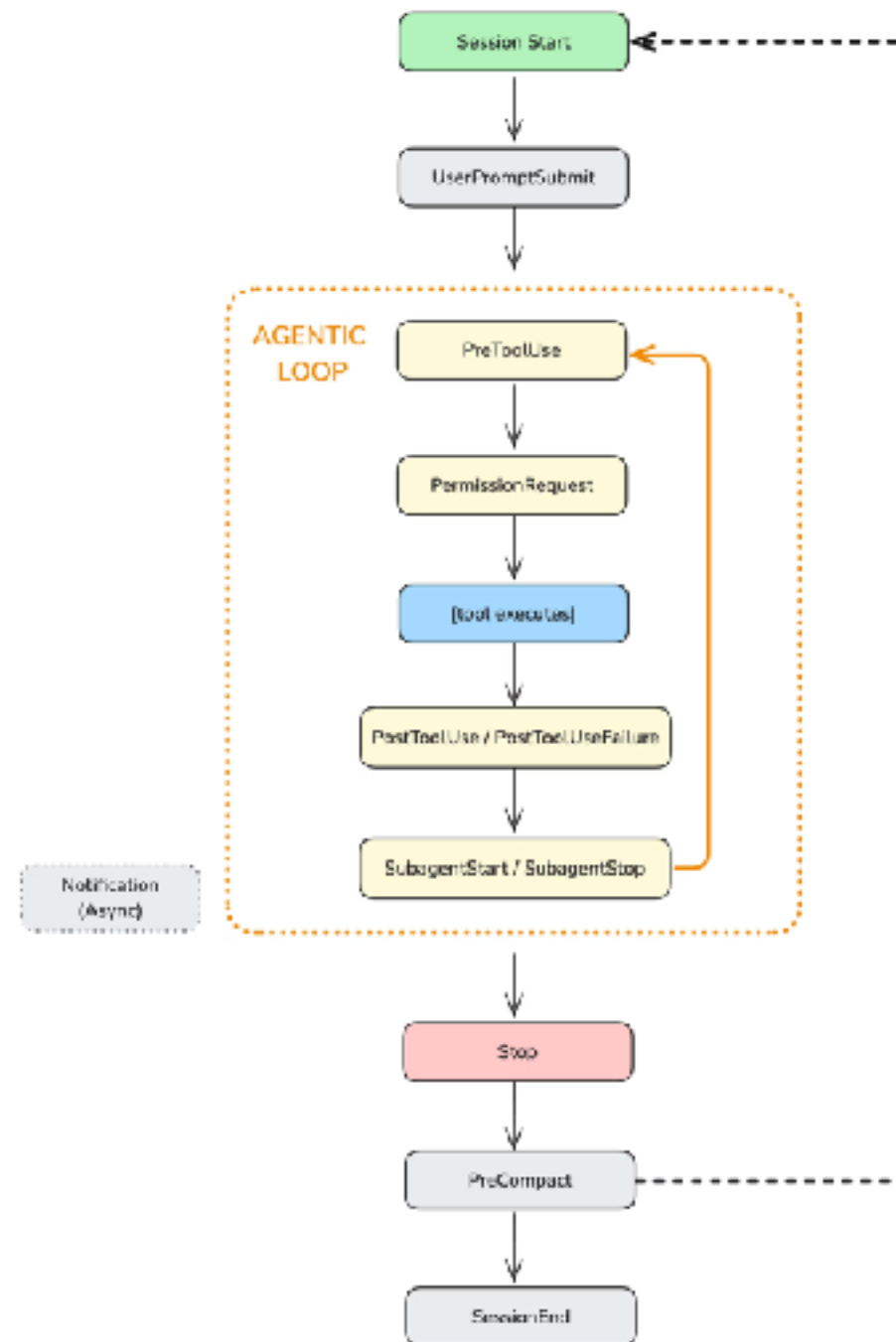
Customize and extent behavior of agent
Triggers for actions before/after conversation

Setup environment, run test, cleanup resources,
Notification, automated pre-commit check ...

<https://code.claude.com/docs/en/hooks>



Hooks in Agent's Lifecycle

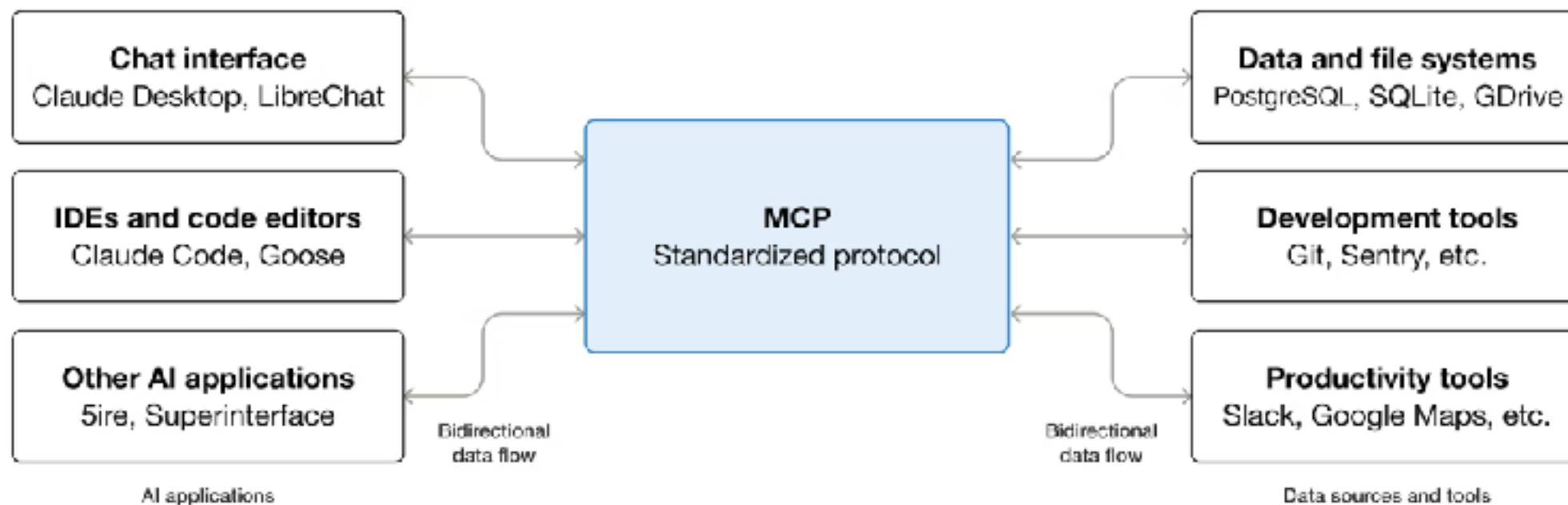


<https://code.claude.com/docs/en/hooks>



MCP (Model Context Protocol)

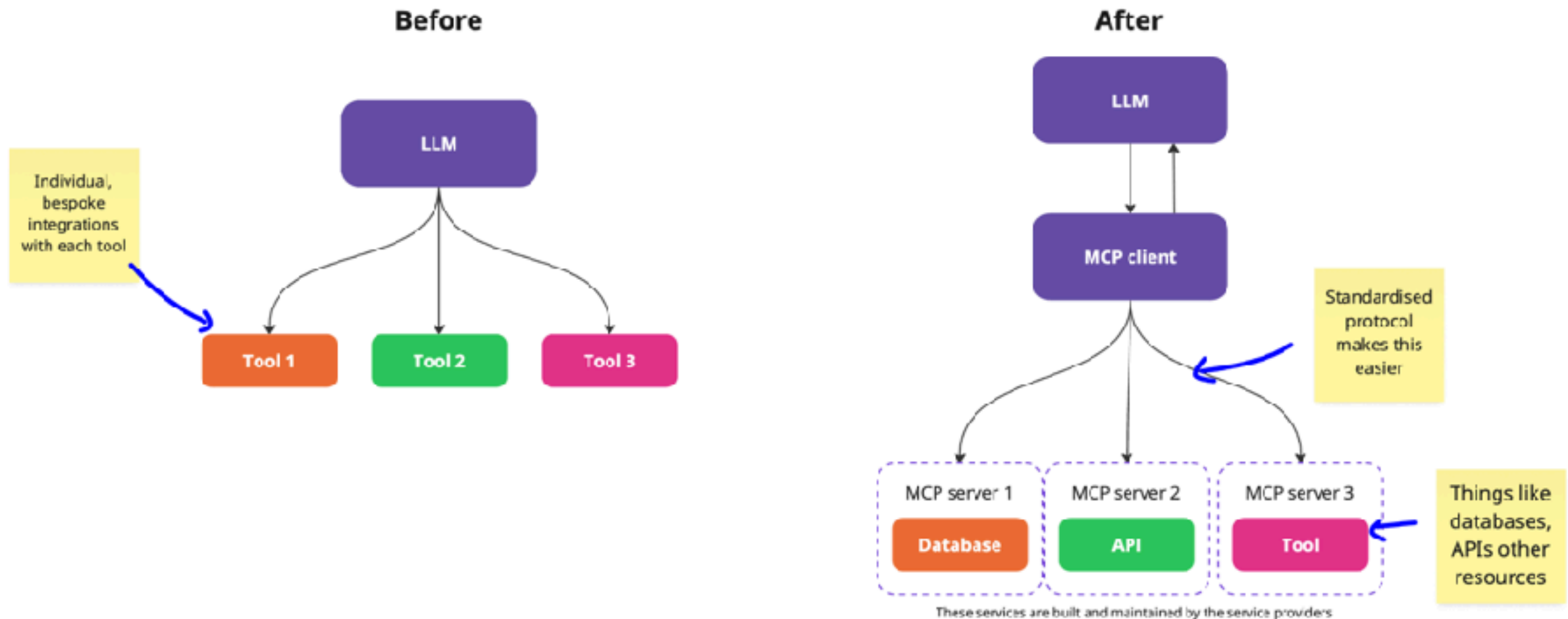
Open standard for connecting AI app to external system



<https://modelcontextprotocol.io/>



MCP (Model Context Protocol)



<https://departmentofproduct.substack.com/p/mcp-explained-a-simple-guide-for>

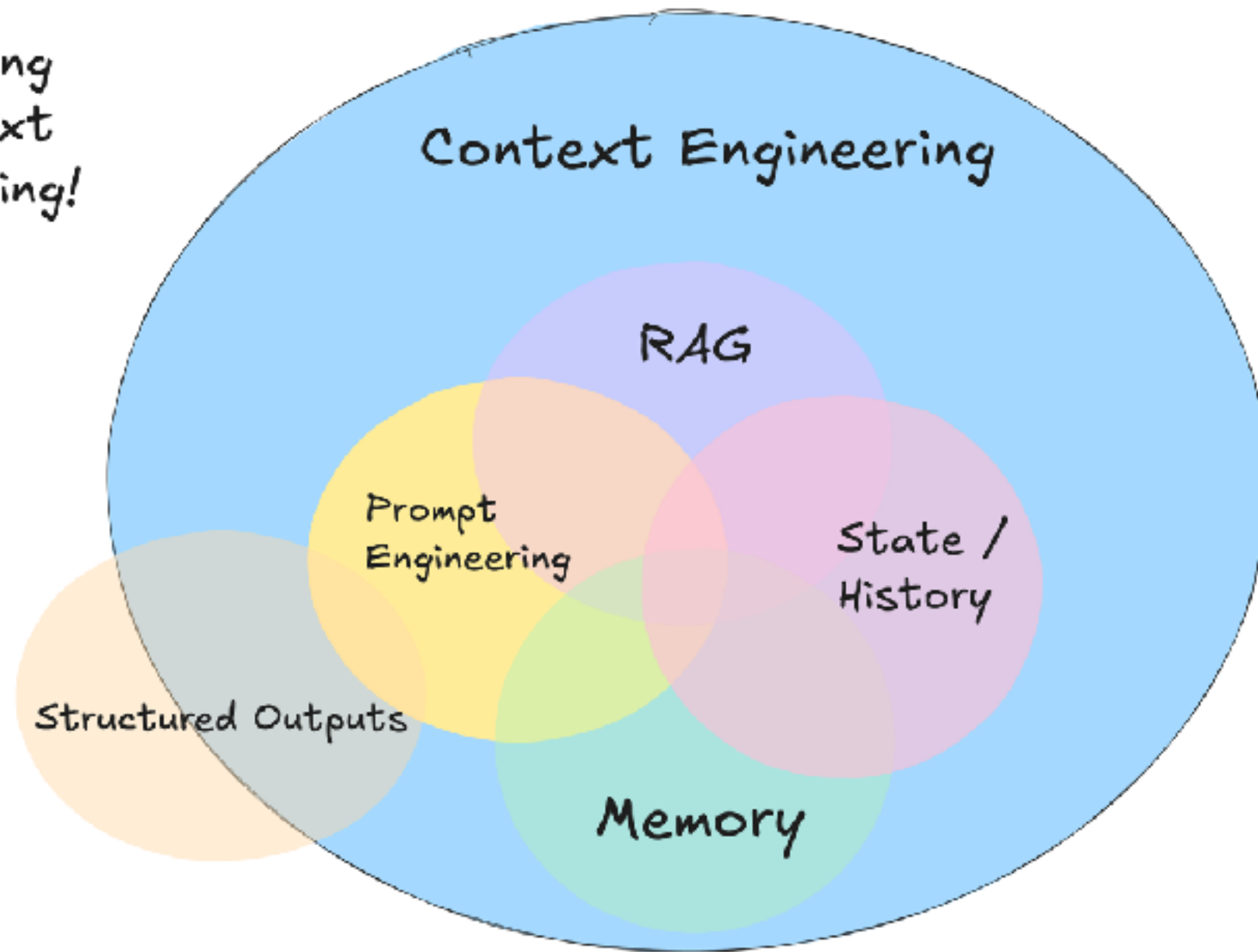


How to manage context ?



Context Engineering

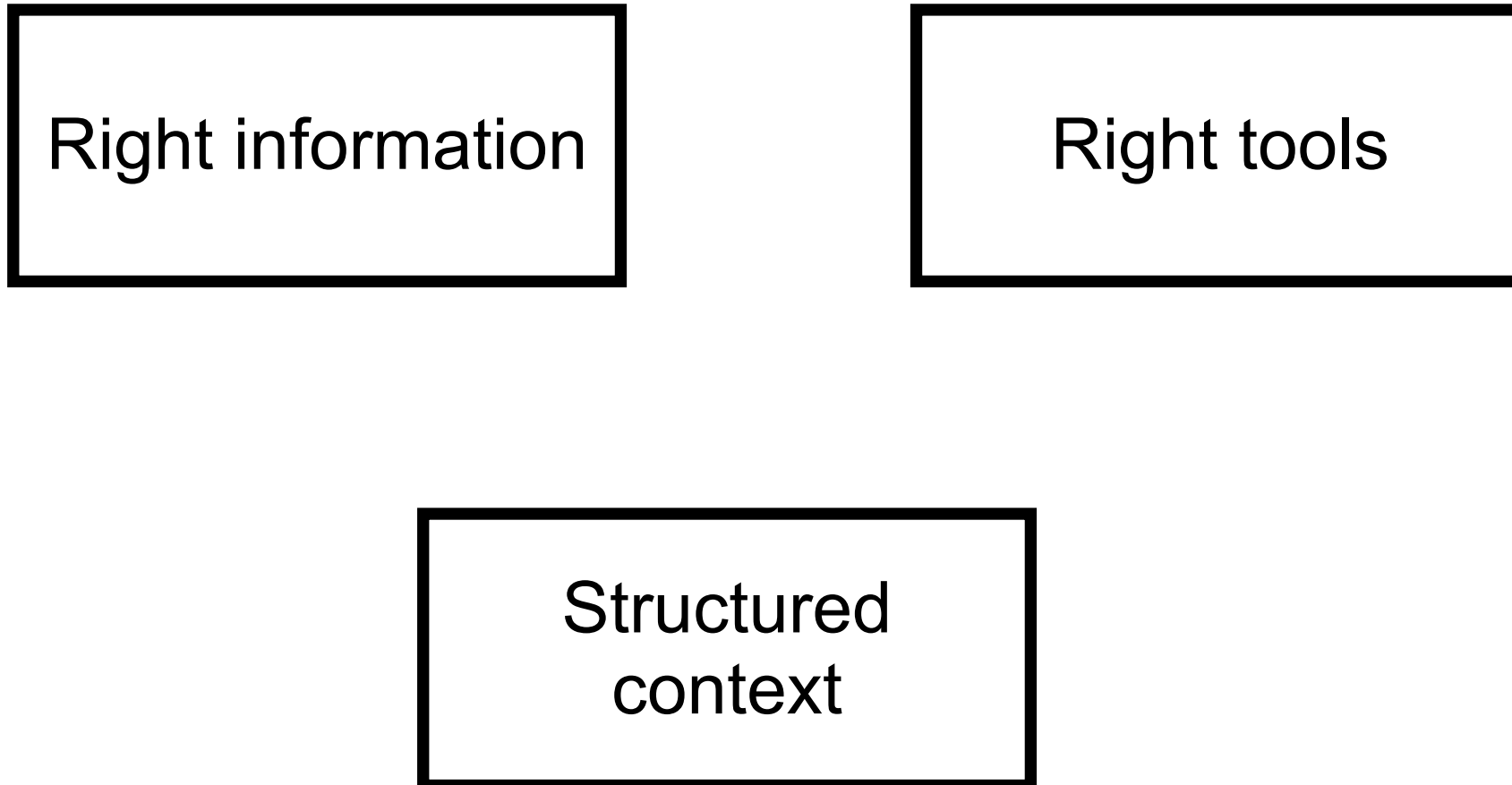
Everything
is Context
Engineering!



<https://www.promptingguide.ai/guides/context-engineering-guide>



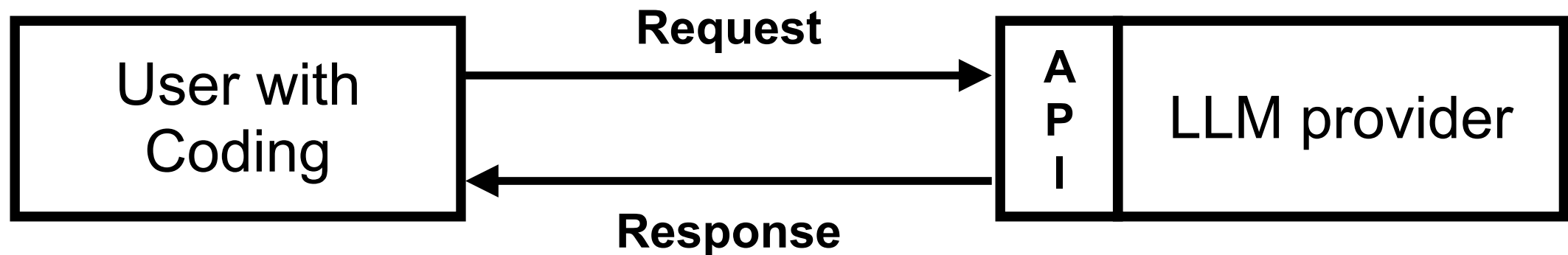
Context Engineering



Context window size ?

Amount of text a model can process and remember
at once time

Measure with **token**



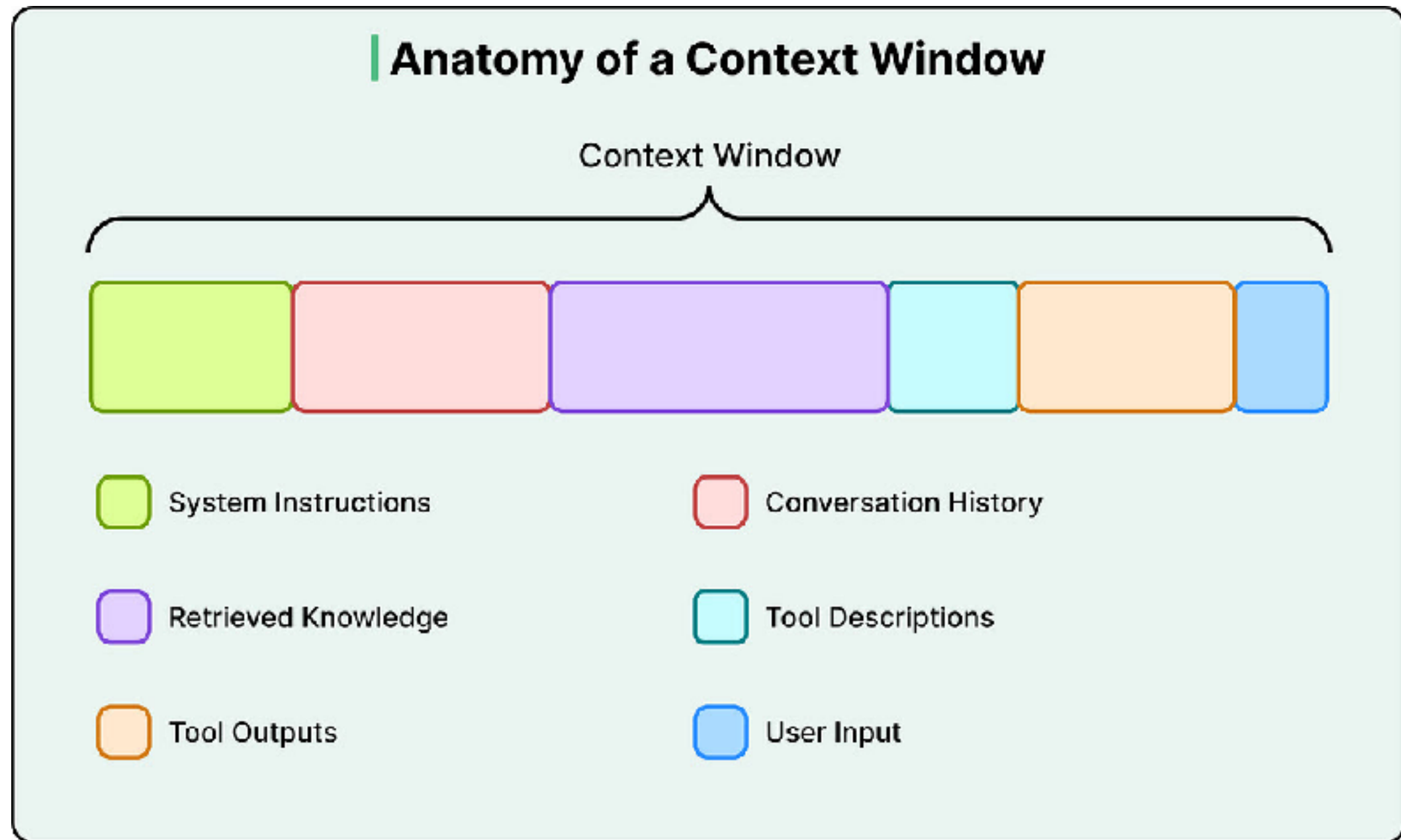
Context window size of models

Different models may have different window size

Model name	Context window size
GPT 4.1	1M
Gemini 1.5, 2.5	1M
Llama 4	1M
Claude Sonnet 3.7	200,000
o4	200,000



Anatomy of Context Window

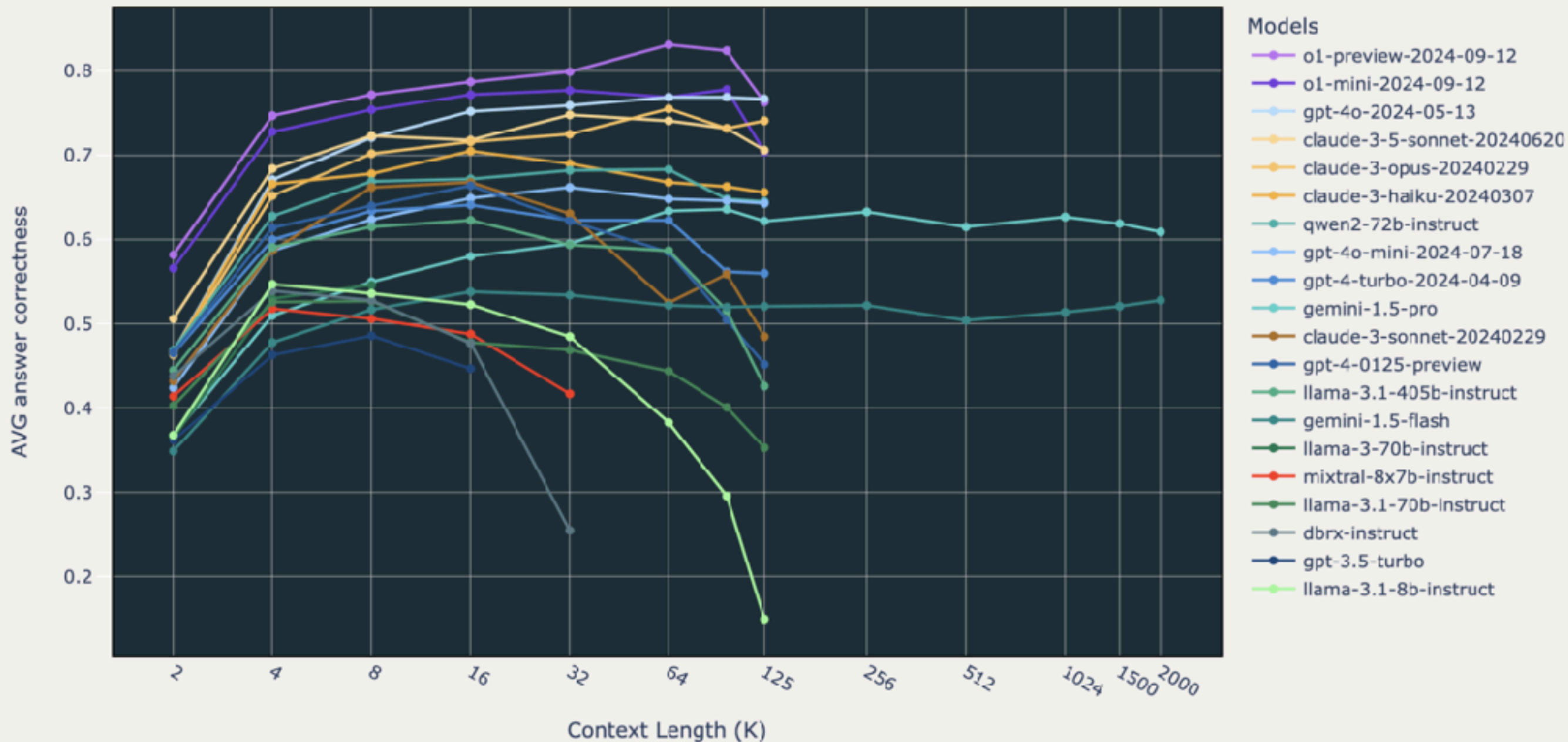


Long context == Better result ?



Long context LLM is better !!

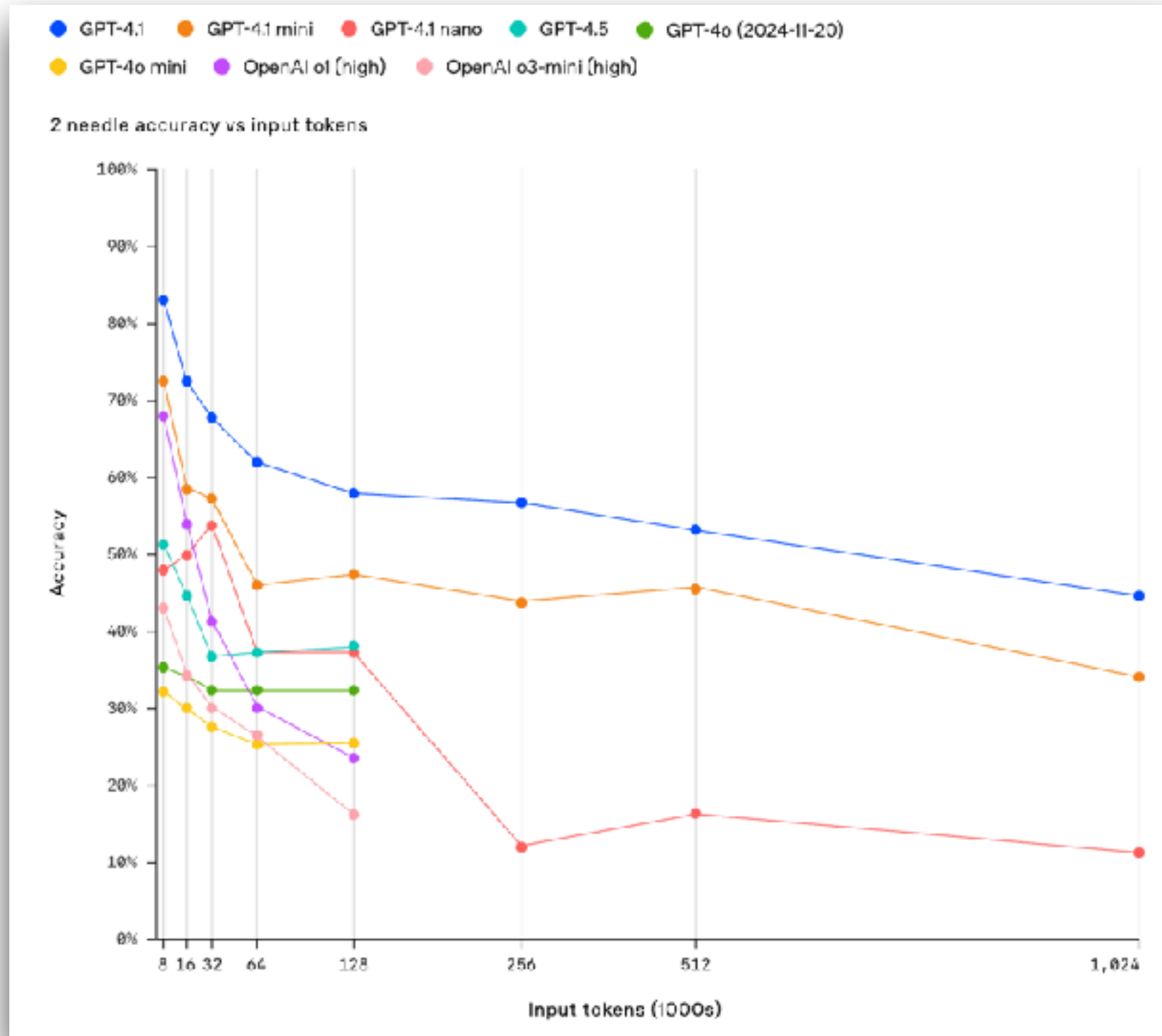
Long Context RAG Performance of LLMs



<https://arxiv.org/abs/2411.03538>



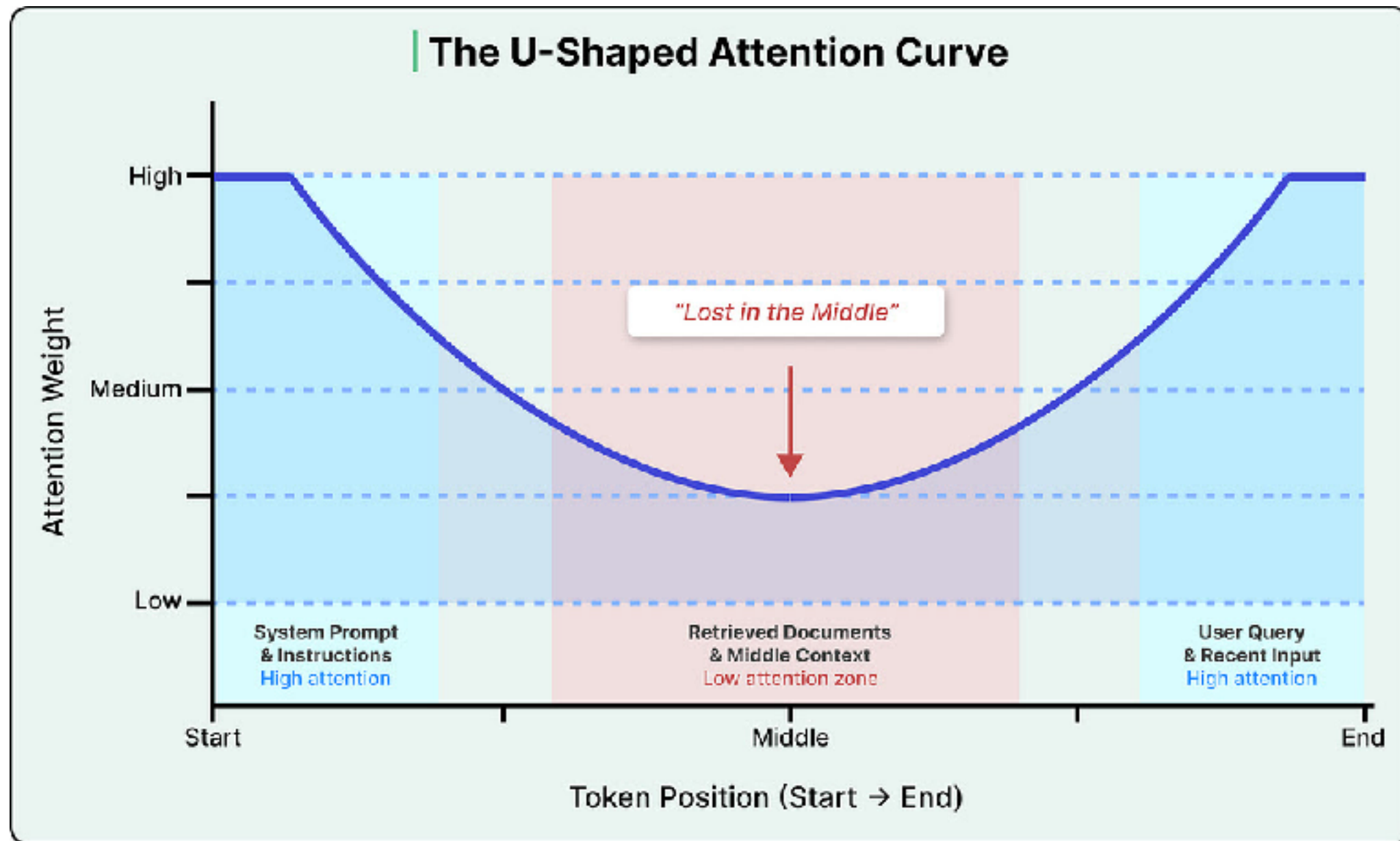
Long context LLM is better !!



<https://openai.com/index/gpt-4-1/>



Loss in the middle of LLM



<https://arxiv.org/abs/2307.03172>

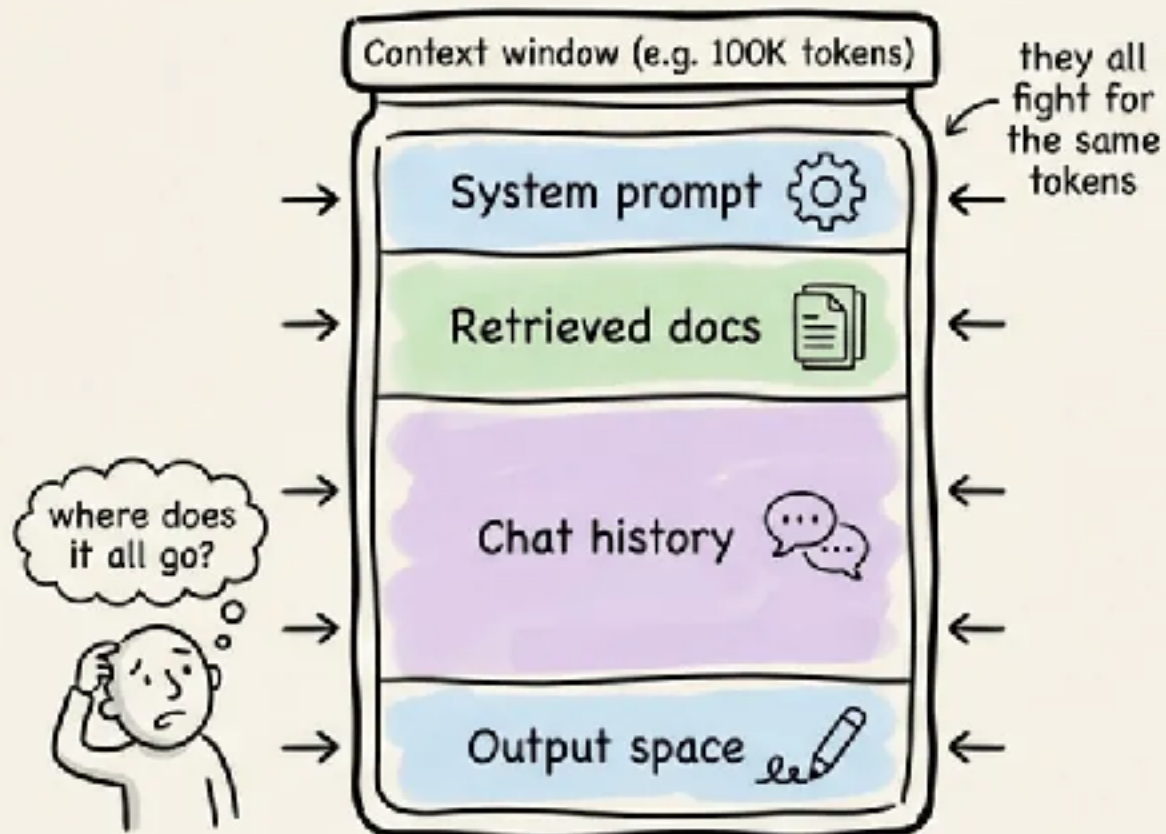


Better context == Better result ?

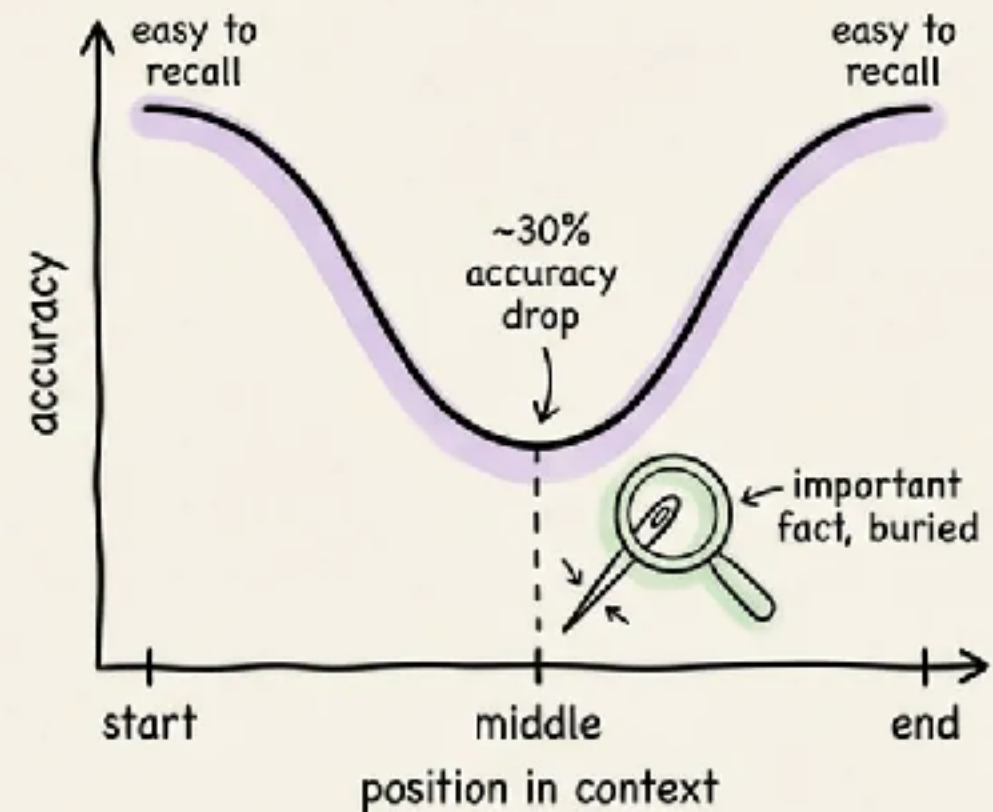


More tokens isn't more memory.

Context = a shared budget



Lost in the middle



Bigger windows don't beat structured memory.

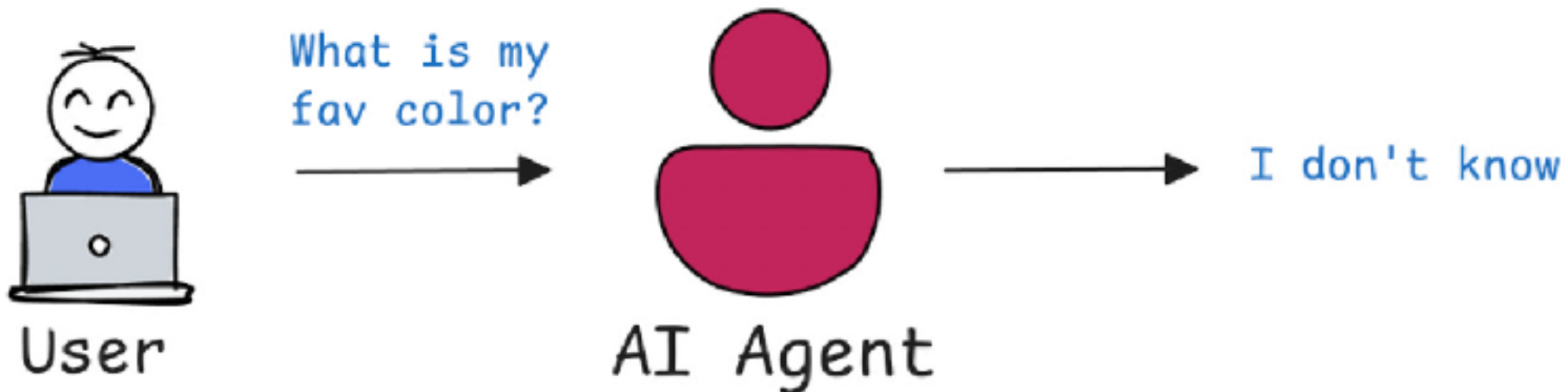
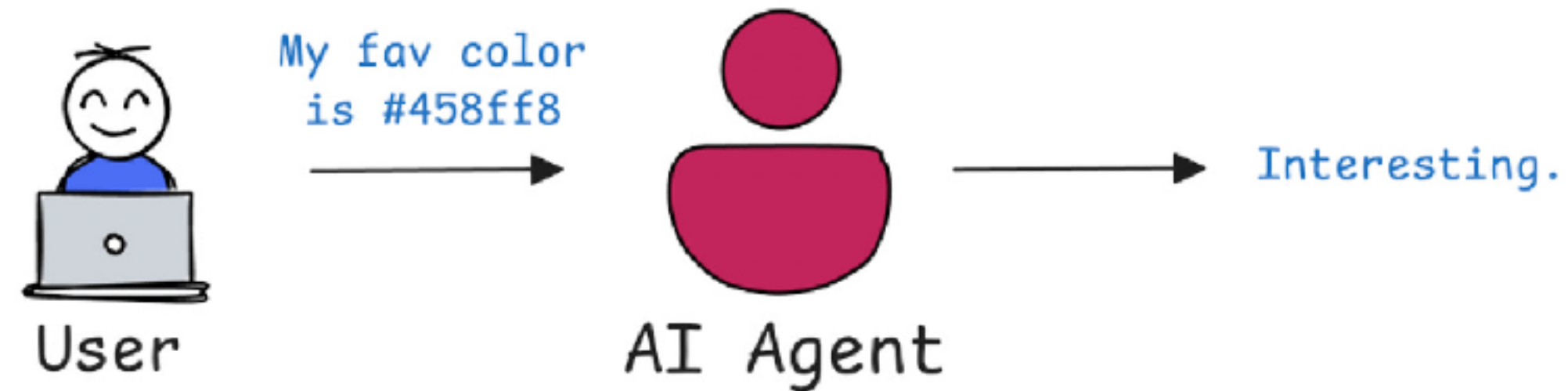
<https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from>



Agent Memory Memory Bank



Interaction without memory

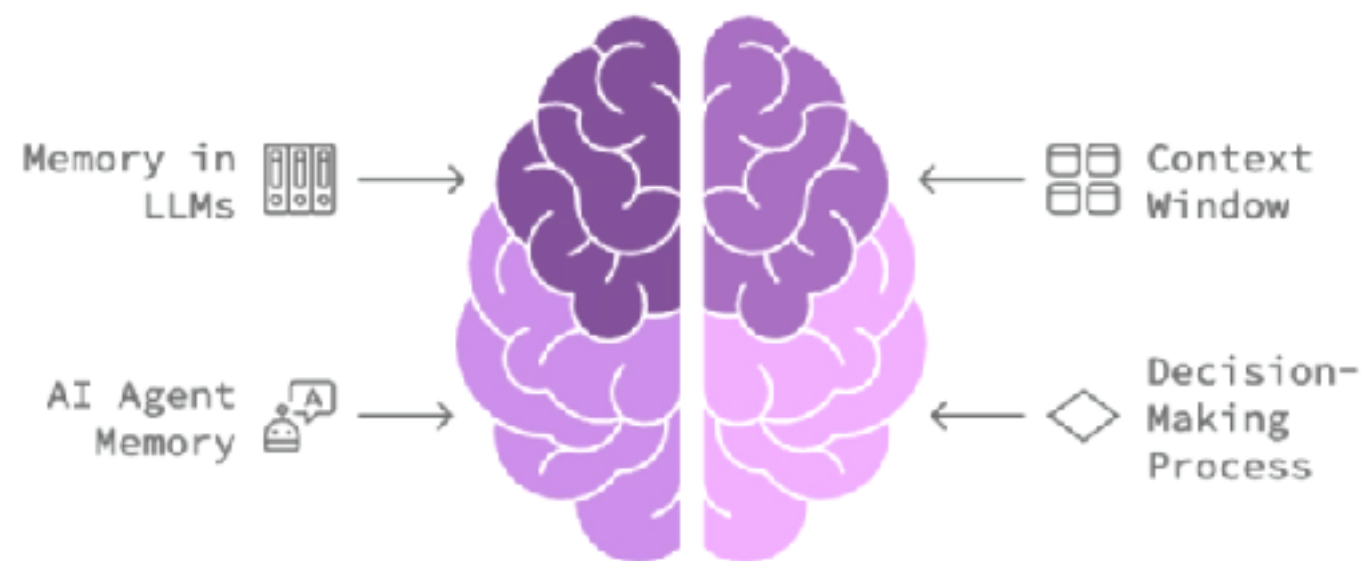


Why AI Agent memory important ?

Enable personalized and context-aware interactions

Improve decision-making and adaptability

Enhance LLM's performance and response



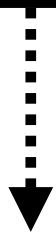
<https://www.falkordb.com/blog/ai-agents-memory-systems/>



Types of Memory

To make intelligent decisions
Agents combine different types of memory

Short-term



Current goals and steps
File system (markdown, JSON)
In-memory database

Episodic



Logs of past interactions
Success and failure

Semantic

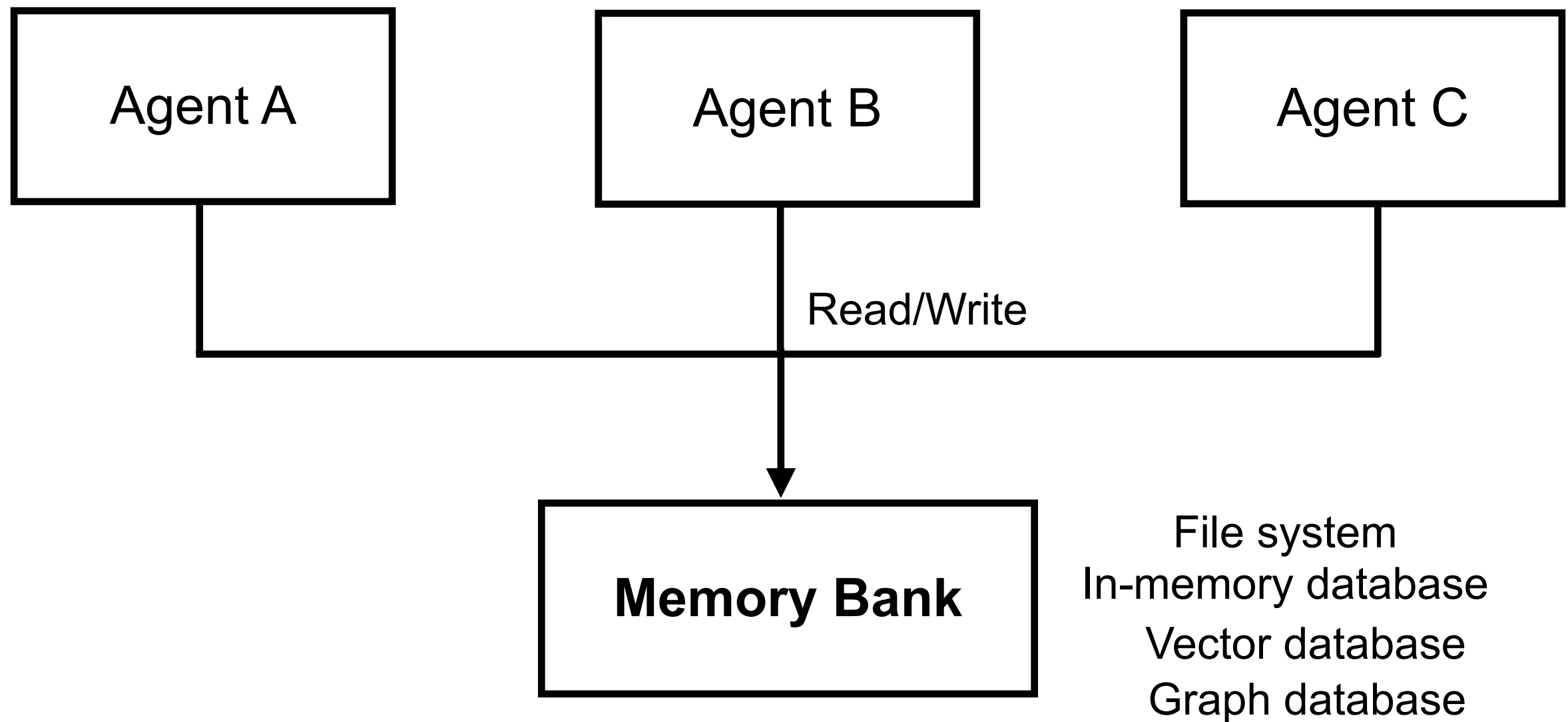


Long-term knowledge
Fact and strategies
RAG
Vector and graph database

<https://bhavishyapandit9.substack.com/p/how-memory-works-in-agentic-ai-a>

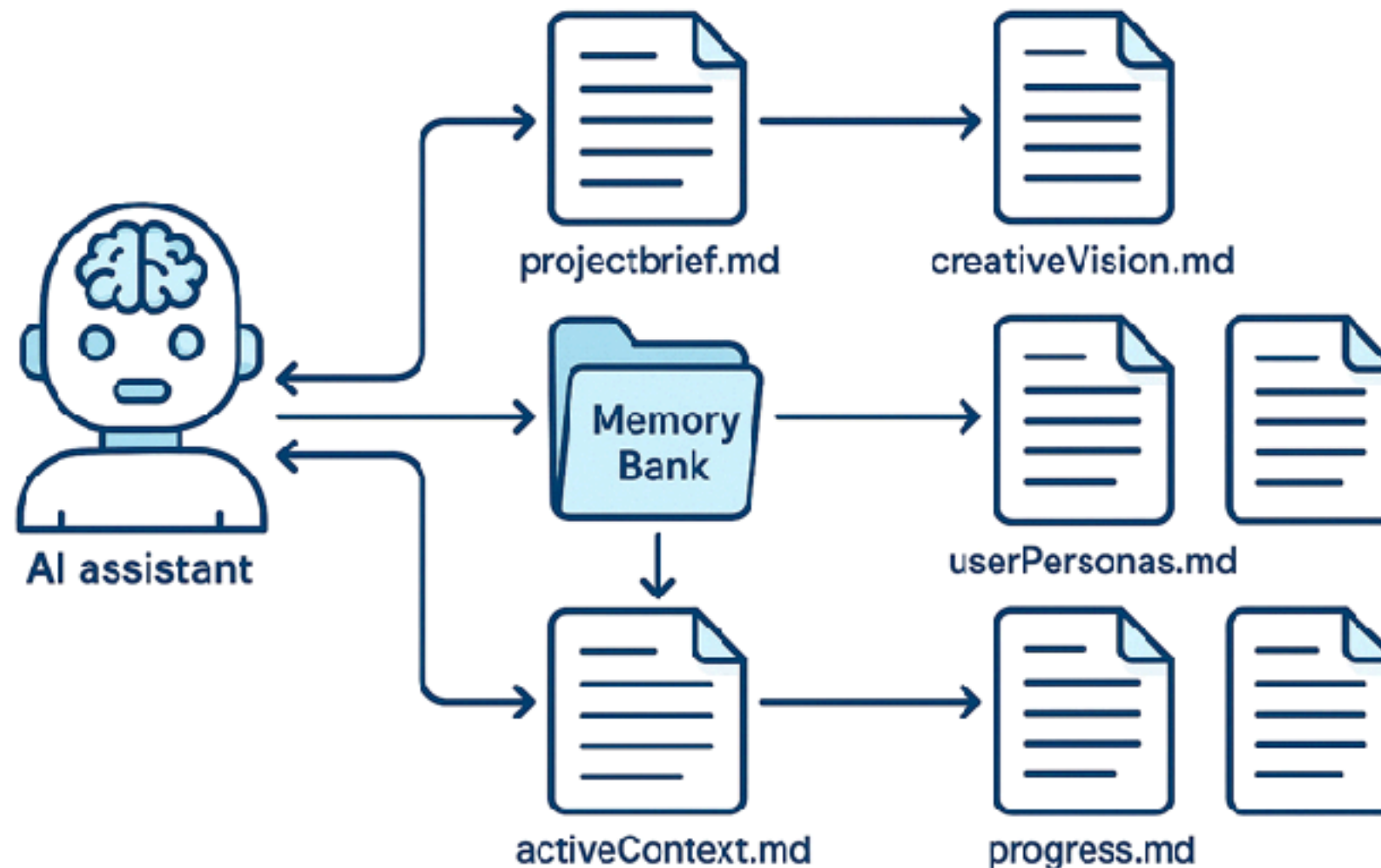


Shared memory between Agent



Cline Memory Bank

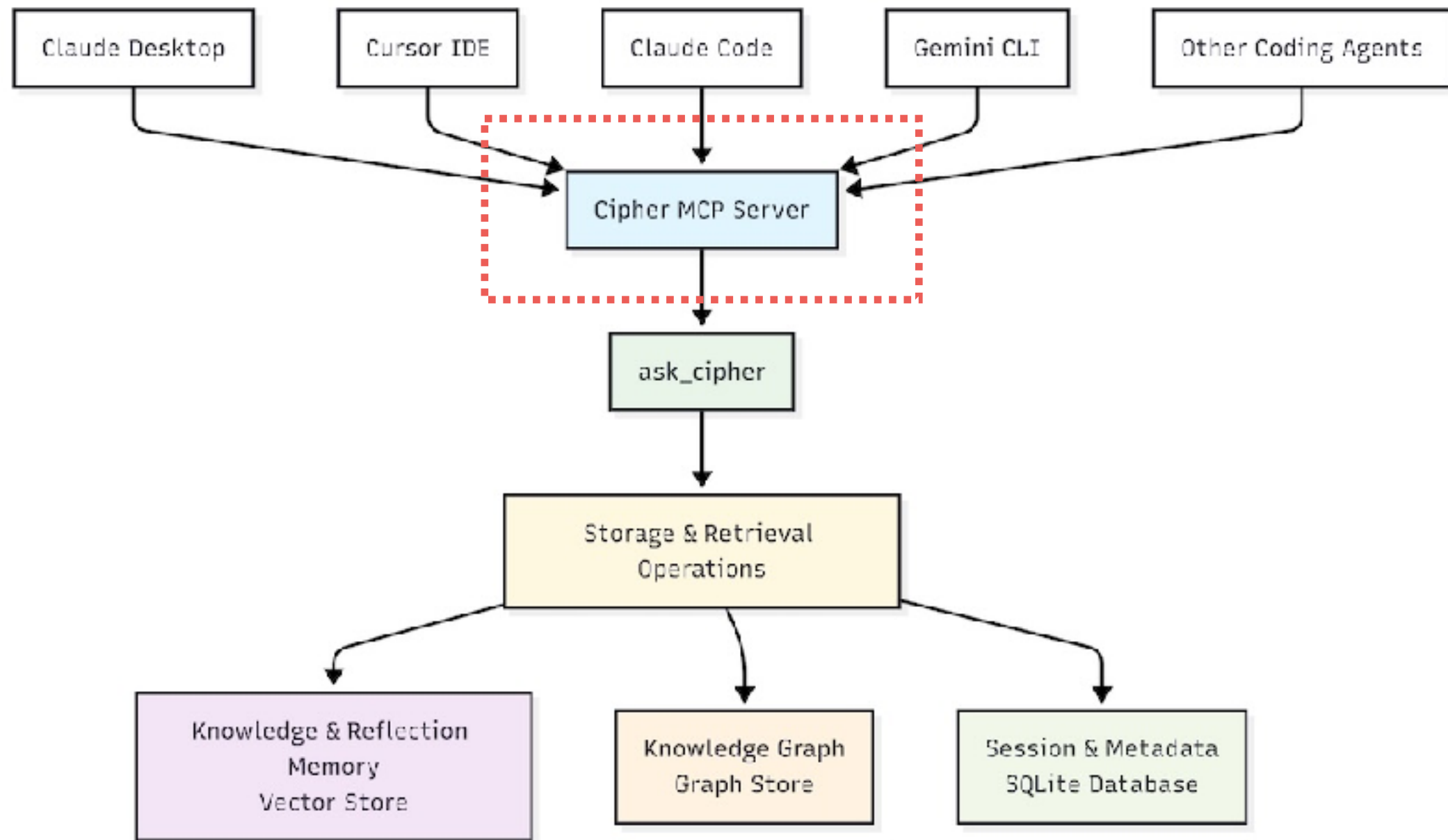
Working with file system



<https://docs.cline.bot/prompting/cline-memory-bank>



Memory Bank via MCP



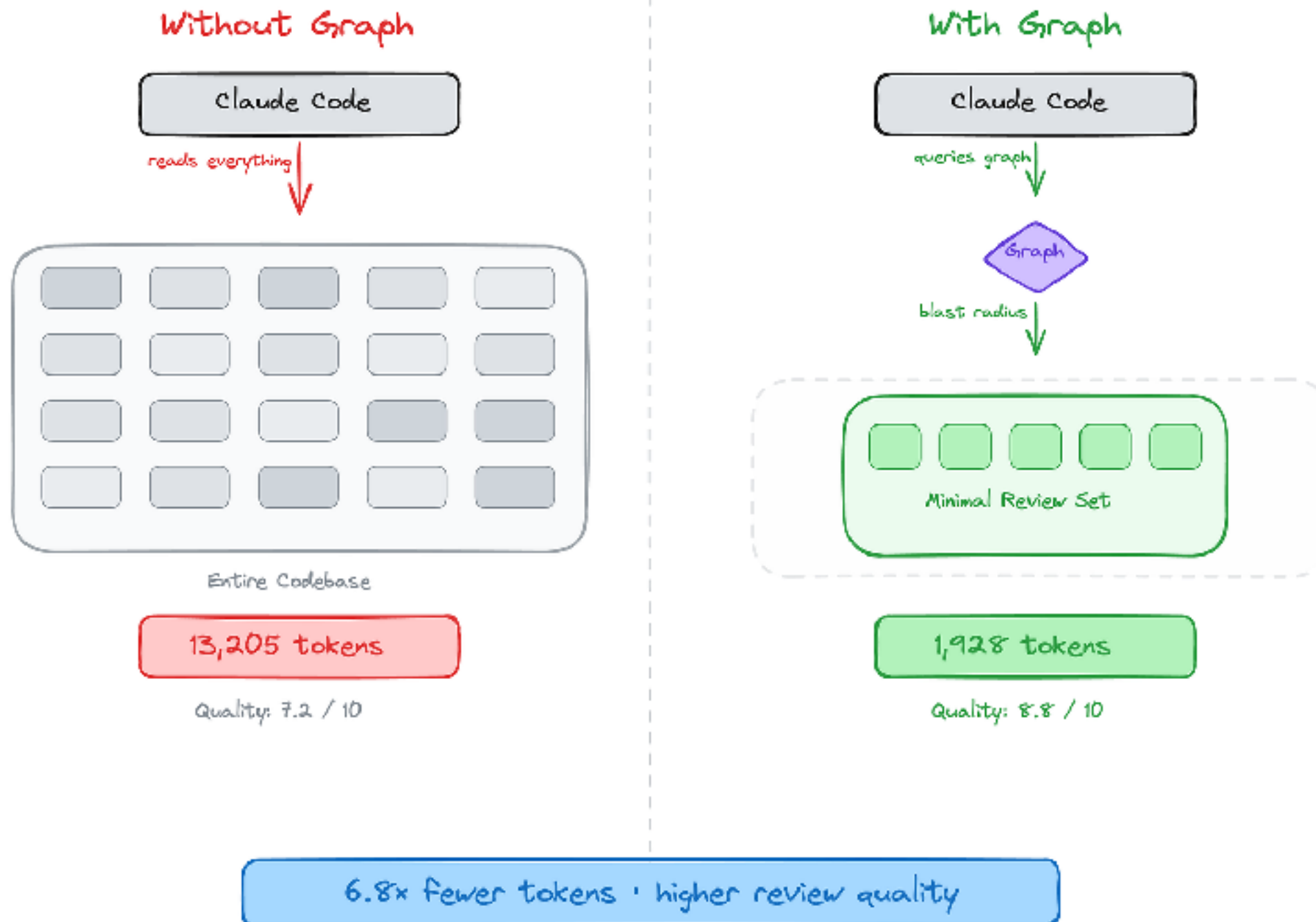
<https://www.somkiat.cc/share-memory-for-ai-ide/>



How to reduce Token usage ?



The Token Problem



<https://code-review-graph.com/>





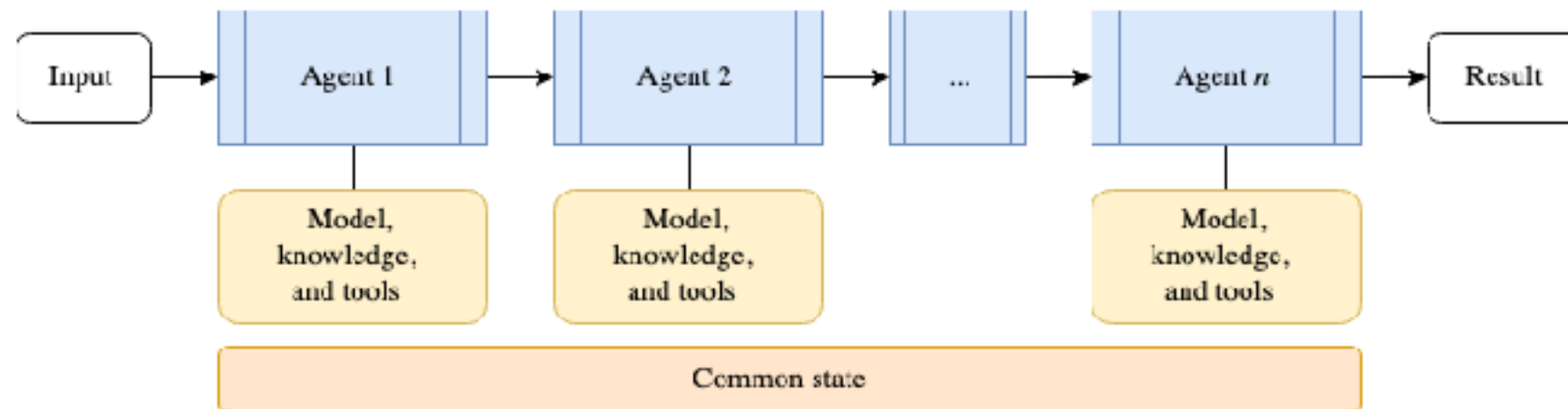
Sub-agent and Multi-agent !!

Reduce size of knowledge !!



Monolith to Multi-agent

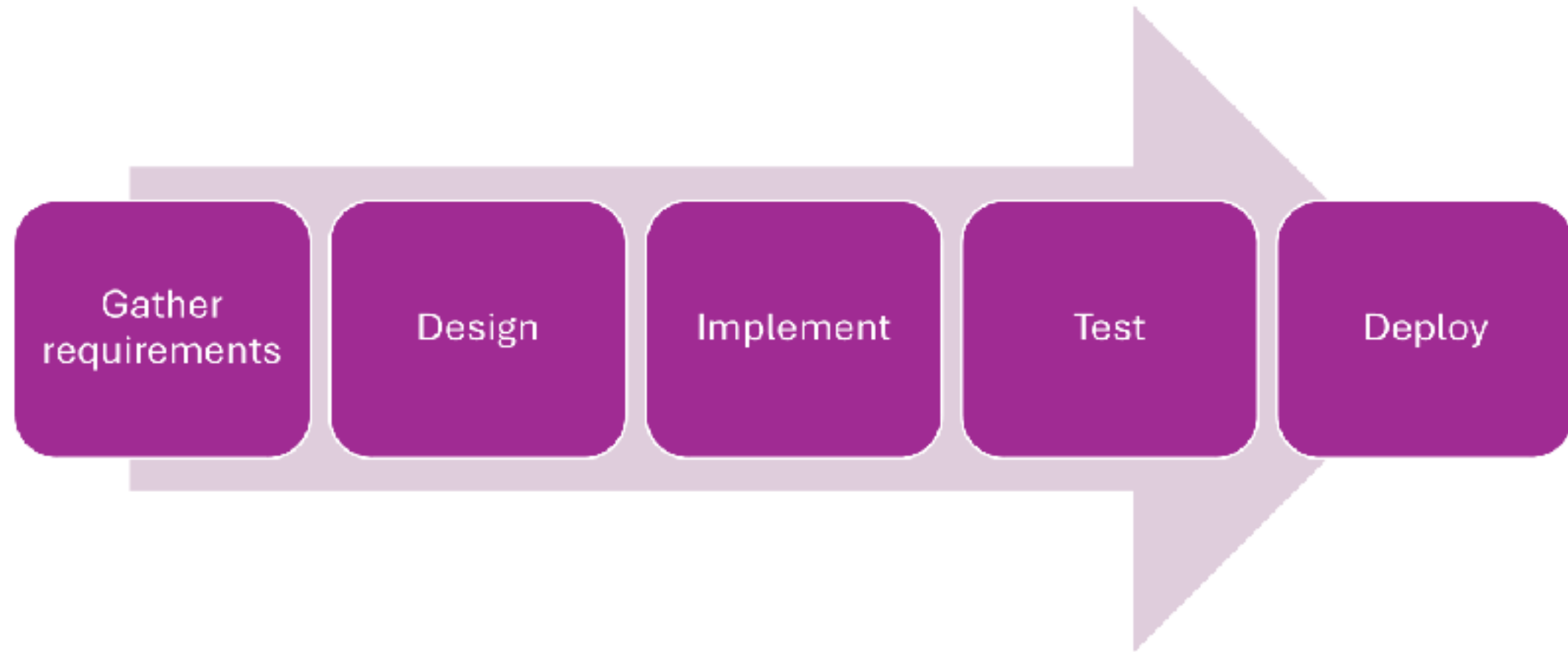
Specialization
Scalability
Maintainability
Optimization



<https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns>



Design your Agents ?



<https://devblogs.microsoft.com/dotnet/introducing-microsoft-agent-framework-preview/>



Design your Agents ?

Each step may contain subtasks
Different specialists have different steps
Progress isn't always linear

Bugs found during testing may send back to
implementation



Design your Agents ?

Full Stack Agent !!

Frontend

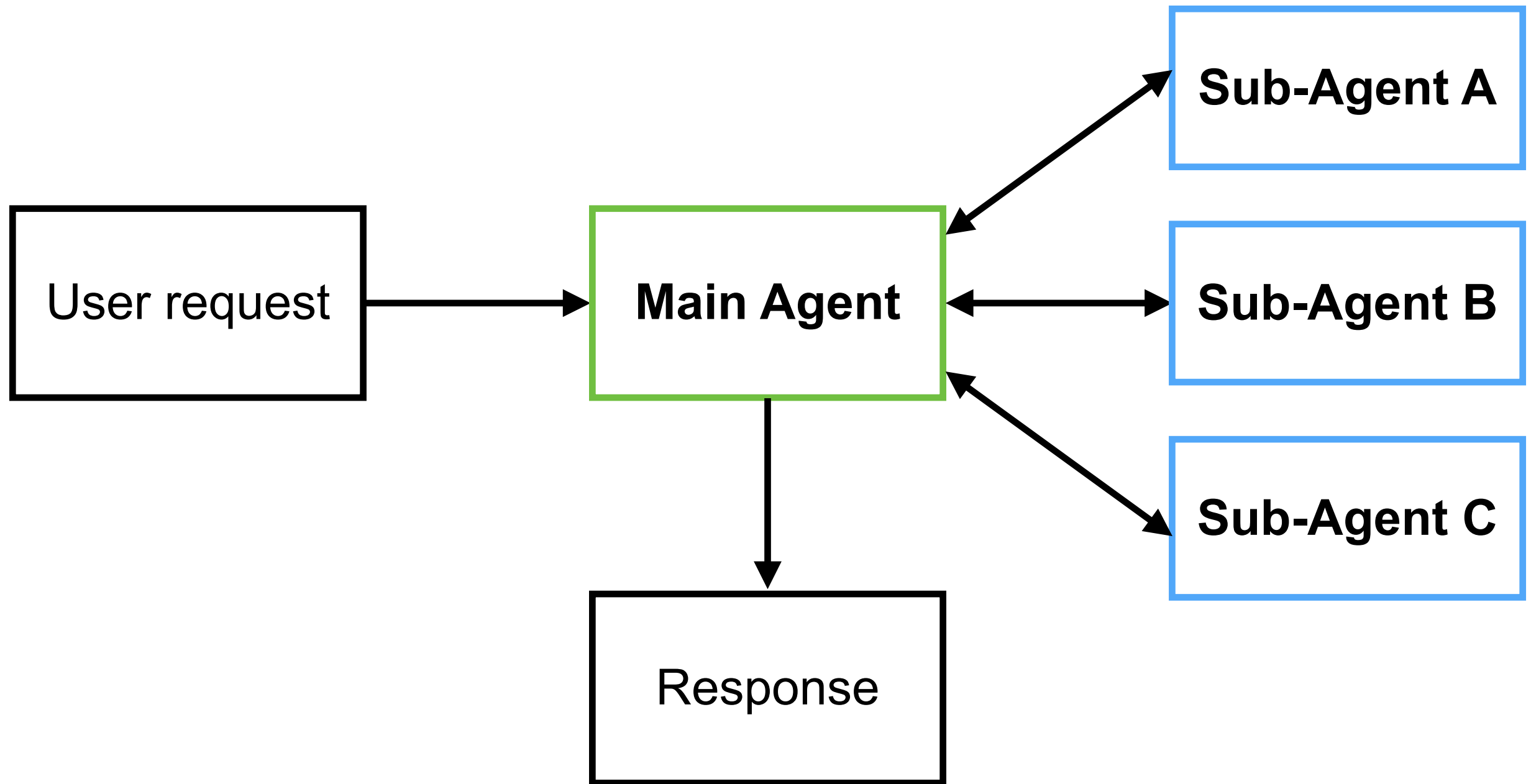
Backend

Database



Multi AI Agents !!

Sub-agent with **centralized** orchestration

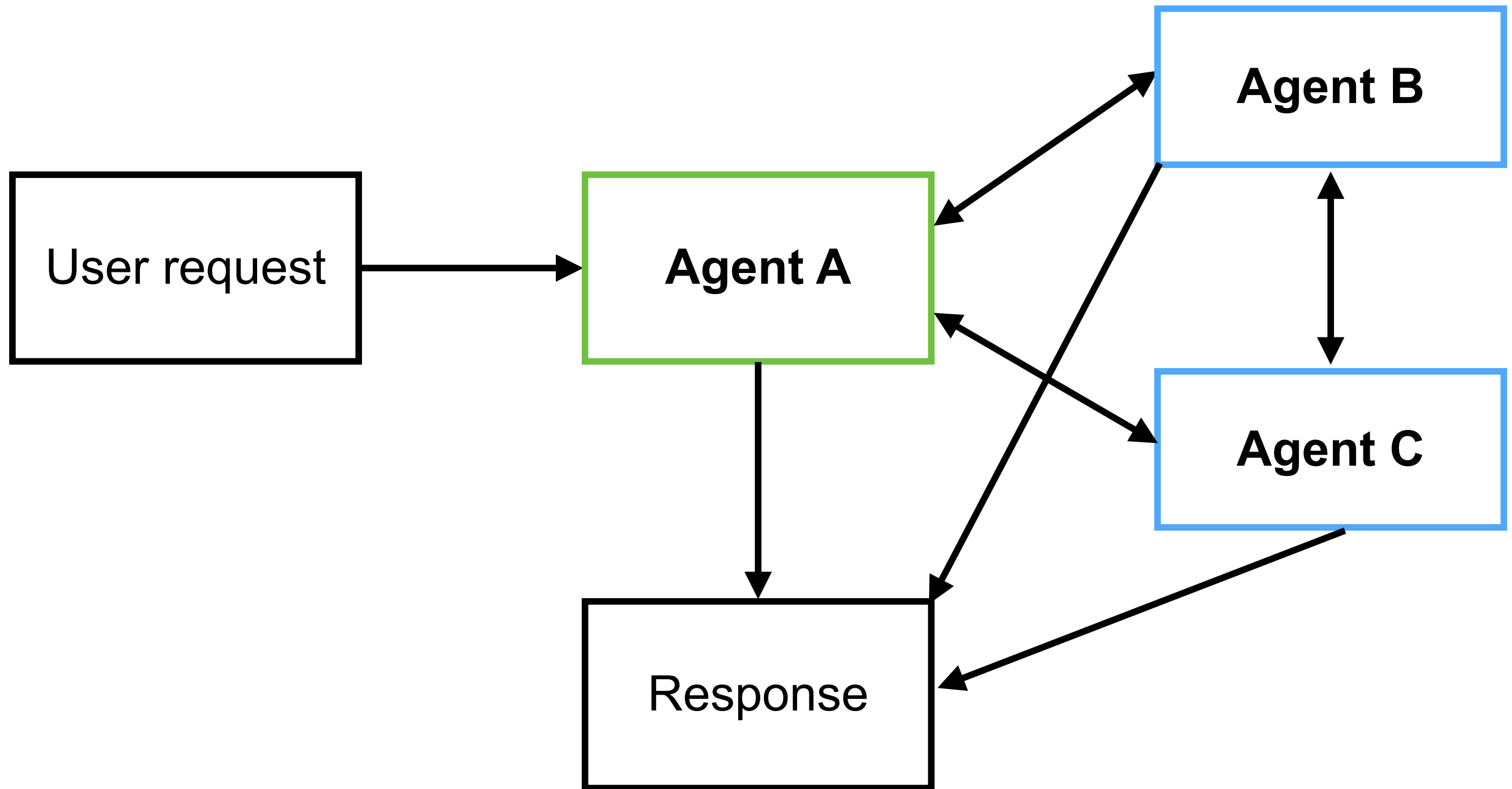


<https://www.blog.langchain.com/choosing-the-right-multi-agent-architecture/>



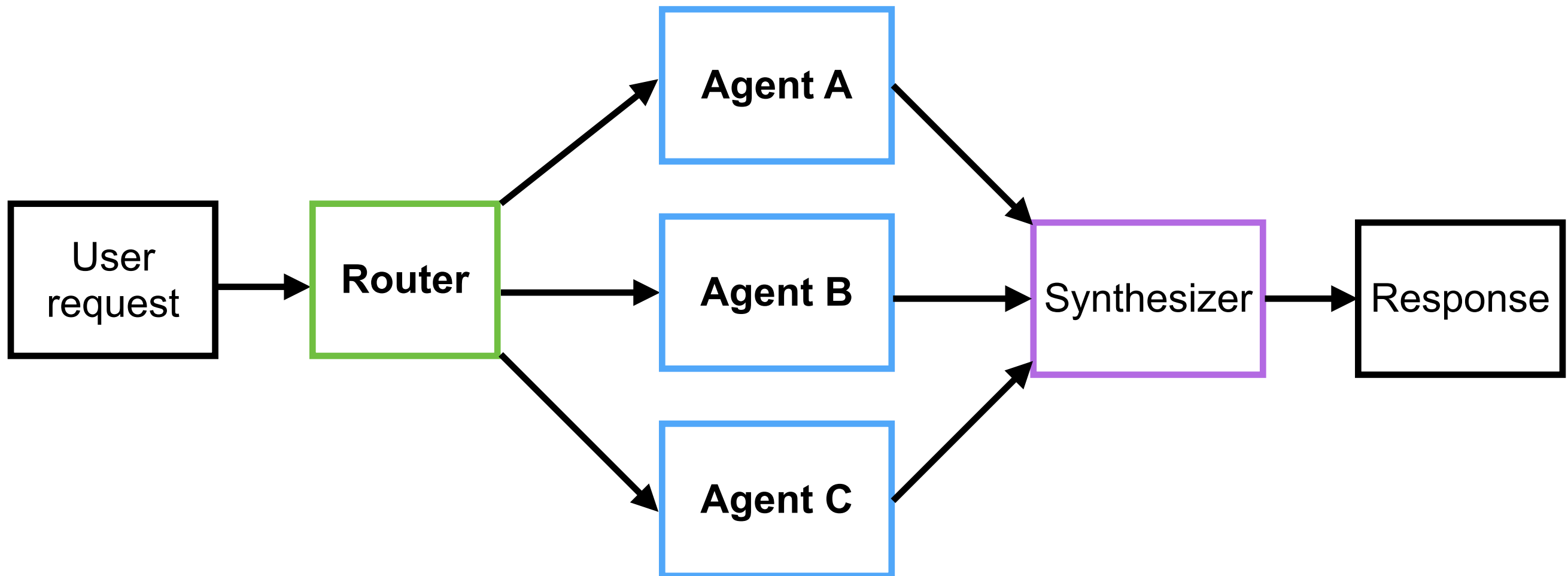
Multi AI Agents !!

State-driven transitions based-on conversation context



Multi AI Agents !!

Router and parallel run



Prompt vs Agent Skills ?



Command

<https://agentskills.io/>



Agent Skills

Moving from a paradigm of
“prompt” to “programming by instruction”

Agent loads specialized prompts and knowledge on-demand

Specialized
task

Reduce
repetition

Compose
capabilities

Efficient
loading

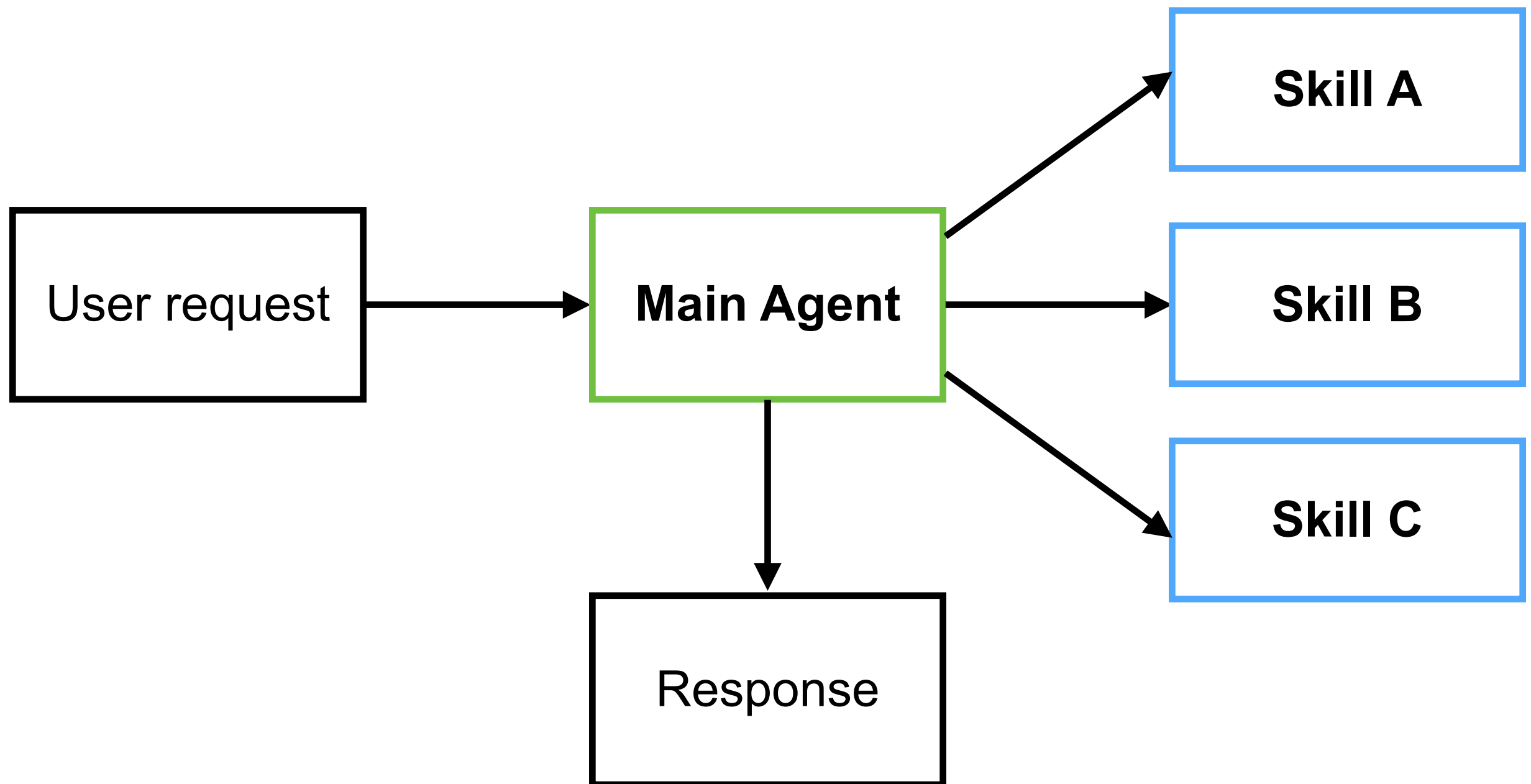
Interoperability

<https://agentskills.io/>



Agent with Skills

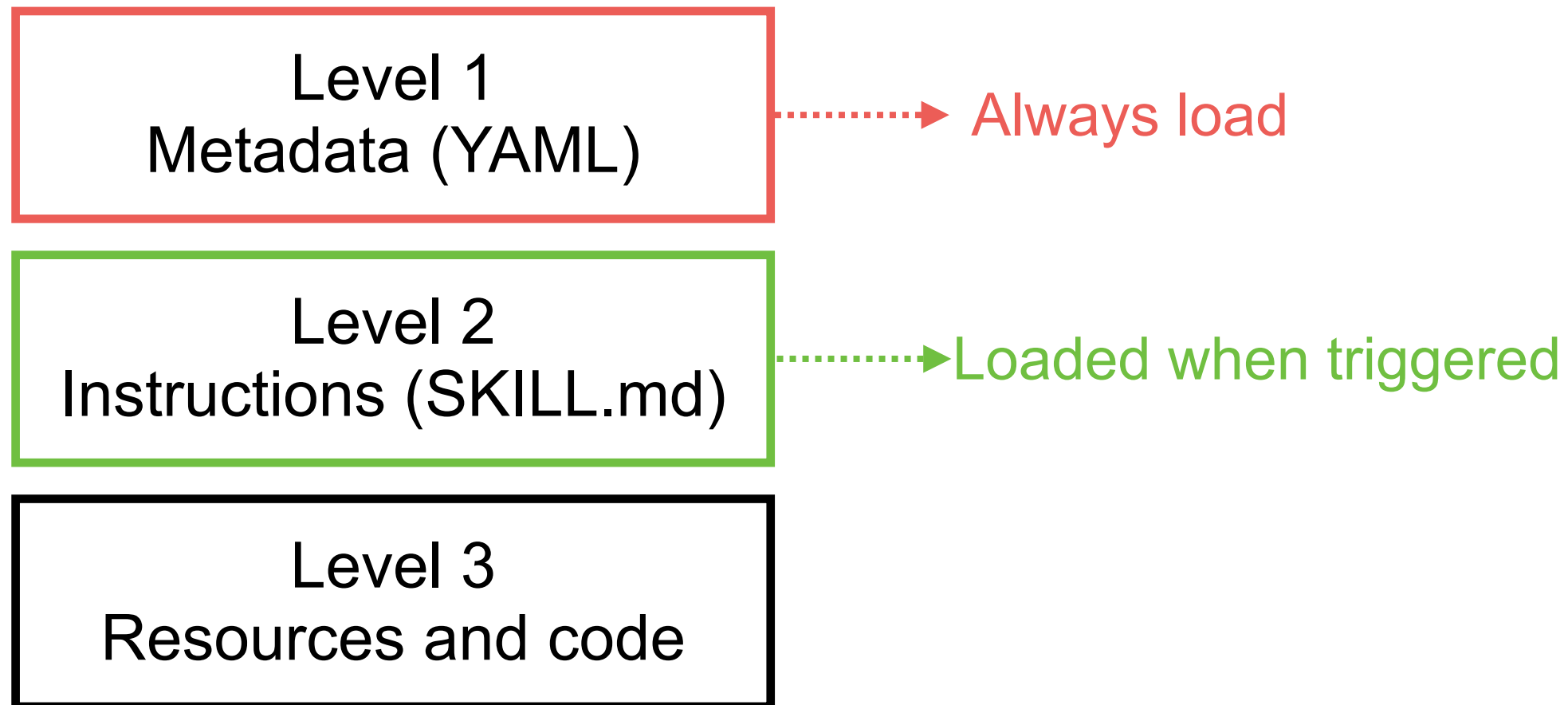
Load skills on-demand from context



<https://www.blog.langchain.com/choosing-the-right-multi-agent-architecture/>



Structure of Agent Skills



<https://agentskills.io/>



Prompt vs Skill

Features	Prompt	Agent Skills
Persistence	Ephemeral (chat-specific)	Persistent (Useable cross conversation)
Activation	Manual by user	Automatic (Triggered by context)
Structure	Unstructured text/ instruction	Structured markdown File system-based
Context load	Full instruction	Load only when needed
Token cost	Higher (Repeated usage)	Lower (Load on-demand)
Use case	One-time, creative quick query	Repeatable workflows Specialized tasks

<https://agentskills.io/>



Awesome Agent Skills

The screenshot shows the Agent-Skills.md website interface. At the top left is the logo 'Agent-Skills.md' and a 'Submit' button. The main heading is 'Find awesome Agent Skills'. Below it is a description: 'Agent-Skills.md is a agent skills marketplace, to find the right agent skills for you.' A search bar contains the text 'c#'. Below the search bar, three skill cards are displayed:

- csharp-developer**
Use when building C# applications with .NET 8+, ASP.NET Core APIs, or Blazor webapps. Invoke for Entity Framework Core, minimal APIs, async patterns, CQRS with MediatR. Keywords: C#, .NET, ASP.NET...
Tags: c#, .net, aspnet-core, blazor
Author: Jeff Allan (24 stars, 5 forks)
- async-expert**
Expert in asynchronous programming patterns across languages (Python asyncio, JavaScript/TypeScript promises, C# async/await, Rust futures). Use for concurrent programming, event loops, async patterns...
Tags: asynchronous-programming, concurrency, performance-optimization, error-handling
Author: martinholovsky (9 stars, 2 forks)
- xUnit Test Framework**
Execute and generate xUnit tests for C#/.NET projects with FluentAssertions and Moq support.
Tags: dotnet, xunit, test-framework, moq
Author: FortiumPartners (5 stars, 1 fork)

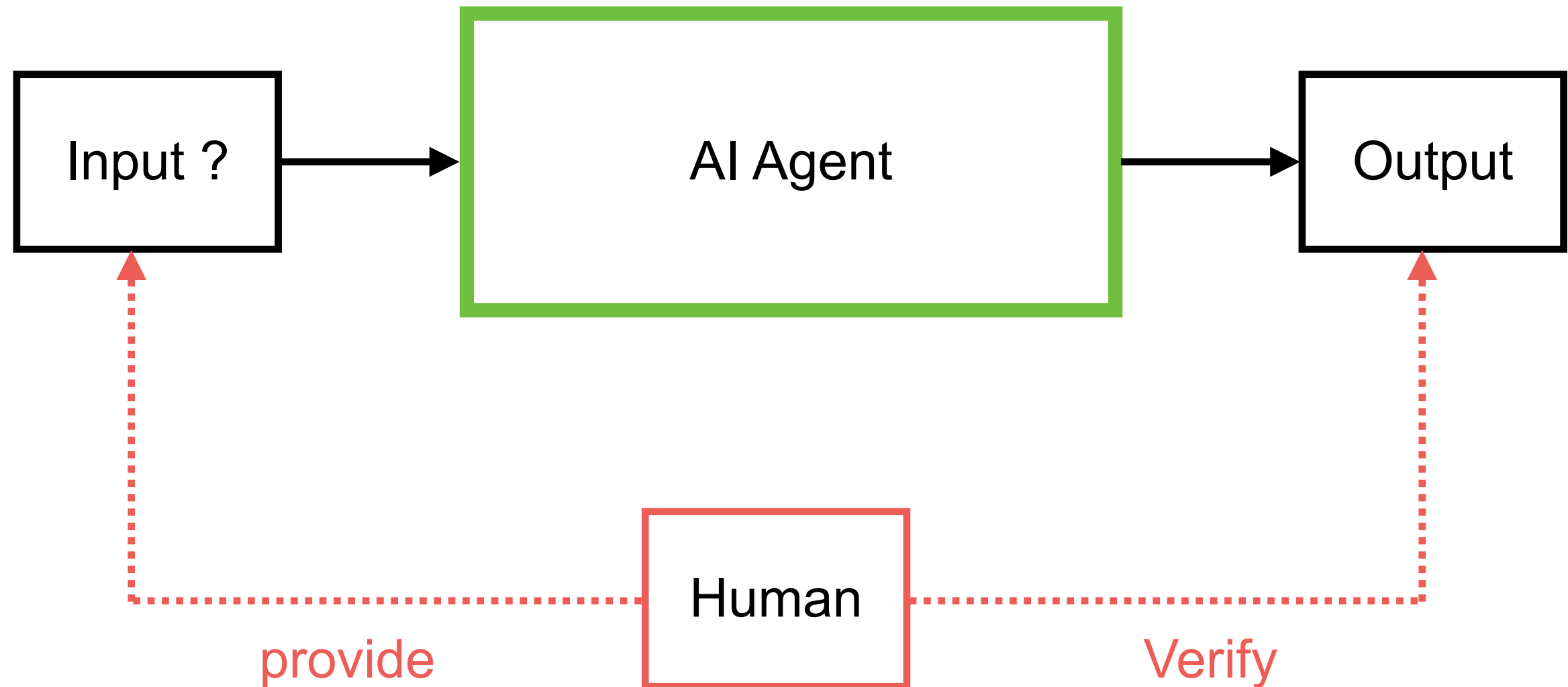
<https://agent-skills.md/>



How to provide inputs ?



How to provide inputs ?



Manage context !! In Software development



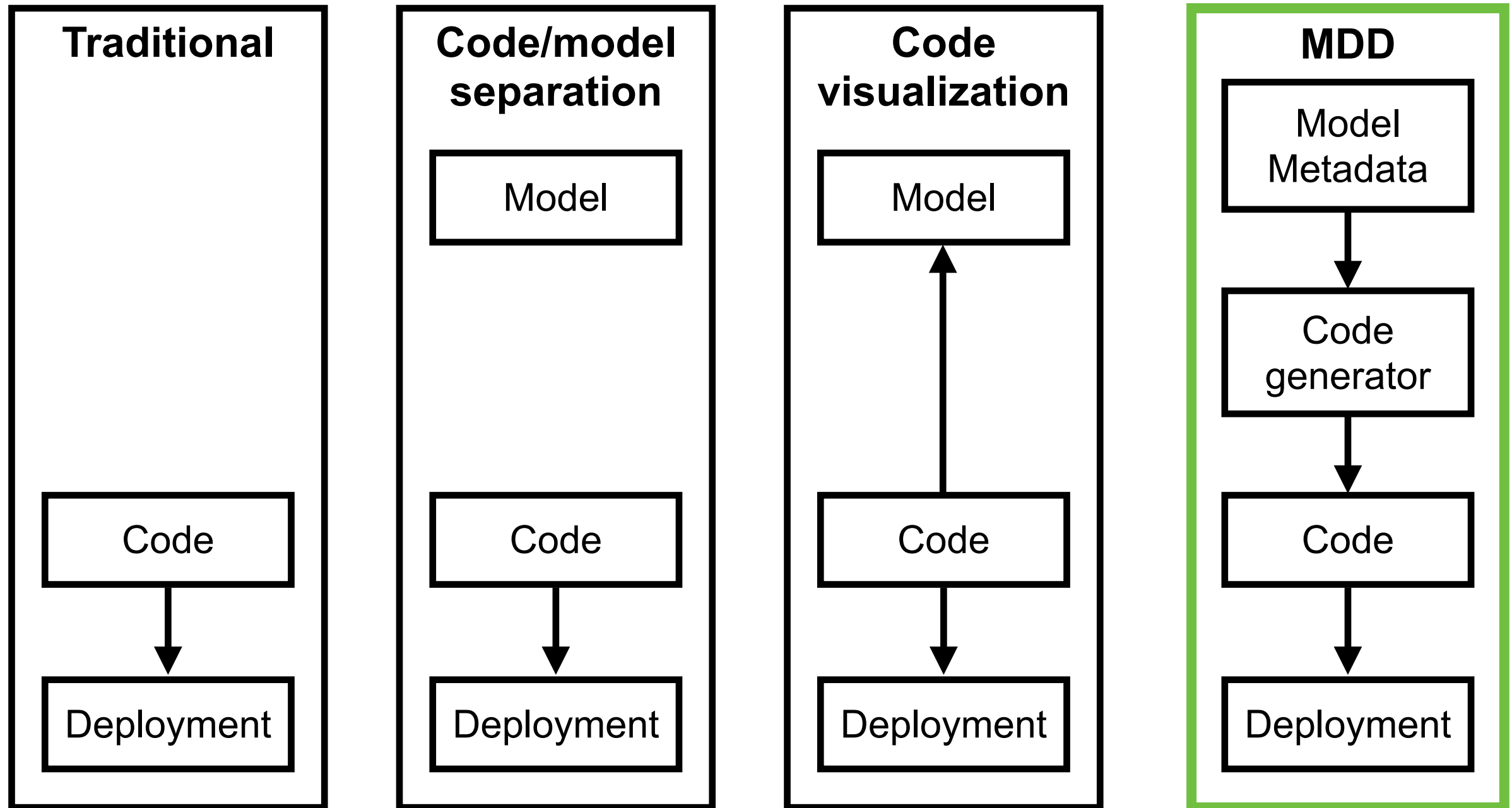
Specification ?
Architecture ?
Modeling ?
Coding convention ?
Project structure ?
Better practice ?
Technology stack ?



Model-Driven Development (MDD)



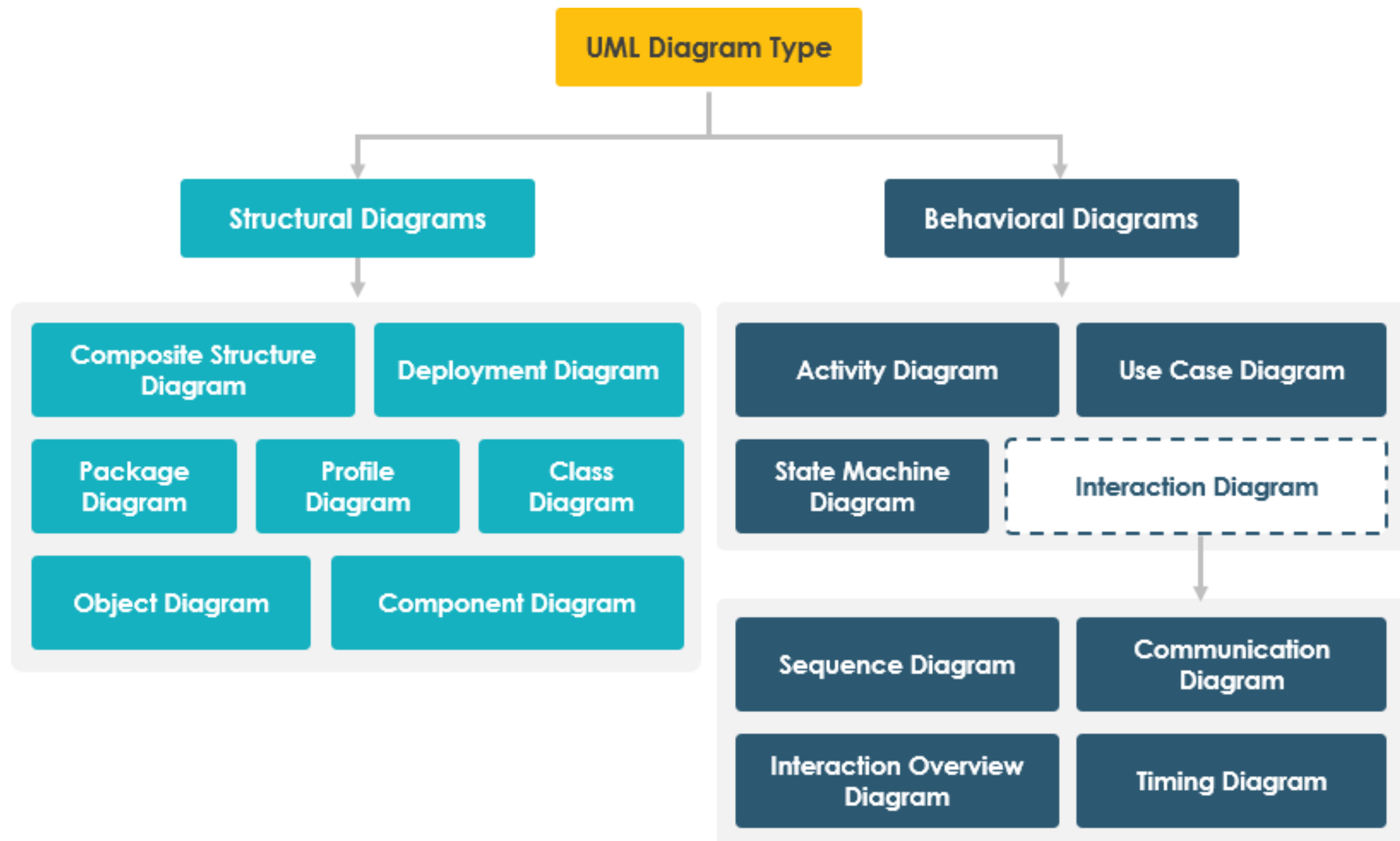
Traditional coding to MDD



<https://arxiv.org/abs/2410.18489>



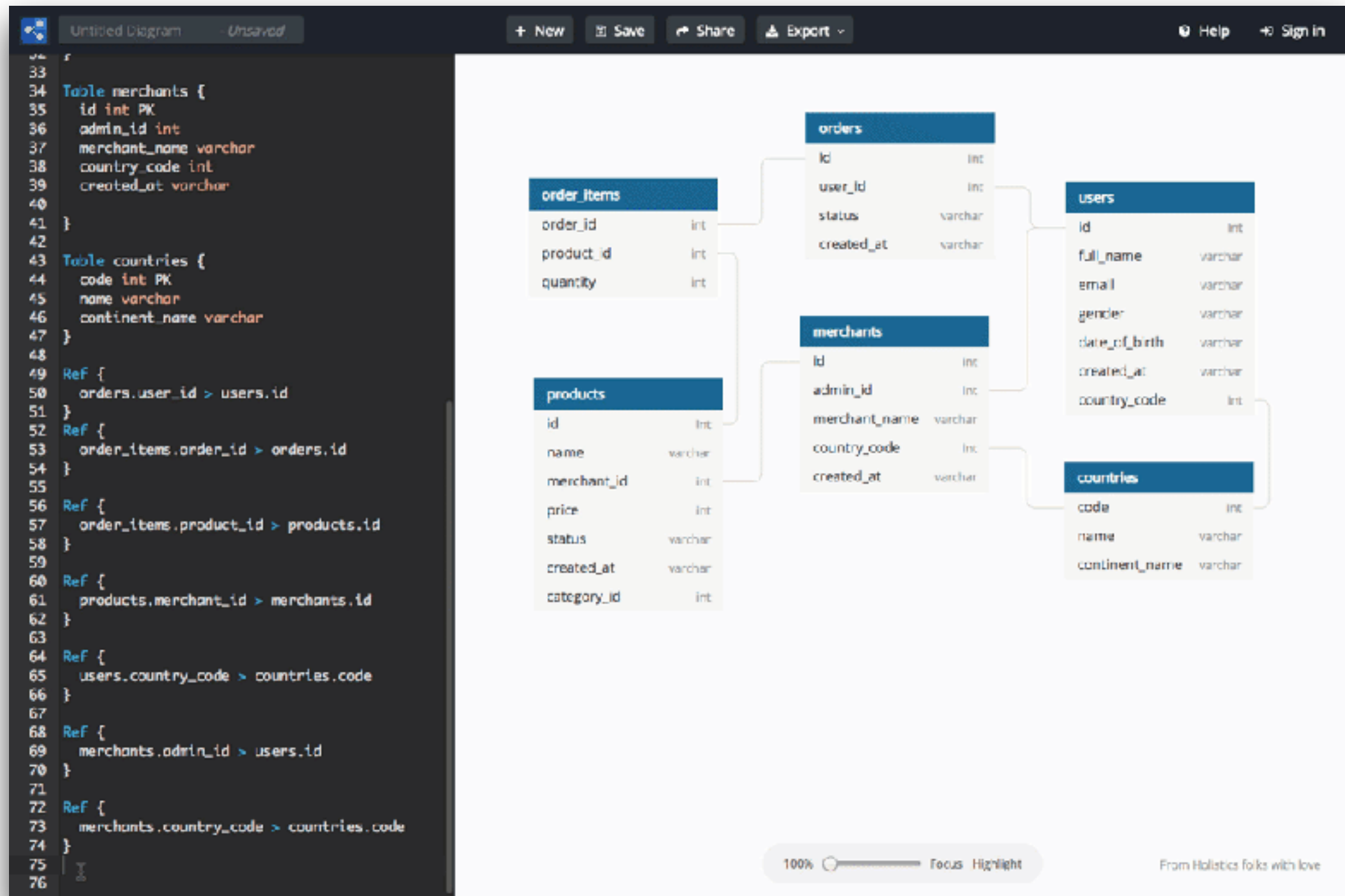
UML Diagrams !!



<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/behavior-vs-structural-diagram/>



ER Diagrams for Database !!



<https://dbml.dbdiagram.io/home/>



Mermaid Diagram

Diagram as a Code (AI-friendly)

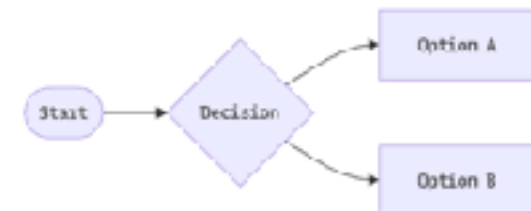
Mermaid Diagramming and charting tool

JavaScript based diagramming and charting tool that renders Markdown-inspired text definitions to create and modify diagrams dynamically.

Try Editor

Get started

```
1 flowchart LR
2   A(["Start"])
3   A --> B["Decision"]
4   B --> C["Option A"]
5   B --> D["Option B"]
```



<https://mermaid.js.org/>



Swagger or OpenAPI

Standardized specification for describe REST APIs

The screenshot displays the Swagger.io homepage. On the left, a hero section titled 'The Future of AI Relies on API Quality' features a green highlight on 'API Quality'. Below the title, text describes Swagger's role in API design, governance, and testing. Two buttons, 'Find your tool' and 'Read the docs', are present. On the right, a preview of a 'pet' API specification is shown, listing endpoints like '/pet/{petId}/uploadImage' (POST), '/pet' (POST), '/pet' (PUT), '/pet/findByStatus' (GET), '/pet/{petId}' (GET), '/pet/{petId}' (POST), and '/pet/{petId}' (DELETE).

Swagger.
Powered by SMARTCAR

Products ▾ Community ▾ Learn ▾ Resources ▾

Search Sign In Get started

The Future of AI Relies on API Quality

Swagger enables design, governance, and testing across the full AI-enabled API lifecycle, ensuring quality at every step.

Build APIs ready for humans, LLMs, agents, and continuous innovation.

Find your tool Read the docs →

TRUSTED BY

pet Everything about your Pets

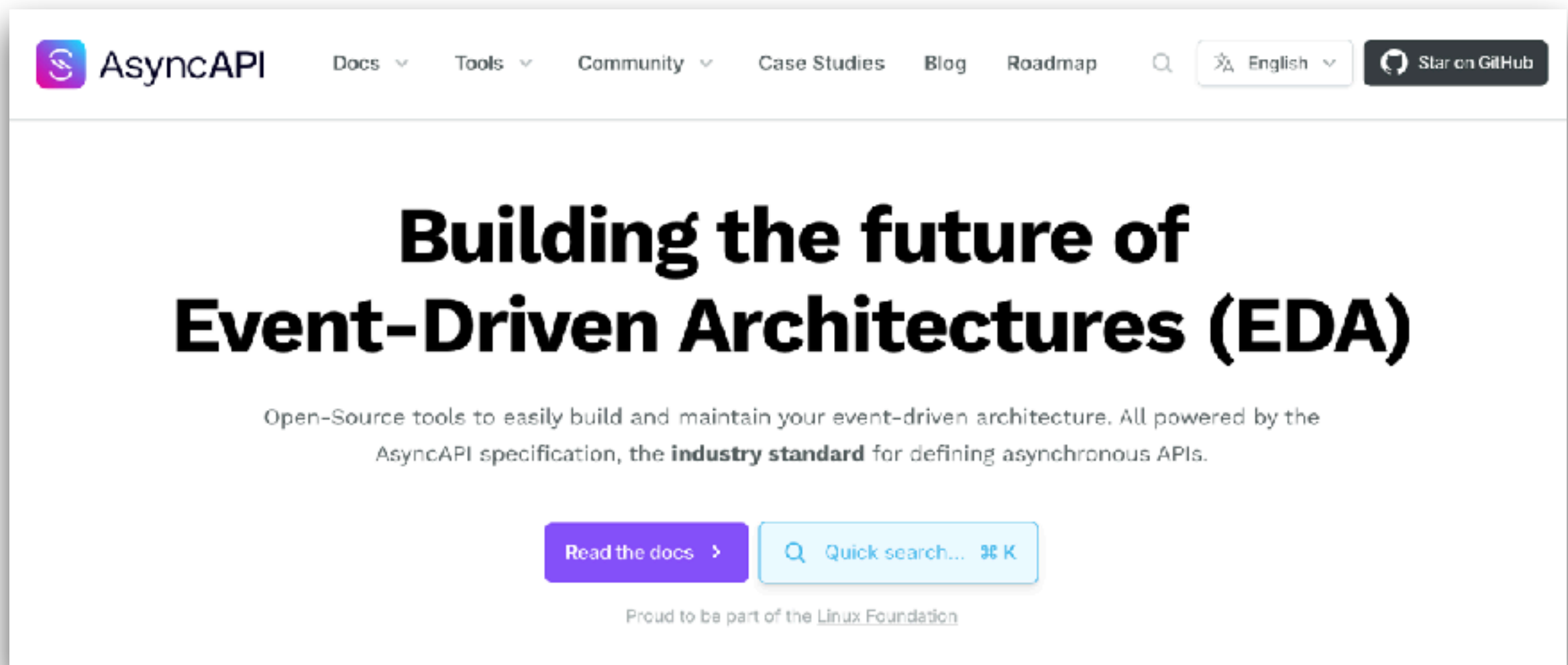
POST	/pet/{petId}/uploadImage	uploads an image	🔒
POST	/pet	Add a new pet to the store	🔒
PUT	/pet	Update an existing pet	🔒
GET	/pet/findByStatus	Finds Pets by status	🔒
GET	/pet/{petId}	Find pet by ID	🔒
POST	/pet/{petId}	Updates a pet in the store with form data	🔒
DELETE	/pet/{petId}	Deletes a pet	🔒

<https://swagger.io/>



AsyncAPI

Standardized specification for describe asynchronous and event-driven APIs (Kafka, AMQP, WebSocket)



<https://www.asyncapi.com/>



JSONSchema

Declarative language used to annotate and validate the structure, constraints and data type of JSON documents

Data validation

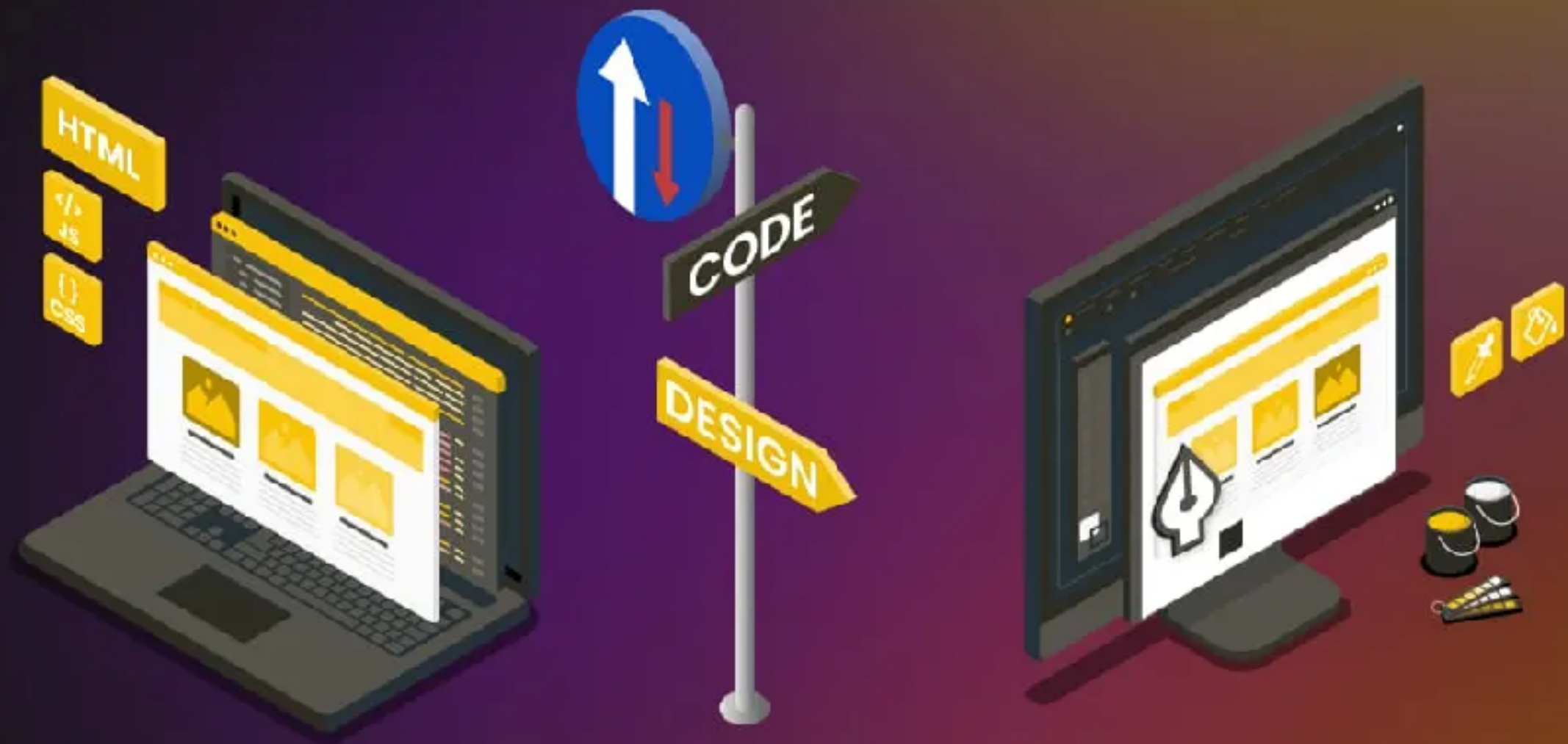
Documentation

Automated
testing

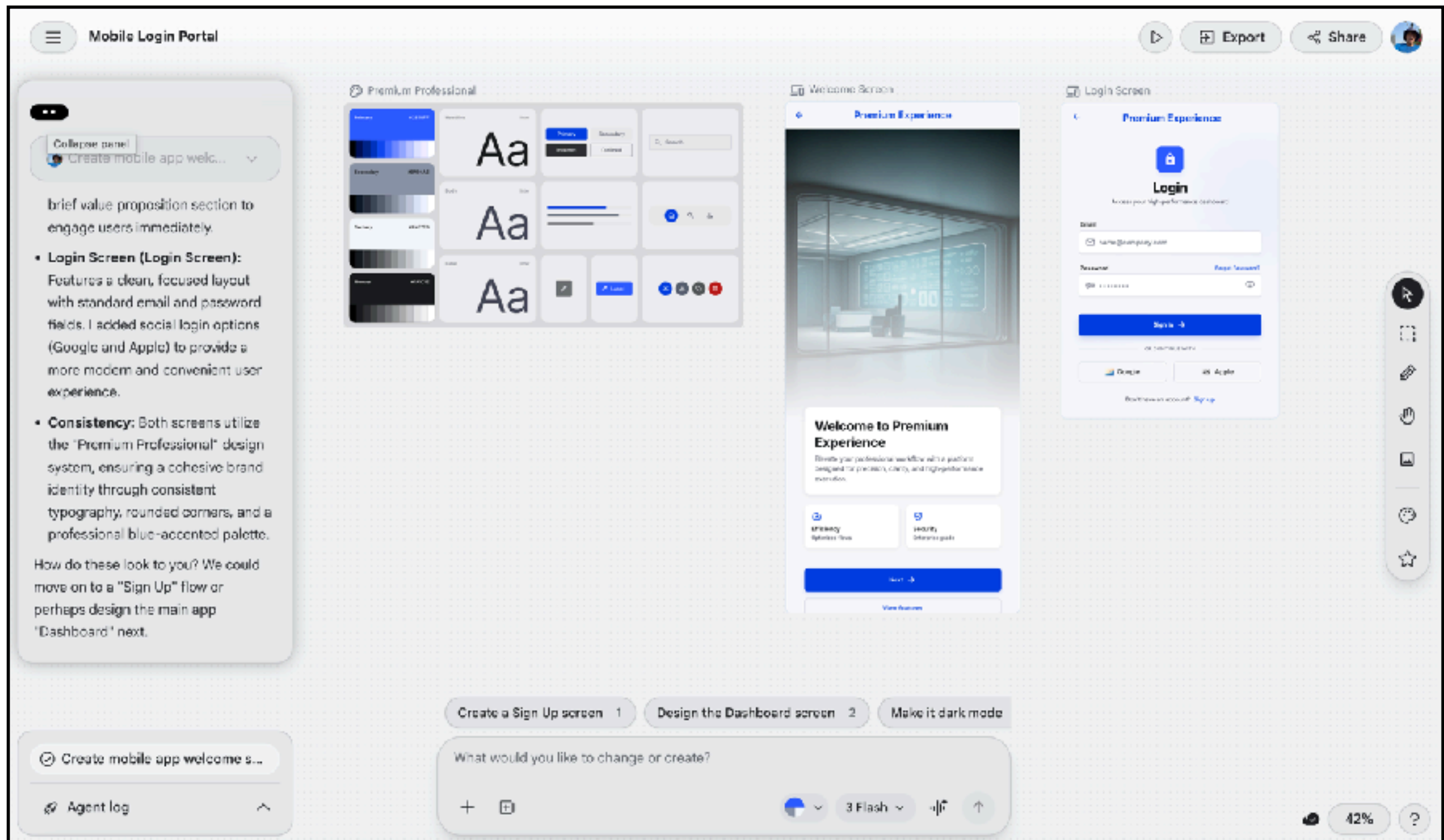
<https://json-schema.org/>



Design to Code Code to Design



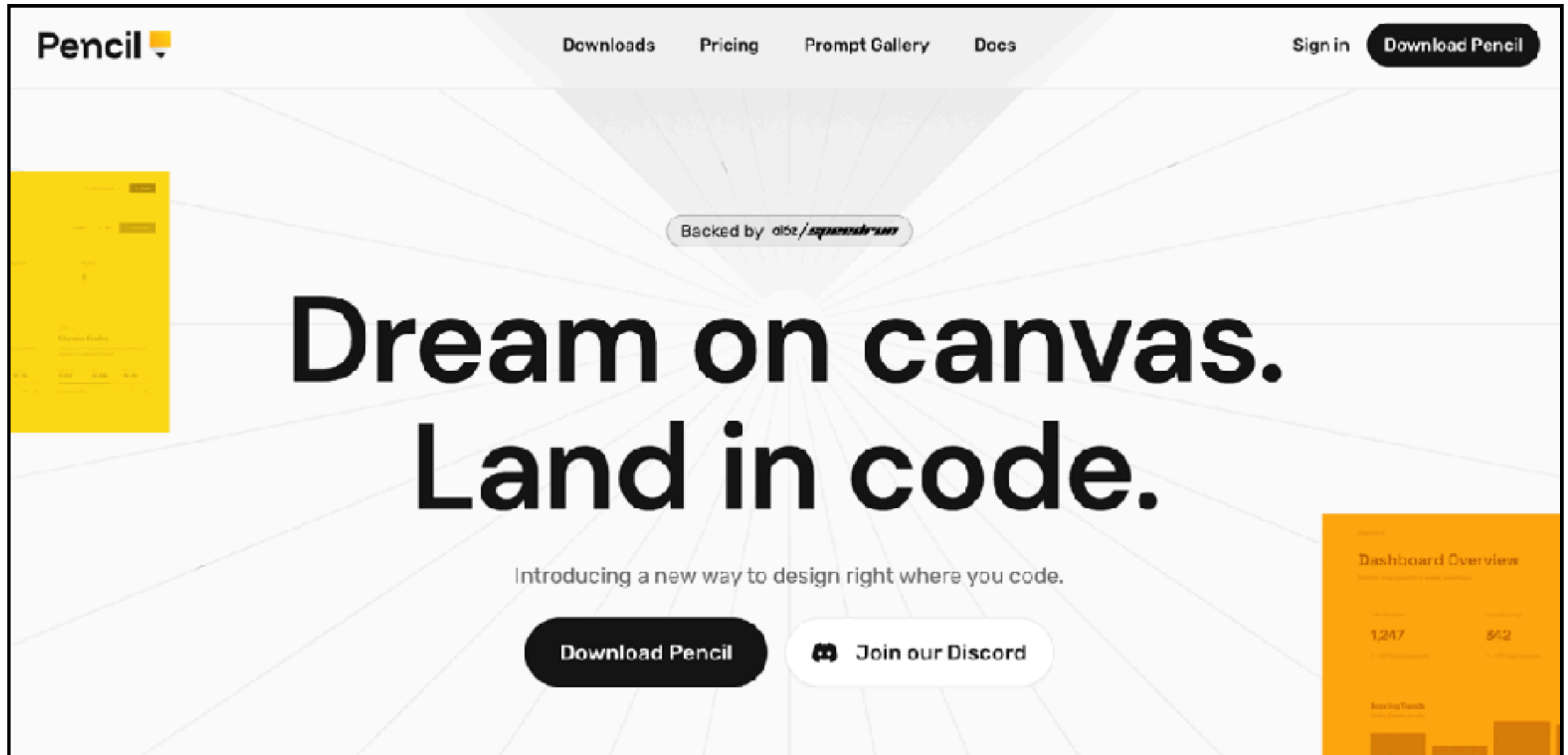
Google Stitch



<https://stitch.withgoogle.com/>



Pencil



<https://www.pencil.dev/>



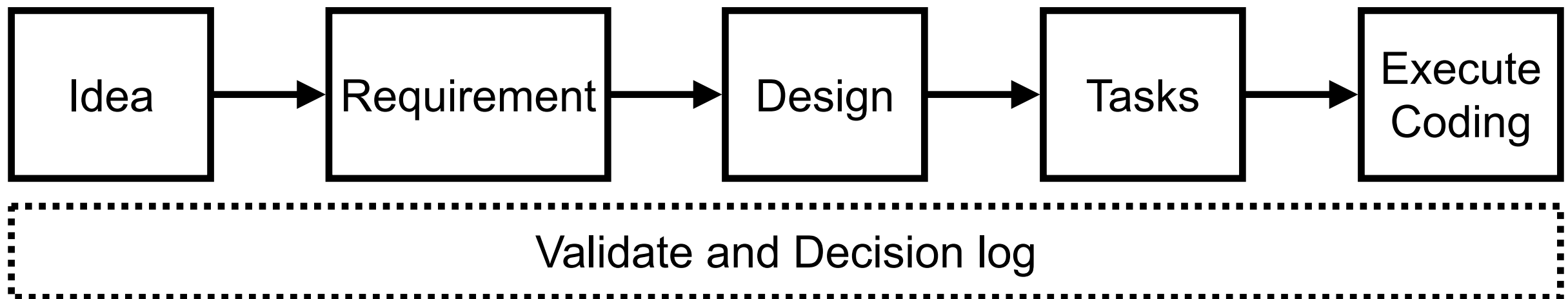
Specification-Driven Development (SDD)



SDD ?

Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code

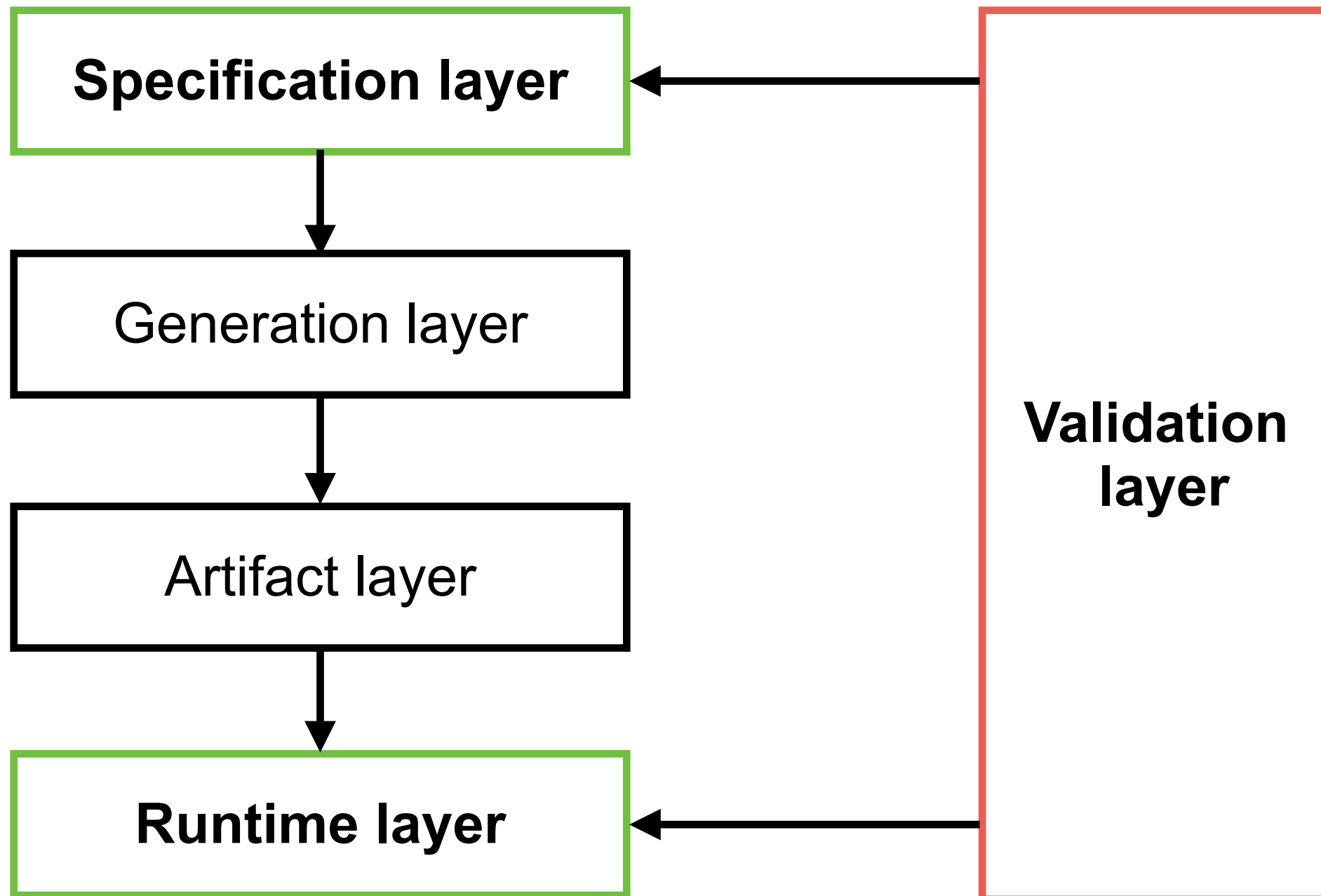


Types of Specification ?

User requirements
Design documents
Interface contracts (UI, API)
Data schemas
Validation rules
Data/business flows
Security policy and constraints
Resources and performance constraints



SSD Architecture



<https://www.infoq.com/articles/spec-driven-development>



MDD + SDD == Better result



When to use SDD ?

No one-size-fit-all !!

Low complexity

High complexity

<https://medium.com/google-cloud/benefits-and-challenges-of-spec-driven-development-and-how-antigravity-is-changing-the-game-3343a6942330>



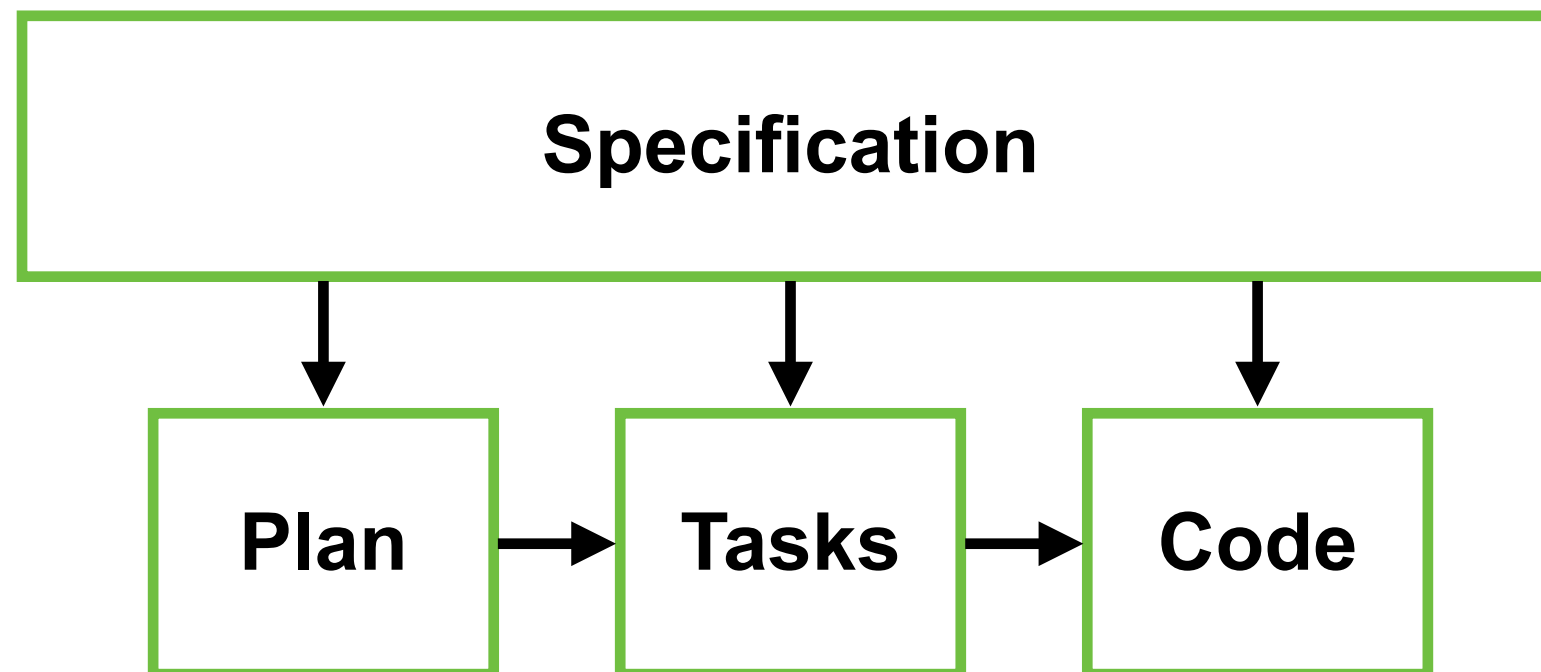
SDD Tools ?

GitHub Spec-kit

Open Spec

Kiro.dev

Tessl



Try by yourself !!



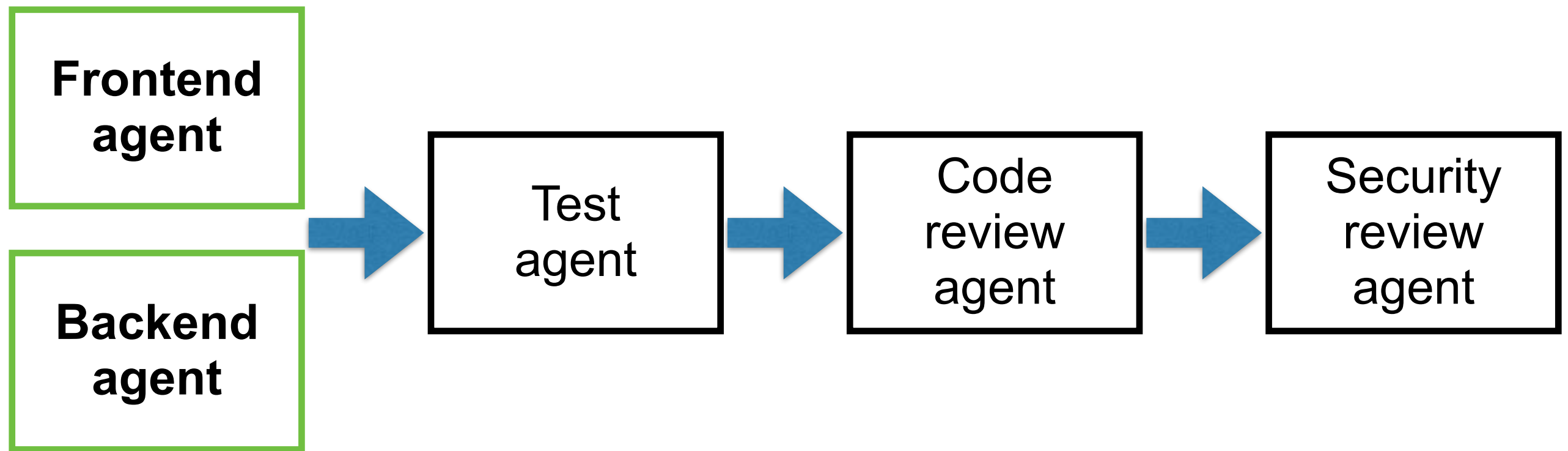


Workshop with VS Code + GitHub Copilot Chat

<https://code.visualstudio.com/docs/copilot/chat/getting-started-chat>



Design your Agents



Manage memory and knowledge of project !!



Frontend
Mobile application

Backend
RESTful API + Database

Write specification

Coding with AI Agent

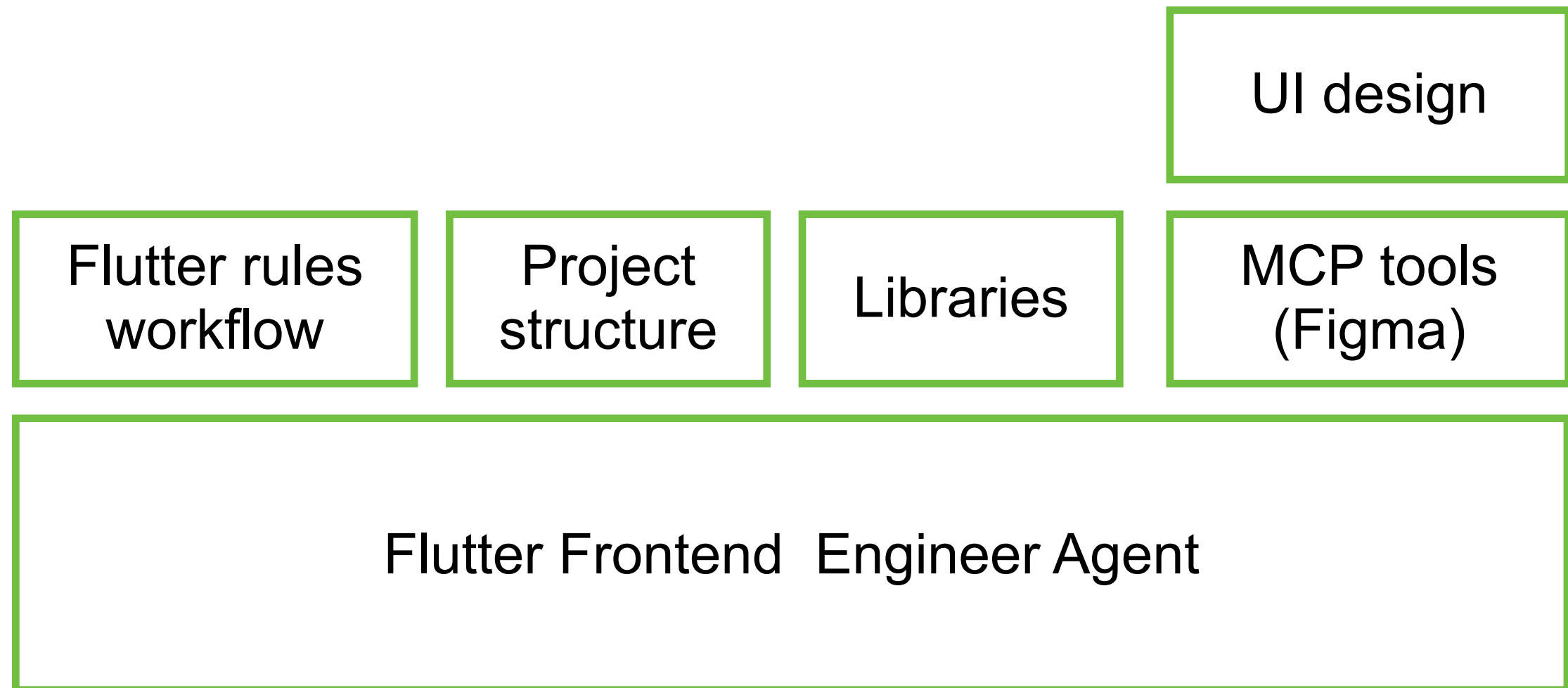
Run, Test and Verify !!

Version management system

<https://github.com/up1/demo-sharing-ai-agent-202605>

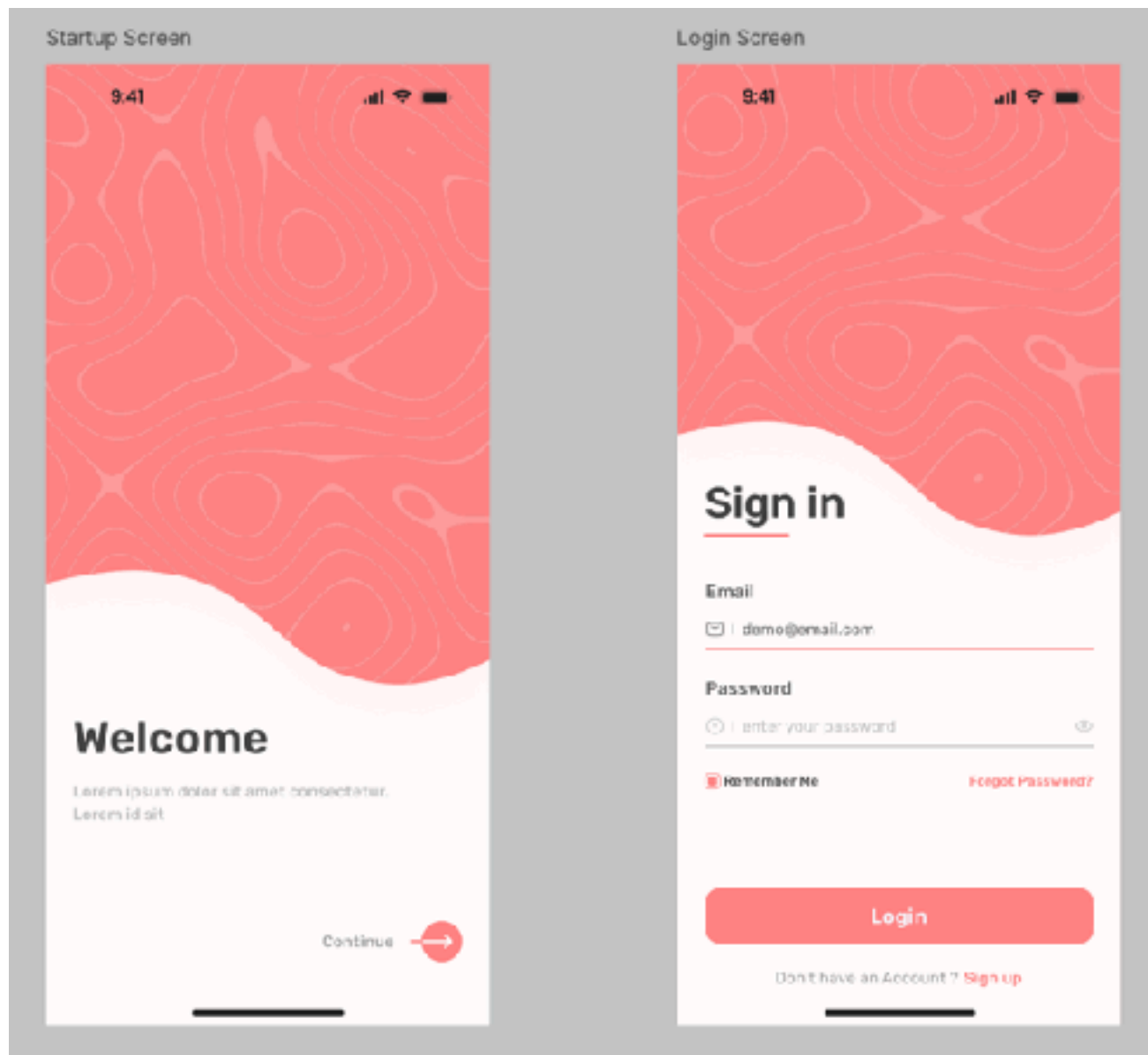


Frontend Agent (1)



User Interface Design

Design to Code technique



<https://bit.ly/3P35iNU>



Frontend Agent (2)

Write specification

UI design

Flutter rules
workflow

Project
structure

Libraries

MCP tools
(Figma)

Flutter Frontend Engineer Agent



Backend Agent

Write specification

NodeJS rules
workflow

Project structure

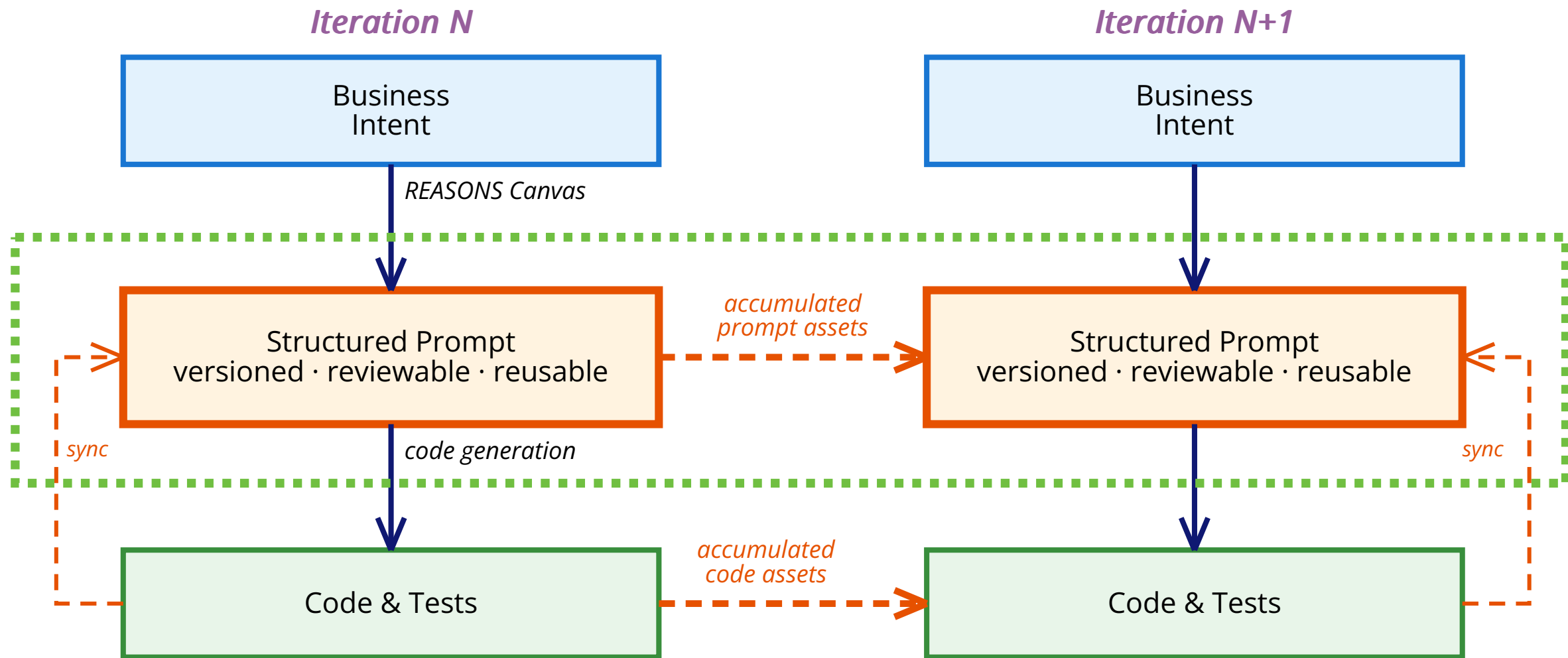
Libraries

NodeJS Backend Engineer Agent



Structure of Specification

Reliable, Repeatable and Versioning



<https://martinfowler.com/articles/structured-prompt-driven/>



Developer Survival Checklists



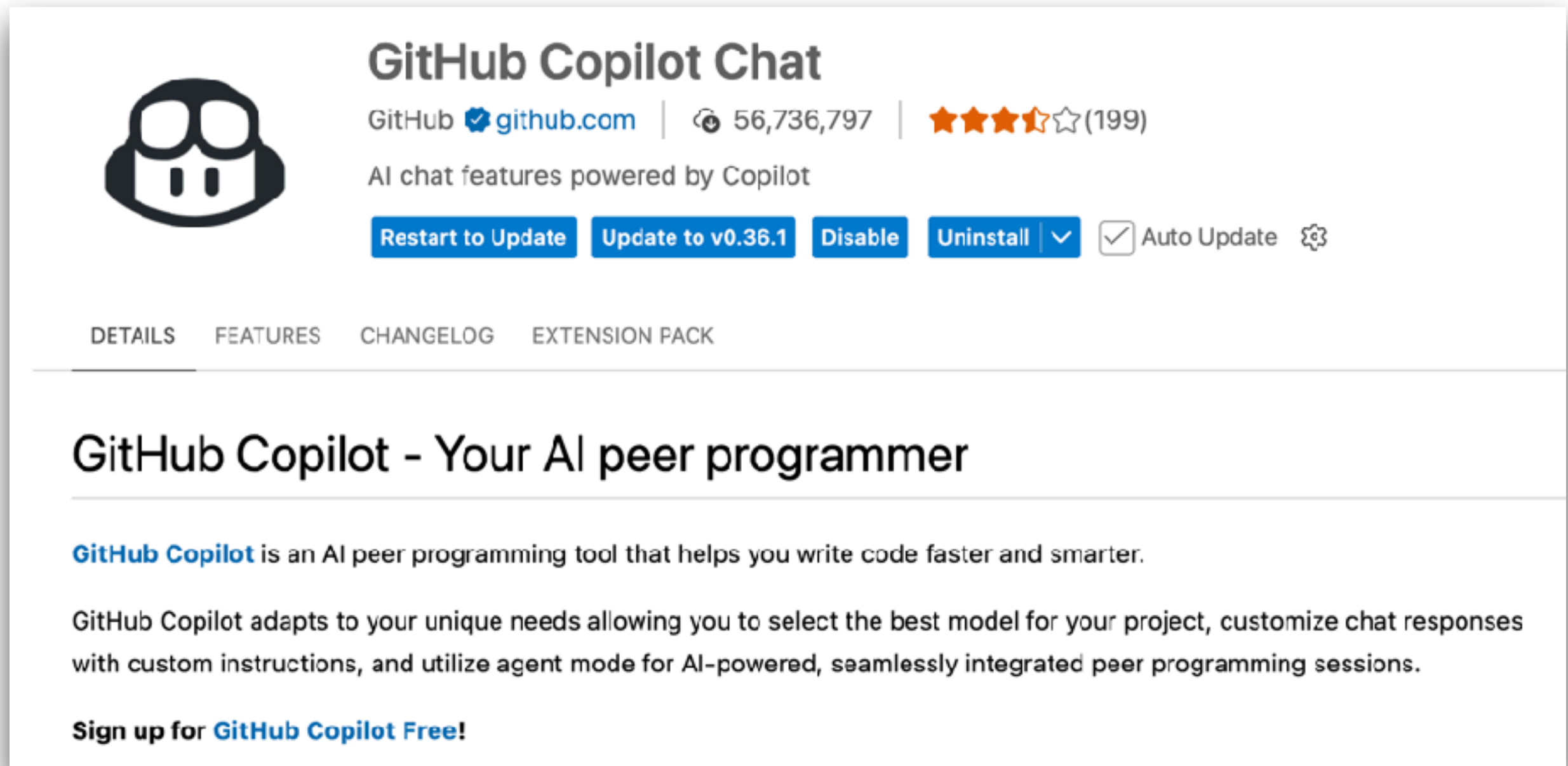
Developer Survival Checklists

1. Focus on logic (good or bad)
2. Audit AI Output (functional and non-functional)
3. Continuous Learning

**Focus on foundation knowledge
to effectively work with and guide AI system**



Install extension in VS Code



The image shows the GitHub Copilot Chat extension page in the Visual Studio Code marketplace. At the top, there's a header with the extension's name 'GitHub Copilot Chat', its publisher 'GitHub' with a verified badge and 'github.com', a download count of 56,736,797, and a star rating of 4.5 out of 5 stars based on 199 reviews. Below this, a description states 'AI chat features powered by Copilot'. A row of buttons includes 'Restart to Update', 'Update to v0.36.1', 'Disable', 'Uninstall' with a dropdown arrow, and a checked 'Auto Update' checkbox with a settings gear icon. A navigation bar below the buttons has links for 'DETAILS', 'FEATURES', 'CHANGELOG', and 'EXTENSION PACK'. The main content area has a heading 'GitHub Copilot - Your AI peer programmer' followed by a paragraph describing it as an AI peer programming tool. Another paragraph explains how it adapts to user needs. At the bottom, there's a call to action to 'Sign up for GitHub Copilot Free!'.

GitHub Copilot Chat
GitHub  github.com |  56,736,797 |      (199)
AI chat features powered by Copilot

[Restart to Update](#) [Update to v0.36.1](#) [Disable](#) [Uninstall](#)  ☒ Auto Update 

[DETAILS](#) [FEATURES](#) [CHANGELOG](#) [EXTENSION PACK](#)

GitHub Copilot - Your AI peer programmer

GitHub Copilot is an AI peer programming tool that helps you write code faster and smarter.

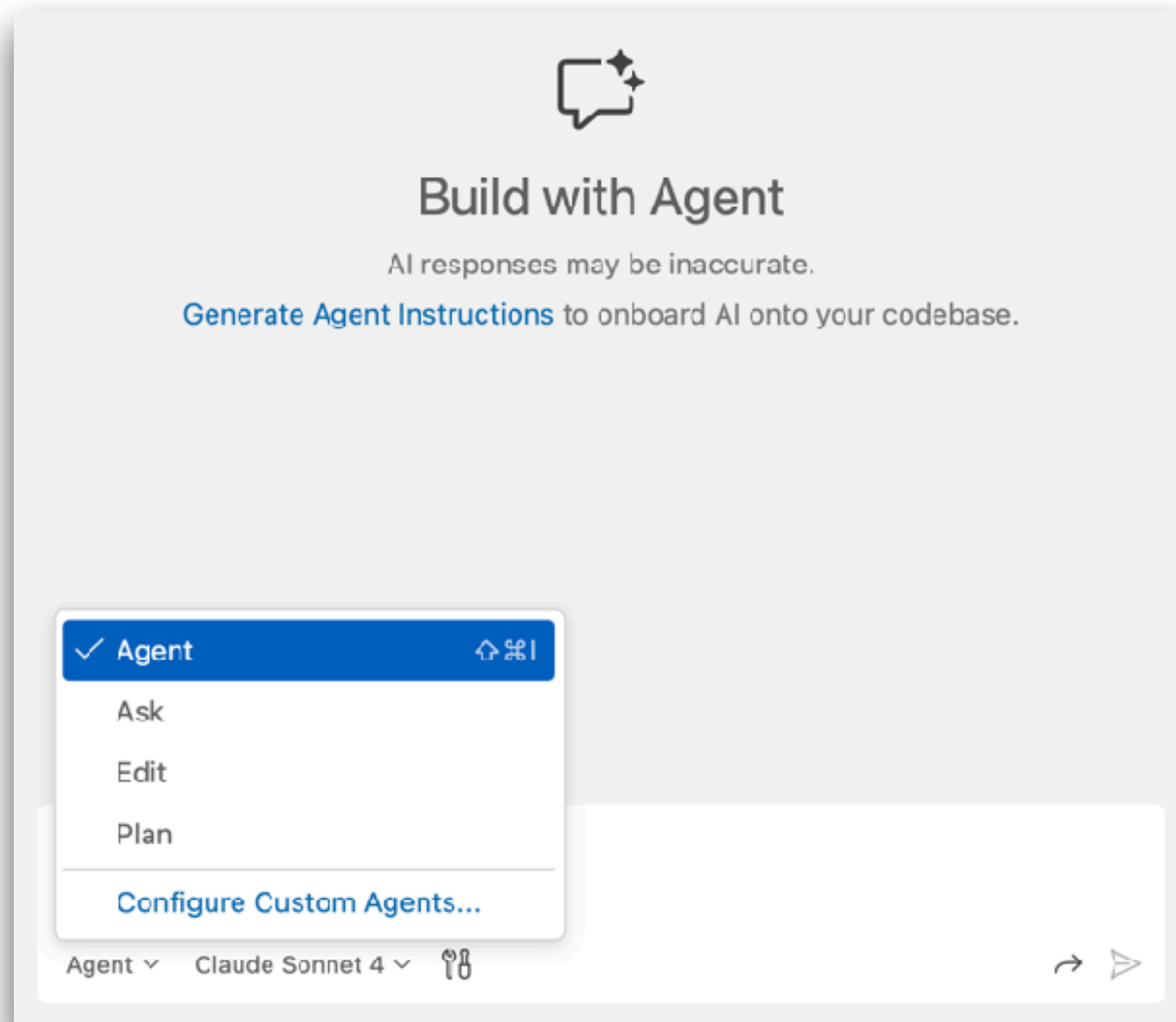
GitHub Copilot adapts to your unique needs allowing you to select the best model for your project, customize chat responses with custom instructions, and utilize agent mode for AI-powered, seamlessly integrated peer programming sessions.

Sign up for [GitHub Copilot Free!](#)

<https://marketplace.visualstudio.com/items?itemName=GitHub.copilot-chat>



Hello Copilot Chat in VS Code



Copilot Chat Modes

Agent

Ask

Edit

Plan

**Custom
Agent**

<https://code.visualstudio.com/docs/copilot/customization/custom-agents>



Custom instructions !!

.github/
copilot-instructions.md

Global



.github/instructions
NAME.instructions.md

Specific by types



AGENTS.md

Disable by default in VS Code

<https://docs.github.com/en/copilot/how-tos/configure-custom-instructions/add-repository-instructions>



Customization !!

Agent

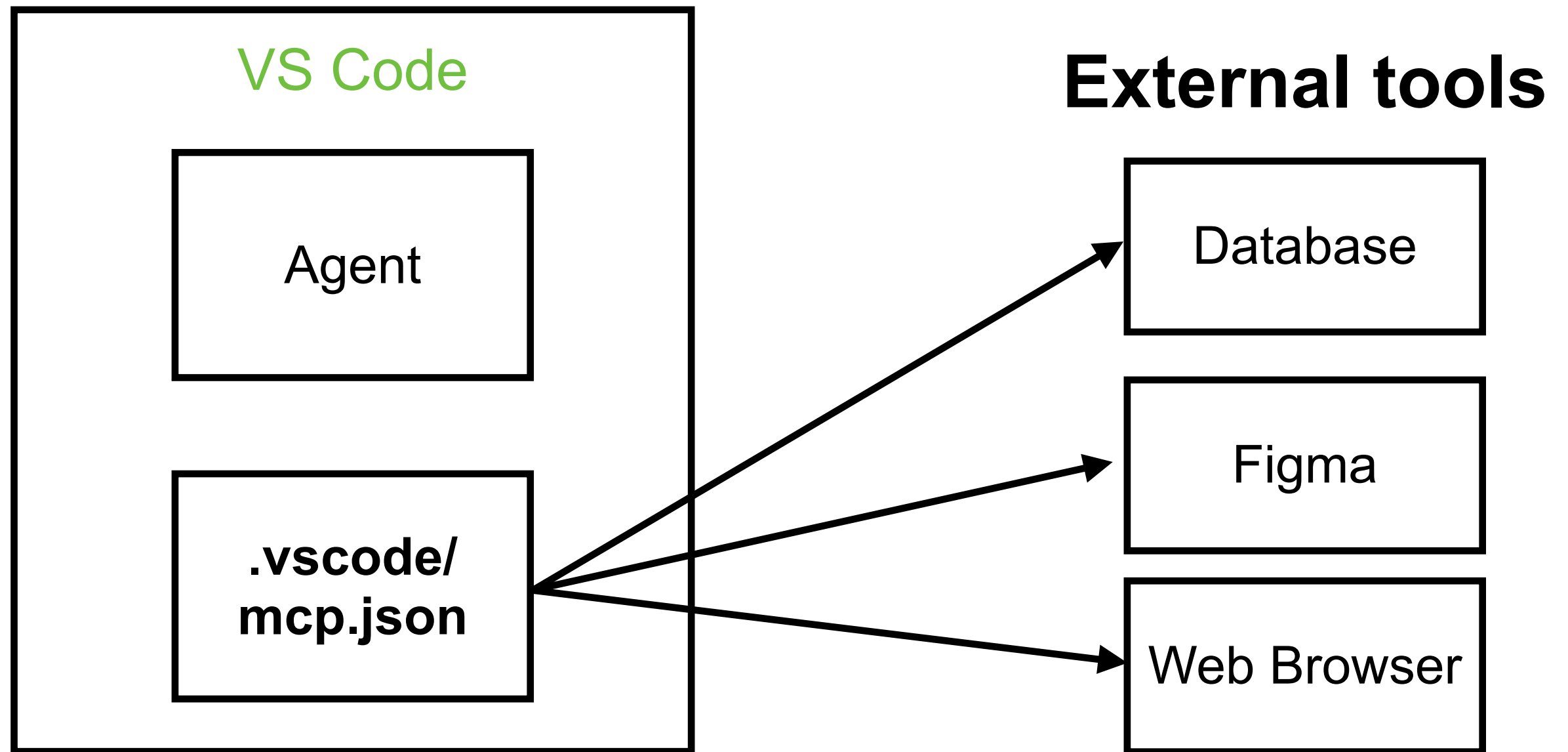
Instruction

Prompts

Skills



Working with MCP



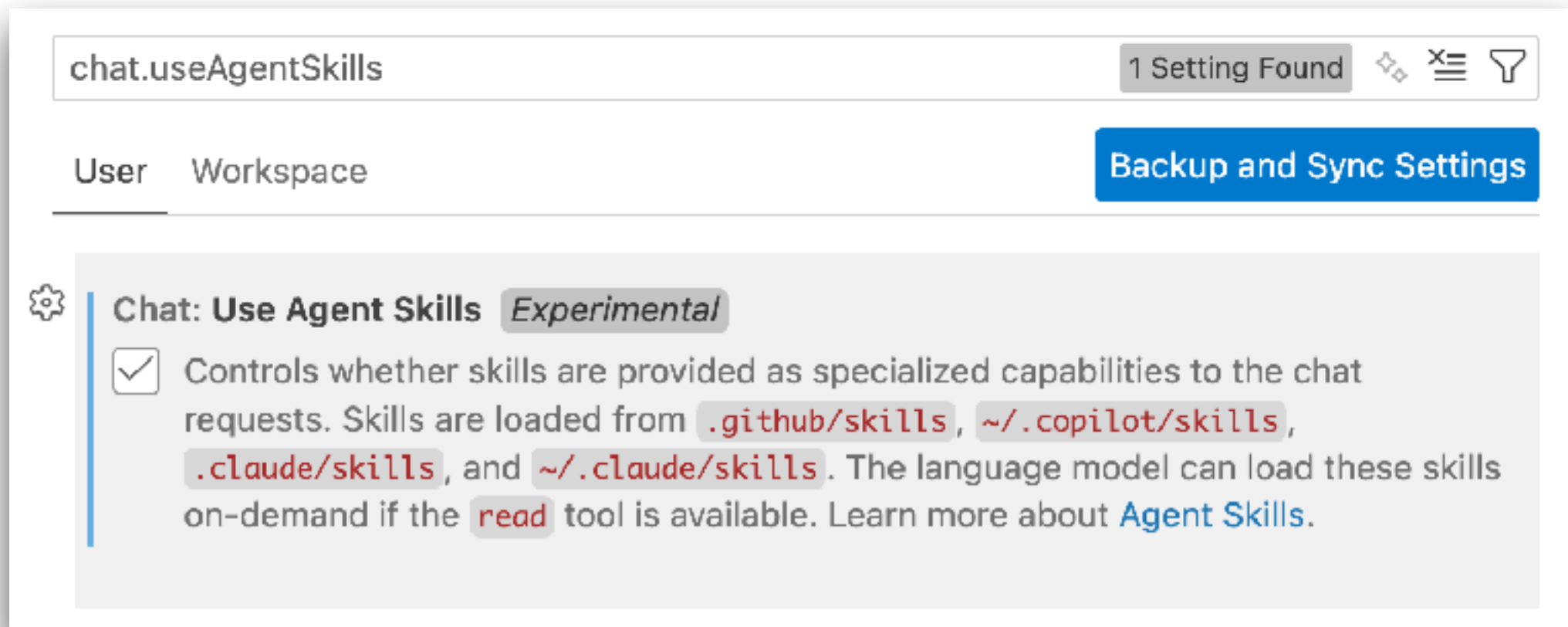
<https://code.visualstudio.com/docs/copilot/customization/custom-agents>



Enable Agent Skills in VS Code

Setting => “**chat.useAgentSkills**”

Create skills in “.github/skills/name/SKILL.md”



<https://code.visualstudio.com/docs/copilot/customization/agent-skills>



Workshop with Claude Code

<https://claude.com/product/claude-code>





`curl -fsSL https://claude.ai/install.sh | bash`



Workshop with VS Code + Cline

<https://cline.bot/>



Cline ?

Opensource AI coding agent
Development partner

Create/edit
files

Run in
terminal

Browse
website

Multi-step
tasks

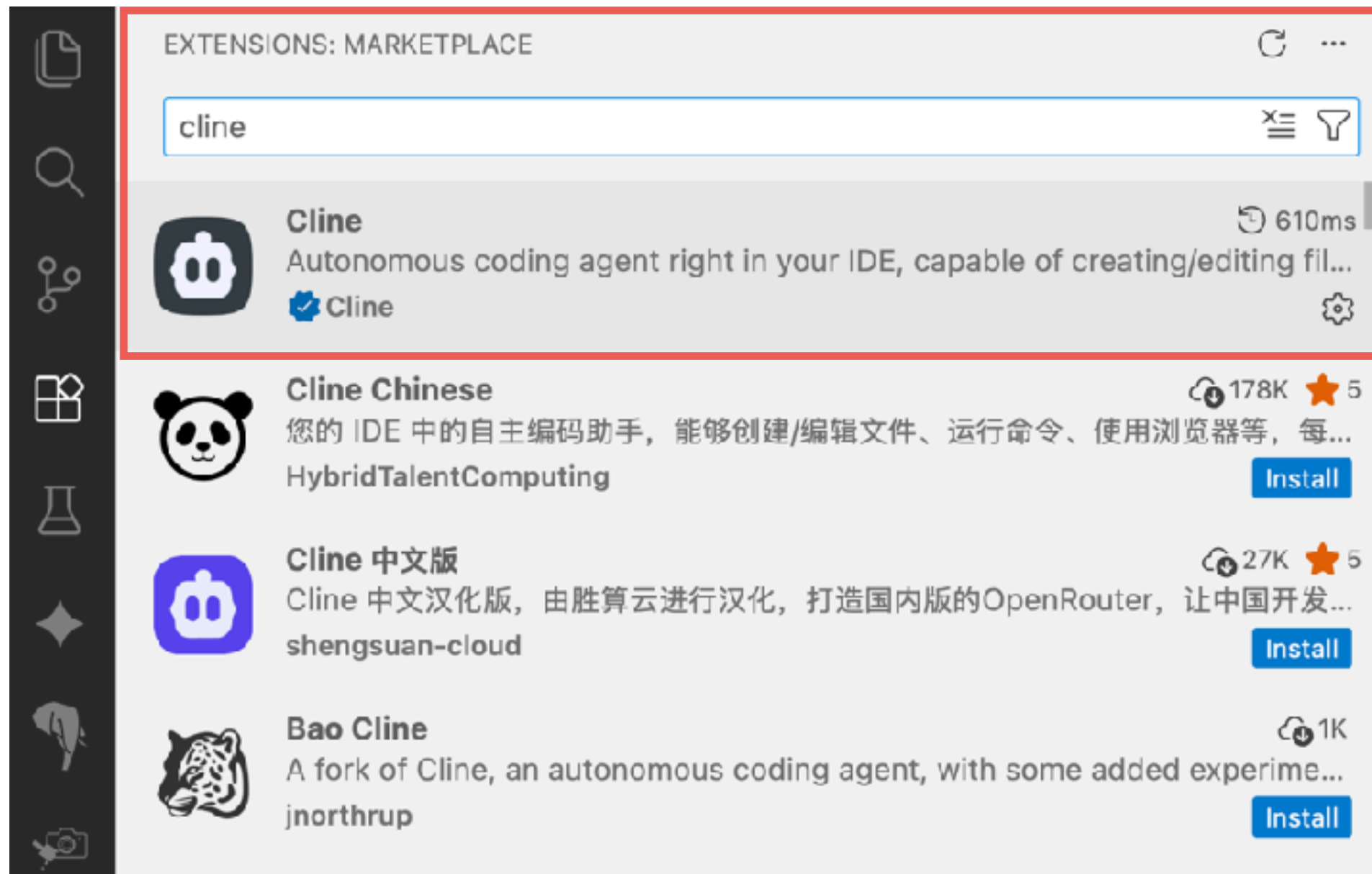
Human-in-the
loop

Support tools

<https://cline.bot/>



Install extension in VS Code



<https://marketplace.visualstudio.com/items?itemName=saoudrizwan.claude-dev>



Install extension in VS Code



Cline

Cline  [cline.bot](#) |  2,902,099 |  (264)

Autonomous coding agent right in your IDE, capable of creating/editing files, running commands,

[Disable](#)

[Uninstall](#)



Auto Update



[DETAILS](#)

[FEATURES](#)

[CHANGELOG](#)

Meet Cline, an AI assistant that can use your **CLI aNd Editor**.

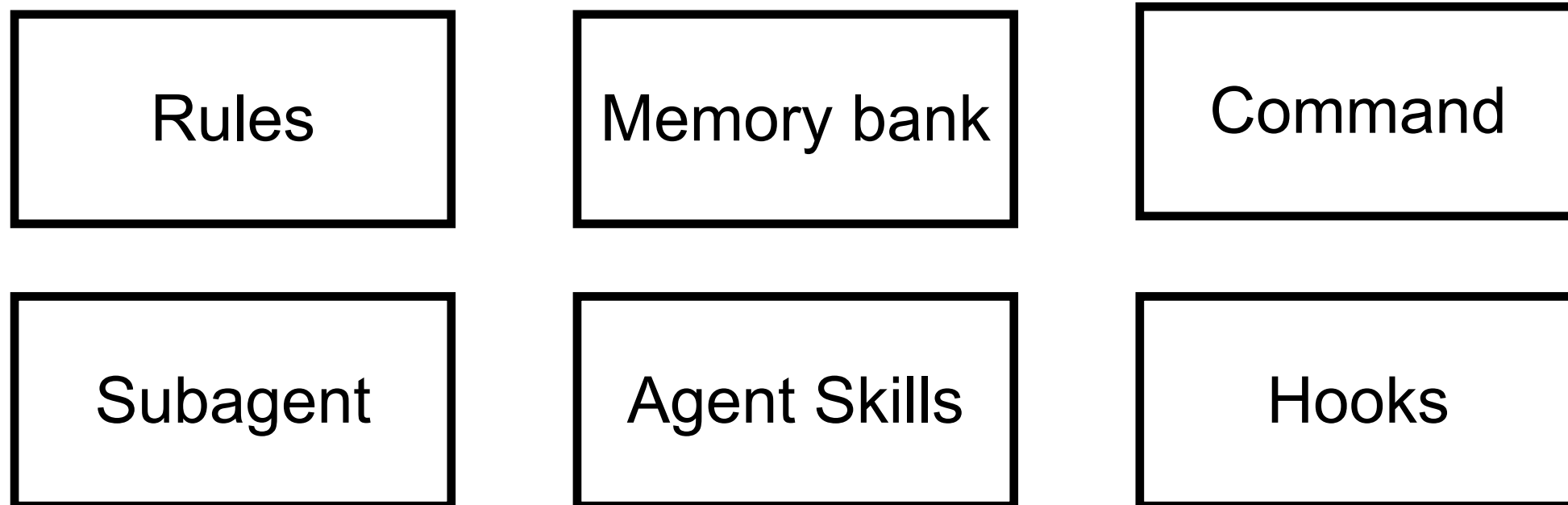
Thanks to [Claude Sonnet's agentic coding capabilities](#), Cline can handle complex software development tasks step-by-step. With tools that let him create & edit files, explore large projects, use the browser, and execute terminal commands (after you grant permission), he can assist you in ways that go beyond code completion or tech support. Cline can even use the Model Context Protocol (MCP) to create new tools and extend his own capabilities. While autonomous AI scripts traditionally run in sandboxed environments, this extension provides a human-in-the-loop GUI to approve every file change and terminal command, providing a safe and accessible way to explore the potential of agentic AI.

<https://marketplace.visualstudio.com/items?itemName=saoudrizwan.claude-dev>

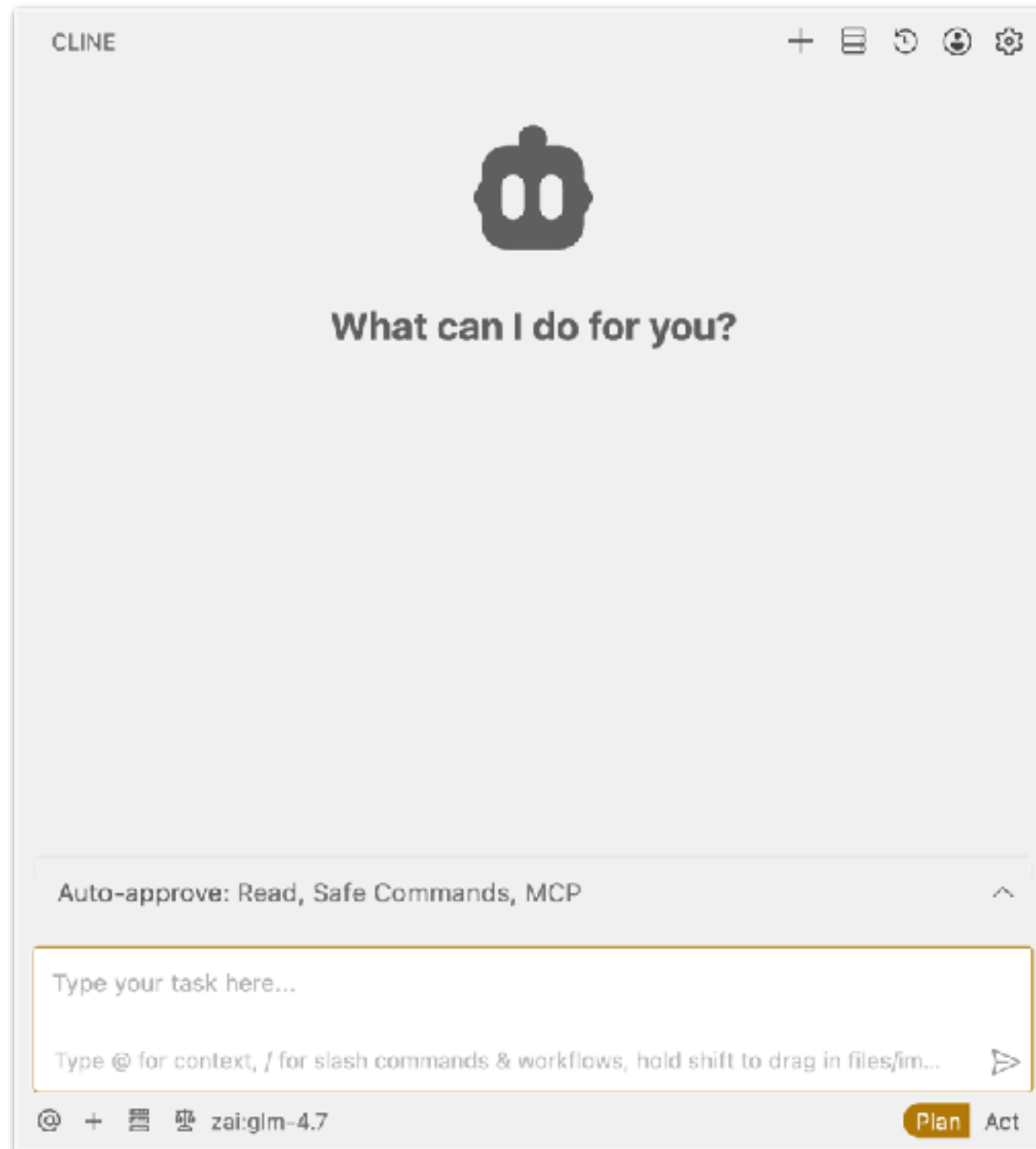


Working with Cline

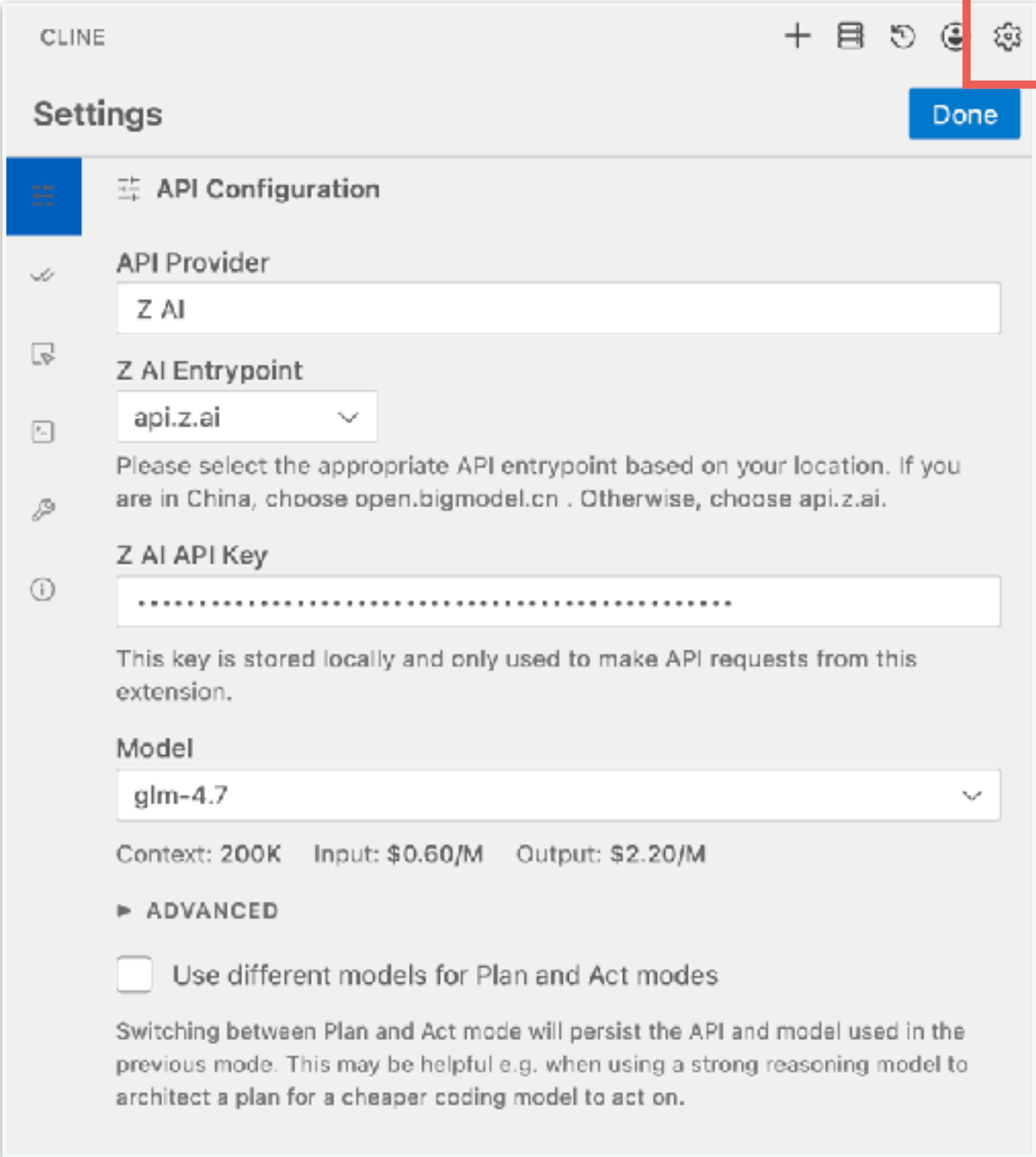
Plan and Act modes
Toggle-able rules (.clinerules)
Checkpoint management
Memory bank system



Chat with Cline



Select your model provider



CLINE

Settings Done

API Configuration

✓ API Provider
Z AI

📁 Z AI Entrypoint
api.z.ai

Please select the appropriate API endpoint based on your location. If you are in China, choose open.bigmodel.cn . Otherwise, choose api.z.ai.

🔑 Z AI API Key
.....

This key is stored locally and only used to make API requests from this extension.

Model
glm-4.7

Context: 200K Input: \$0.60/M Output: \$2.20/M

▶ ADVANCED

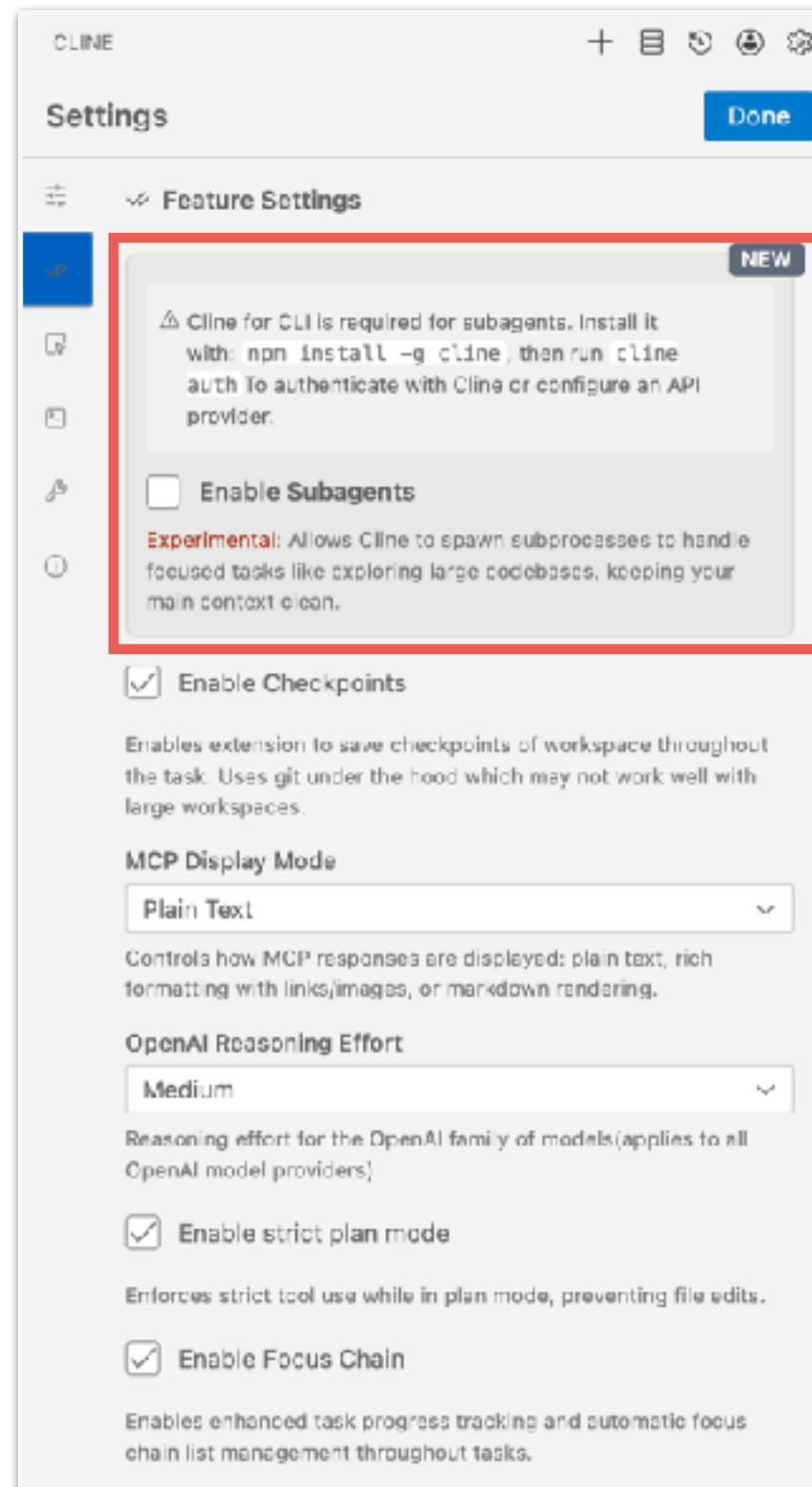
☐ Use different models for Plan and Act modes

Switching between Plan and Act mode will persist the API and model used in the previous mode. This may be helpful e.g. when using a strong reasoning model to architect a plan for a cheaper coding model to act on.

← Click to config



Feature Settings



Enable subagents
with Cline CLI



Cline CLI

`npm install -g cline
client --version`

```
Cline CLI  
Cline CLI Version: 1.0.10  
Cline Core Version: 3.52.0  
Commit: 66d7664  
Built: 2026-01-22T20:32:21Z  
Built by: runner  
Go version: go1.24.12  
OS/Arch: darwin/arm64
```

Support only Mac and Linux !!

<https://docs.cline.bot/cline-cli/overview>

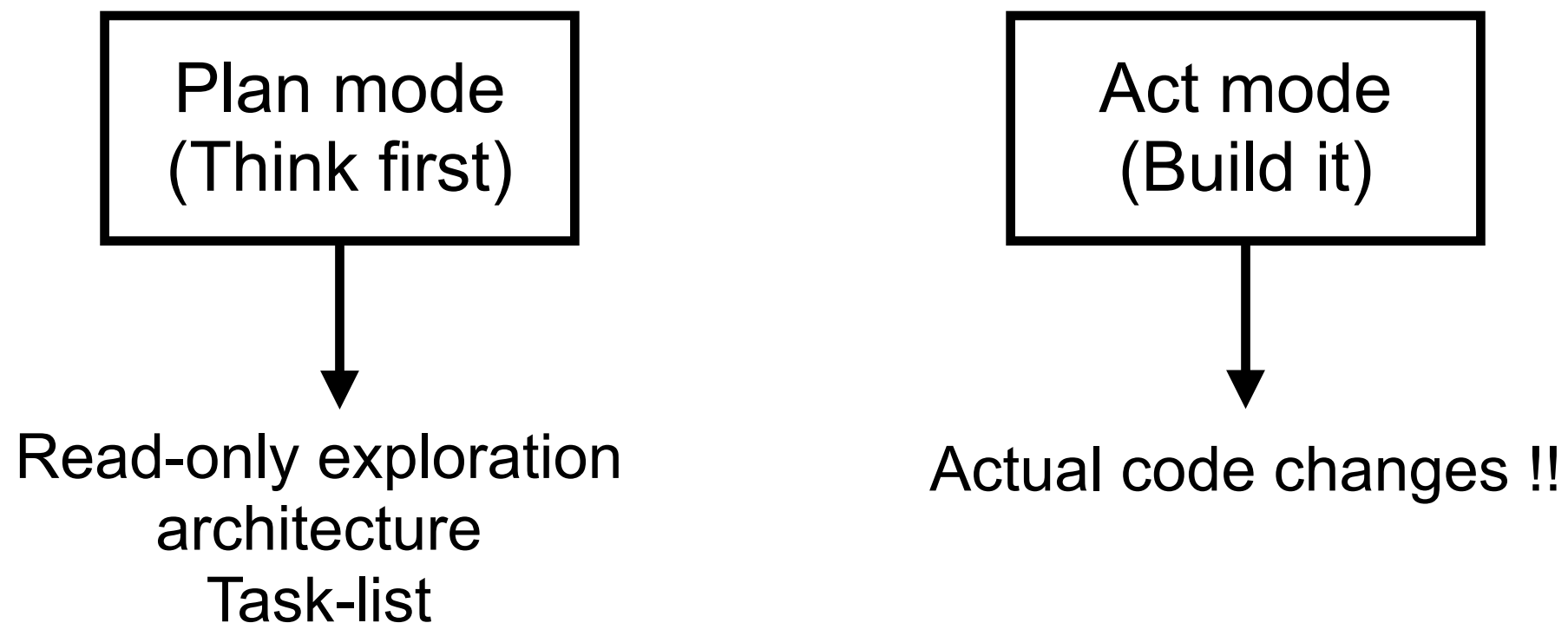


How to use Cline ?



Plan and Act mode

Separate strategic thinking from implementation

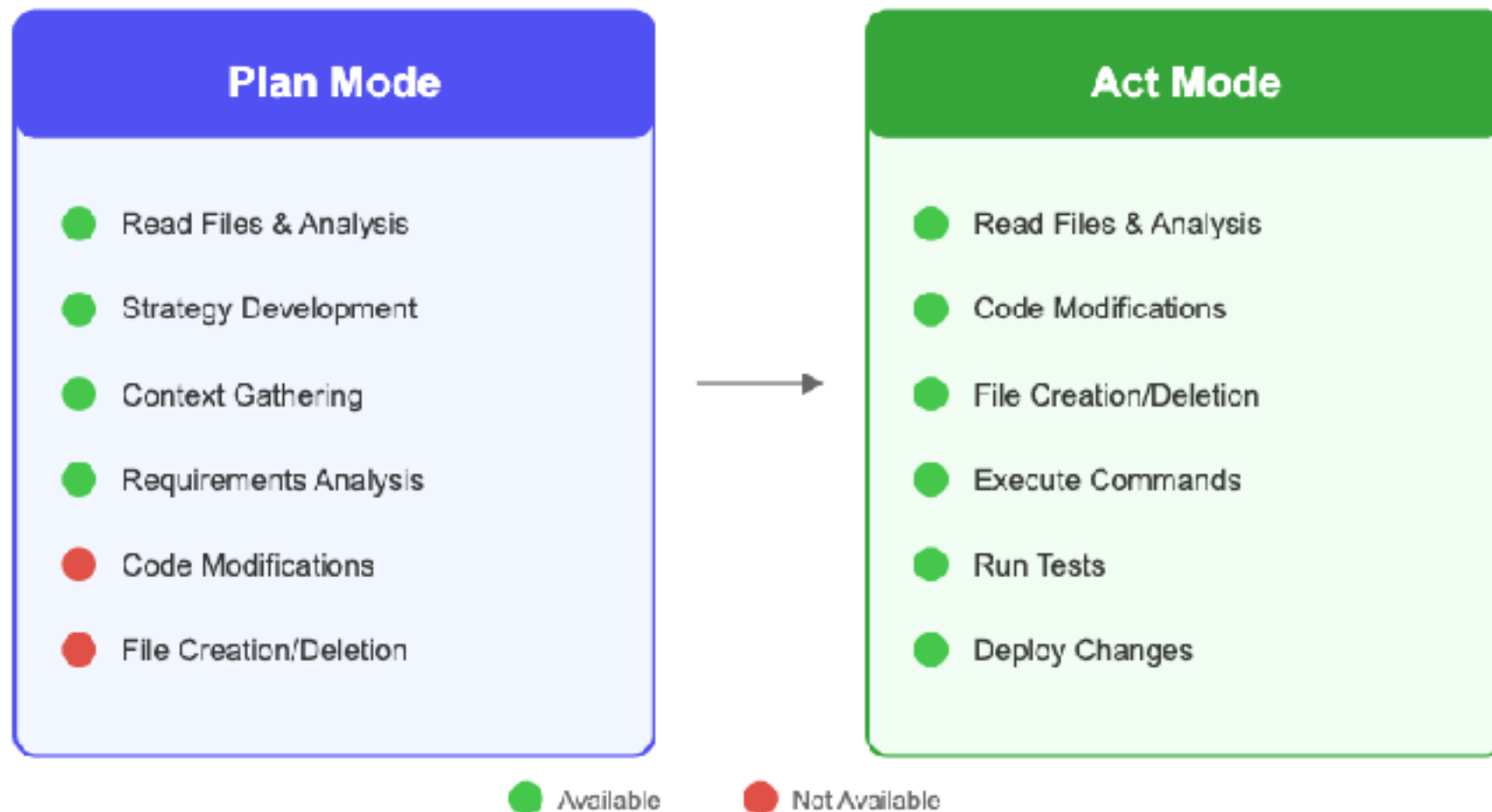


<https://docs.cline.bot/features/plan-and-act>



Plan and Act mode

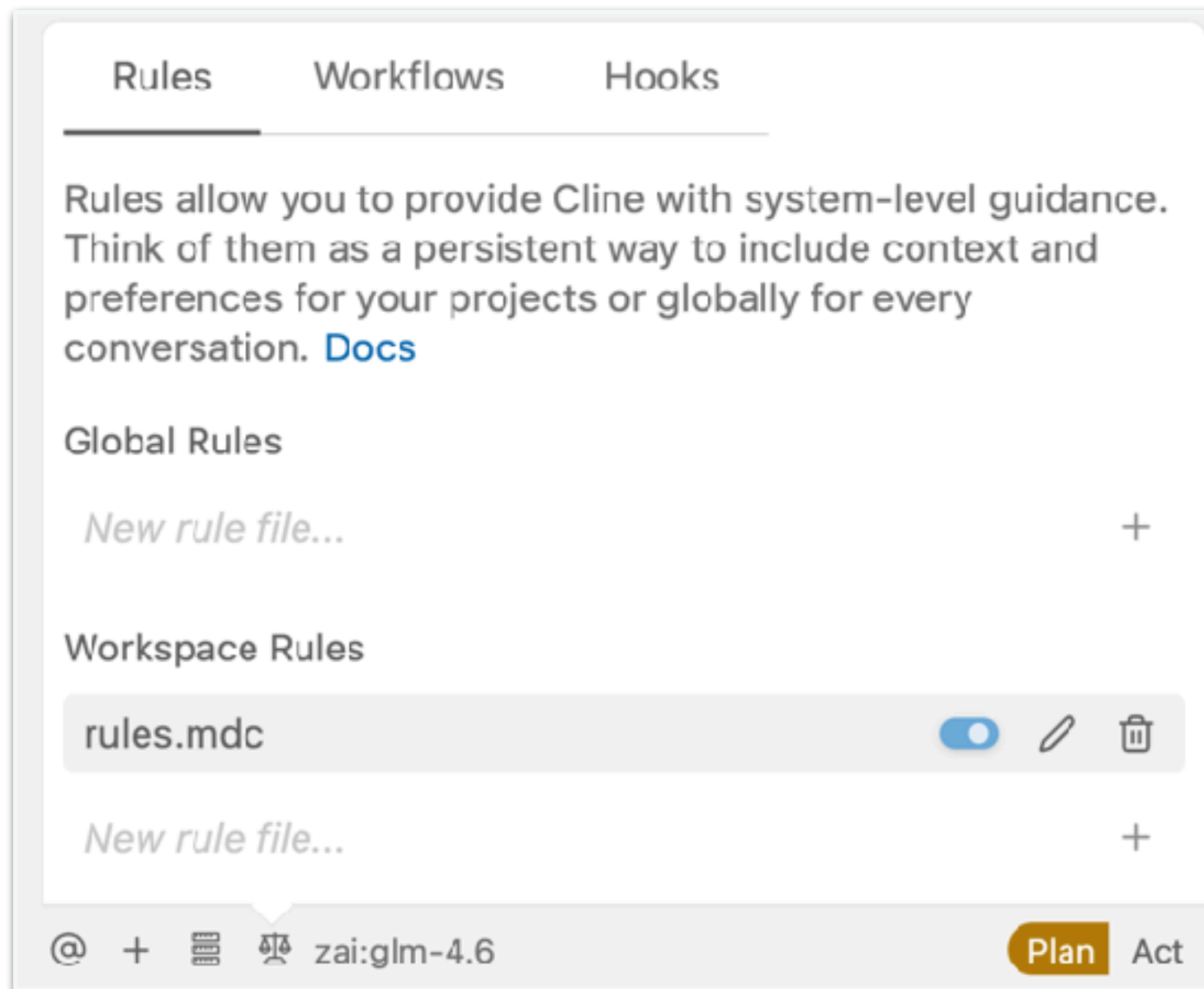
Plan vs Act Mode Capabilities



<https://docs.cline.bot/features/plan-and-act>

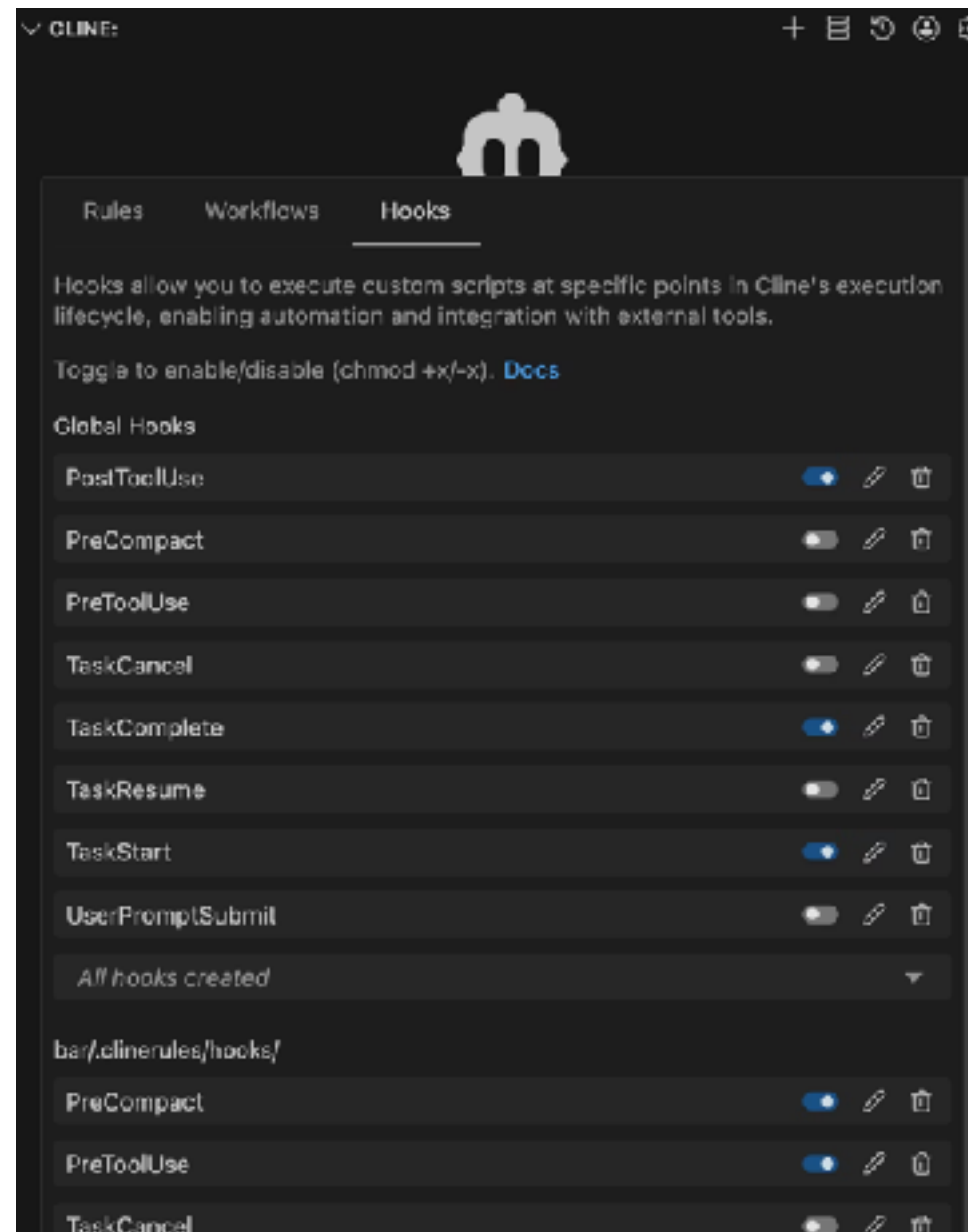


Manage rules/workflow/hooks



Hooks

Inject custom logics into Cline's workflow



<https://docs.cline.bot/features/hooks>



Auto-approve !!

Read and write files
Execute safe commands

Auto-approve: Read (all), Edit (all), All Commands, MCP

Let Cline take these actions without asking for approval. [Docs](#)

☒

 Read project files

☒

 Read all files

☒

 Edit project files

☒

 Edit all files

☒

 Execute safe commands

☒

 Execute all commands

☐

 Use the browser

☒

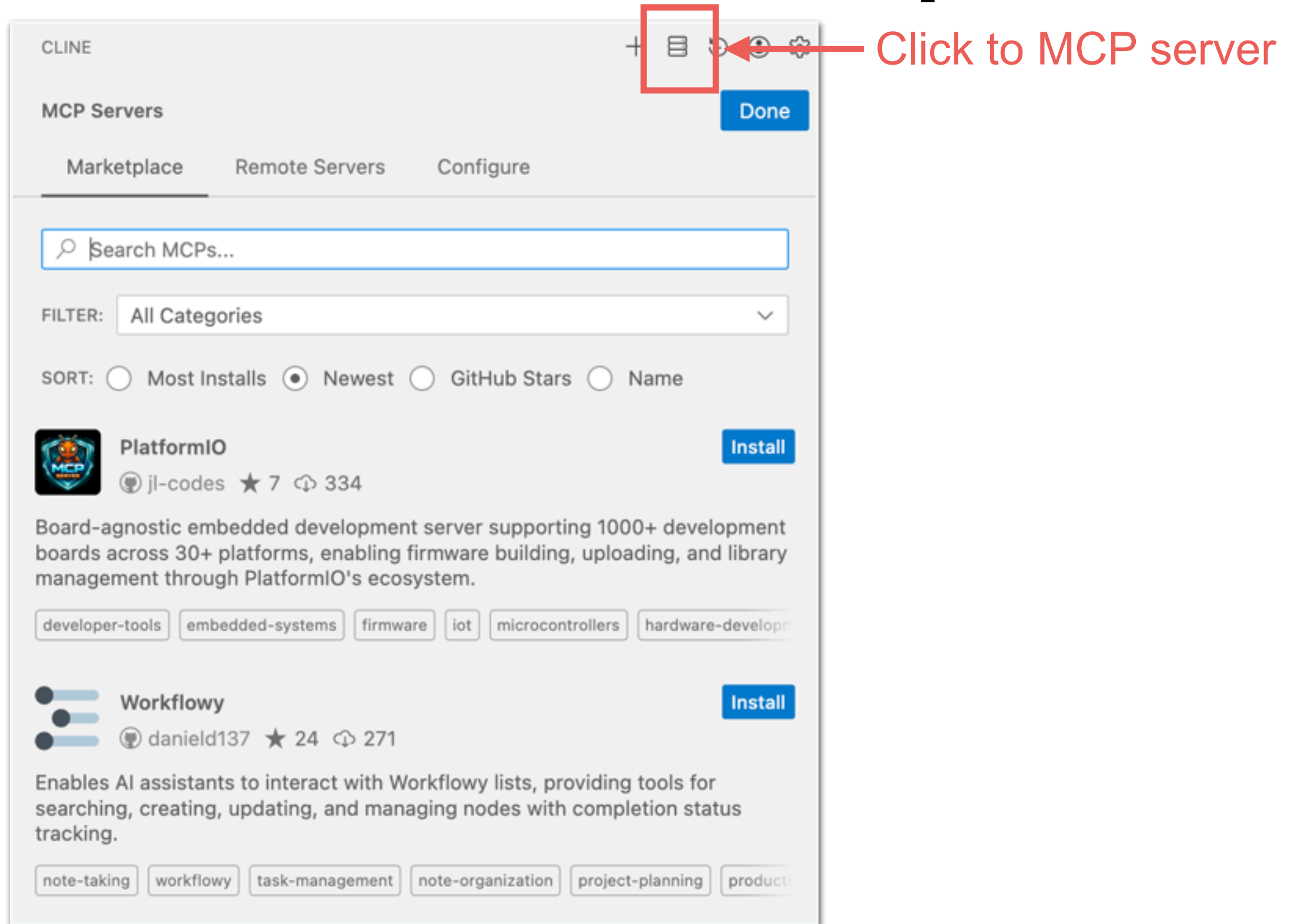
 Use MCP servers

☐

Enable notifications



Enable MCP in workspace



The screenshot shows the CLINE MCP Servers interface. At the top right, there is a red box around a menu icon (three horizontal lines) with a red arrow pointing to it and the text "Click to MCP server". Below this is a "Done" button. The interface has three tabs: "Marketplace", "Remote Servers", and "Configure". The "Marketplace" tab is active. It features a search bar labeled "Search MCPs...", a filter dropdown set to "All Categories", and sort options: "Most Installs", "Newest" (selected), "GitHub Stars", and "Name". Two MCP servers are listed: "PlatformIO" by "jl-codes" (7 stars, 334 forks) and "Workflowy" by "danield137" (24 stars, 271 forks). Each server has an "Install" button. The PlatformIO server description is "Board-agnostic embedded development server supporting 1000+ development boards across 30+ platforms, enabling firmware building, uploading, and library management through PlatformIO's ecosystem." The Workflowy server description is "Enables AI assistants to interact with Workflowy lists, providing tools for searching, creating, updating, and managing nodes with completion status tracking." Both servers have tags below their descriptions.

CLINE

MCP Servers

Done

Marketplace Remote Servers Configure

Search MCPs...

FILTER: All Categories

SORT: ☐ Most Installs ☒ Newest ☐ GitHub Stars ☐ Name

 **PlatformIO** 
jl-codes ★ 7 334

Board-agnostic embedded development server supporting 1000+ development boards across 30+ platforms, enabling firmware building, uploading, and library management through PlatformIO's ecosystem.

developer-tools embedded-systems firmware iot microcontrollers hardware-development

 **Workflowy** 
danield137 ★ 24 271

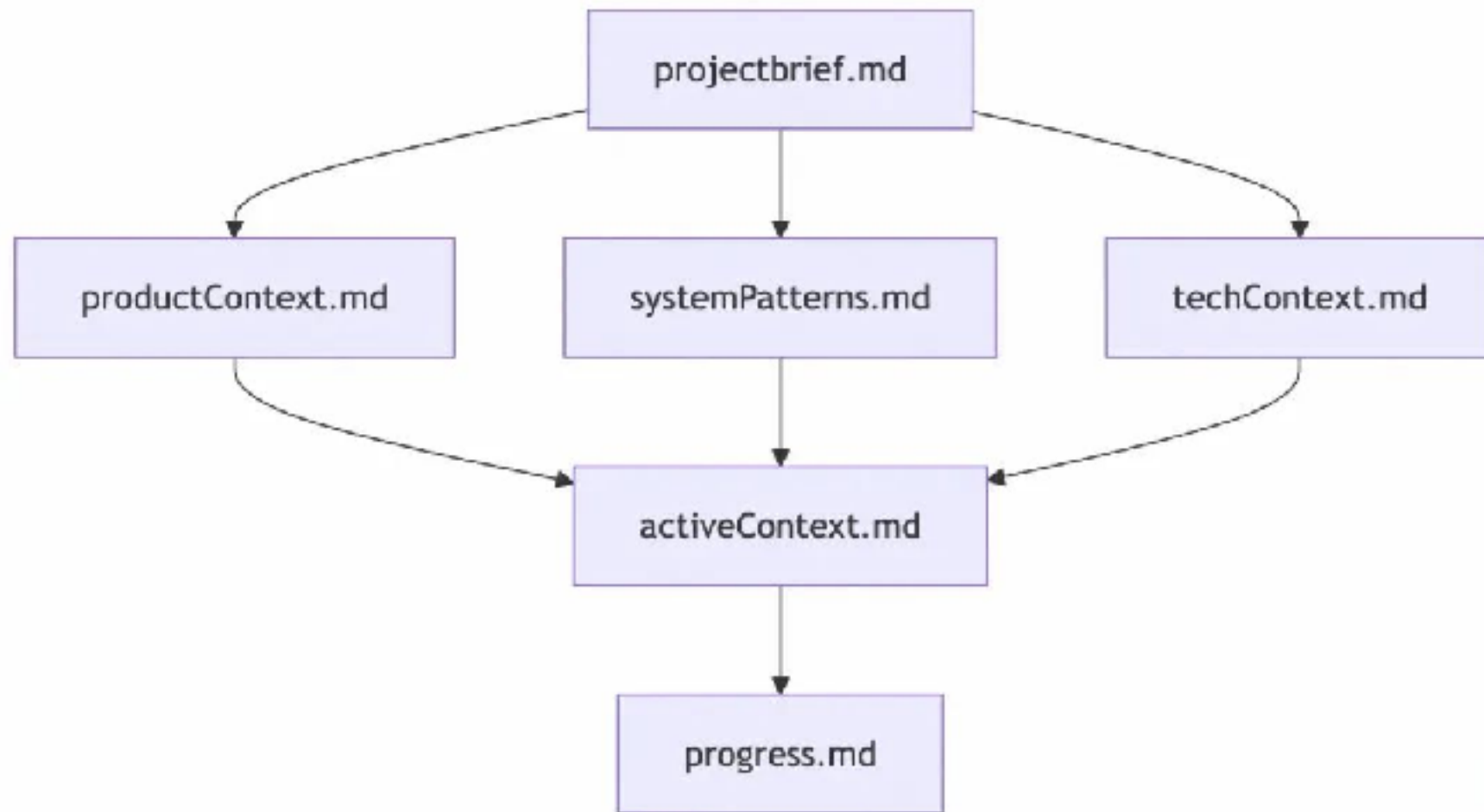
Enables AI assistants to interact with Workflowy lists, providing tools for searching, creating, updating, and managing nodes with completion status tracking.

note-taking workflowy task-management note-organization project-planning producti



Cline's Memory Bank

Manage context for AI across sessions
by using structured document



<https://docs.cline.bot/features/plan-and-act>



Agent Skills in Cline

☐ Enable Background Edit

Experimental: Allows editing files in background without opening the diff view in editor.

☐ Enable Multi-Root Workspace

Experimental: Allows cline to work across multiple workspaces.

☒ Enable Skills

Experimental: Enables Skills for reusable, on-demand agent instructions from `.cline/skills/` directories.

☐ Enable YOLO Mode

EXPERIMENTAL & DANGEROUS: This mode disables safety checks and user confirmations. Cline will automatically approve all actions without asking. Use with extreme caution.

Experimental feature !!



Q/A

